

Complex 2D+3D Systems

Multi-Degree-of-Freedom Equations of Motion

Compendium of Exact Analytical Solutions

Solution Notice:

A compendium of benchmark multi-degree-of-freedom non-linear dynamical systems and analytical kinematic reductions. BY LAAVNAY GUPTA

Contents

1 Problem 1: Dual Rolling Cylinders Coupled to Suspended Pendulums via Three Springs	7
1.1 Problem Diagram	7
1.2 System Description and Constants	7
1.3 Spring Extensions and Kinematic Relations	8
1.4 Velocity Components and Pure Rolling Constraints	8
1.5 Total Kinetic and Potential Energy	9
1.6 Lagrange Equations of Motion	9
1.7 Explicit Equations of Motion for Each Degree of Freedom	9
2 Problem 2: Two Coupled Rotating Discs with Oblique Spring and Suspended Pendulum	11
2.1 Problem Diagram	11
2.2 System Description and Constants	11
2.3 Position Coordinates and Velocity Components	11
2.4 Spring Geometry and Instantaneous Length	12
2.5 Total Kinetic and Potential Energy	12
2.6 Lagrange Equations of Motion	13
2.7 Explicit Equations of Motion for Each Degree of Freedom	13
3 Problem 3: Rolling Cylinder with Eccentric Pendulum on a Spring-Loaded Cart	14
3.1 1. Generalized Coordinates and Parameters	14
3.2 2. Kinematics	14
3.3 3. Kinetic and Potential Energy	15
3.4 4. Euler–Lagrange Equations	15
3.5 5. Explicit Equations of Motion	15

4	Problem 4: Translating Bead on an Elastically Supported Tilting Track	17
4.1	1. Coordinate System and Parameters	17
4.2	2. Kinematics	18
4.3	3. Kinetic and Potential Energy	18
4.4	4. Euler–Lagrange Equations	18
4.5	5. Explicit Equations of Motion	19
5	Problem 5: Rolling Ring on a Parabolic Track with Nested Disc and Elastic Pendulum	21
5.1	1. Generalized Coordinates and Parameters	21
5.2	2. Kinematics	22
5.3	3. Kinetic and Potential Energy	22
5.4	4. Euler–Lagrange Equations	23
5.5	5. Explicit Equations of Motion	23
6	Problem 6: Parallelogram Linked Discs with Eccentric Pendulum on a Parabolic Track	25
6.1	1. Generalized Coordinates and Parameters	25
6.2	2. Kinematics	25
6.3	3. Kinetic and Potential Energy	26
6.4	4. Euler–Lagrange Equations	26
6.5	5. Explicit Equations of Motion	27
7	Problem 7: Two Pure-Rolling Discs with Interconnecting and Anchor Springs	29
7.1	1. Generalized Coordinates and Parameters	29
7.2	2. Kinematics and Spring Geometry	30
7.3	3. Kinetic and Potential Energy	30
7.4	4. Euler–Lagrange Equations	30
7.5	5. Explicit Equations of Motion	31
8	Problem 8: Triple Inverted Rod System Coupled by Multi-Spring Network	32
8.1	1. Generalized Coordinates and Parameters	32
8.2	2. Kinematics and Spring Geometry	33
8.3	3. Kinetic and Potential Energy	33
8.4	4. Euler–Lagrange Equations	33
8.5	5. Explicit Equations of Motion	34
9	Problem 9: Rocking Frame with Spring-Loaded Slider and Suspended Pendulum	35
9.1	1. Generalized Coordinates and Parameters	35

9.2	2. Kinematics	35
9.3	3. Kinetic and Potential Energy	36
9.4	4. Euler–Lagrange Equations	37
9.5	5. Explicit Equations of Motion	37
10	Problem 10: Pure-Rolling Disc with Triple Suspended Pendulums	39
10.1	1. Generalized Coordinates and Parameters	39
10.2	2. Kinematics	40
10.3	3. Kinetic and Potential Energy	40
10.4	4. Euler–Lagrange Equations	40
10.5	5. Decoupled Explicit Equations of Motion	41
11	Problem 11: Regular Octagonal Rocker on Parabolic Track with Elastic Pendulum	42
11.1	1. Generalized Coordinates and Parameters	42
11.2	2. Kinematics	42
11.3	3. Kinetic and Potential Energy	43
11.4	4. Euler–Lagrange Equations	43
11.5	5. Explicit Equations of Motion	44
12	Problem 12: Rocking Semicircular Shell with Nested Rolling Cylinder and T-Bar Pendulum	45
12.1	1. Generalized Coordinates and Parameters	45
12.2	2. Kinematics and Spring Geometry	46
12.3	3. Kinetic and Potential Energy	46
12.4	4. Euler–Lagrange Equations	47
12.5	5. Explicit Decoupled Equations of Motion	48
13	Problem 13: Pure-Rolling Cylinder on a Vertical Wall with Offset Spring and Suspended Pendulum	49
13.1	1. Generalized Coordinates and Parameters	49
13.2	2. Kinematics and Spring Geometry	50
13.3	3. Kinetic and Potential Energy	50
13.4	4. Euler–Lagrange Equations	50
13.5	5. Explicit Decoupled Equations of Motion	51
14	Problem 14: Parallel-Link Suspended Frame with Spring-Loaded Slider and Pendulum	52
14.1	1. Generalized Coordinates and Parameters	52

14.2	2. Kinematics	53
14.3	3. Kinetic and Potential Energy	53
14.4	4. Euler–Lagrange Equations	53
14.5	5. Explicit Decoupled Equations of Motion	54
15	Problem 15: Suspended Platform with Dual Cylinders and Cable Constraint	55
15.1	Coordinate System and Parameters	55
15.2	Kinematics	56
15.3	Kinetic and Potential Energy	56
15.4	Case 1: Unconstrained System Formulation	56
15.5	Case 2: Constrained System via Lagrange Multipliers	57
15.6	Modified Equations of Motion	57
16	Problem 16: Seven-Body Suspended Multi-Link Viscoelastic Network under Harmonic Forcing	59
16.1	Generalized Coordinates and System Parameters	59
16.2	Kinematic Chain and Viscoelastic Internal Forces	60
16.3	Kinetic Energy, Potential Energy, and Dissipation	60
16.4	Euler–Lagrange Equations with Dissipation and External Work	61
16.5	Explicit Equations of Motion	61
17	Problem 17: Parallel-Suspended Central Block with Flanking Viscoelastic Blocks and Dual Pendulums	63
17.1	Generalized Coordinates and Parameters	63
17.2	Kinematics	64
17.3	Kinetic Energy, Potential Energy, and Dissipation	64
17.4	Euler–Lagrange Equations	64
17.5	Explicit Equations of Motion	65
18	Problem 18: Parabolic Track with Nested Cylinder and Link-Coupled External Disc	66
18.1	Generalized Coordinates and Parameters	66
18.2	Kinematics and Constraint Geometry	67
18.3	Kinetic and Potential Energy Formulations	68
18.4	Euler–Lagrange Equations with Constraint Multipliers	68
18.5	Explicit Decoupled Equations of Motion	69
19	Problem 19: Elastically Supported Guiding Bar with Freely Moving Discs	70
19.1	Generalized Coordinates and Parameters	70

19.2 Kinematics and Spring Geometry	70
19.3 Kinetic and Potential Energy	71
19.4 Euler–Lagrange Equations	71
19.5 Explicit Equations of Motion	71
20 Problem 20: Parallel-Suspended E-Frame with Dual Spring-Sliders and Pendulums	73
20.1 Generalized Coordinates and Parameters	73
20.2 Kinematics	73
20.3 Kinetic and Potential Energy	74
20.4 Euler–Lagrange Equations	74
20.5 Explicit Equations of Motion	74
21 Problem 21: Coupled Wheels with Connecting Rod, Spring-Slider, and T-Bar Pendulum	76
21.1 Coordinate Reduction and Kinematic Loop Closure	76
21.2 Kinematics	77
21.3 Kinetic and Potential Energy	77
21.4 Euler–Lagrange Equations	77
21.5 Explicit Equations of Motion	78
22 Problem 22: Rocker-Coupled Flywheel with Spring-Piston and T-Bar Pendulum	79
22.1 Kinematic Reduction and Generalized Coordinates	79
22.2 Kinematic Jacobians and Loop Relations	80
22.3 Kinematics of the Suspended T-Bar Pendulum	80
22.4 Kinetic and Potential Energy	80
22.5 Euler–Lagrange Equations	81
22.6 Explicit Equations of Motion	81
23 Problem 23: Rolling Ring on Parabolic Track with Linked Discs and Elastic Pendulum	82
23.1 Generalized Coordinates and Parameters	82
23.2 Kinematics and Velocity Cascade	83
23.3 Kinetic and Potential Energy Formulations	83
23.4 Euler–Lagrange Equations	84
23.5 Explicit Equations of Motion	85
24 Problem 24: Spring-Supported Platform with Rocking Semicircular Shell, Dual Rolling Discs, and Pendulum	87

24.1 Generalized Coordinates and System Parameters	87
24.2 Kinematics and Spring Geometry	88
24.3 Kinetic and Potential Energy Formulations	89
24.4 Euler–Lagrange Equations	89
24.5 Explicit Equations of Motion	90
25 Problem 25: Cable-Suspended Pendulum Coupled to Dual Spring-Mass Pulleys	91
25.1 Generalized Coordinates and Parameters	91
25.2 Kinematics and Cable Constraint	92
25.3 Kinetic and Potential Energy Formulations	92
25.4 Euler–Lagrange Equations	92
25.5 Explicit Decoupled Equations of Motion	93
26 Problem 26: Spatial Tracking Mechanism on an Inclined Turntable with Rolling Disc and Spring-Loaded Arm	94
26.1 Generalized Coordinates and Parameters	94
26.2 Kinematics	95
26.3 Kinetic and Potential Energy	95
26.4 Euler–Lagrange Equations	95
26.5 Explicit Decoupled Equations of Motion	96

1 Problem 1: Dual Rolling Cylinders Coupled to Suspended Pendulums via Three Springs

1.1 Problem Diagram

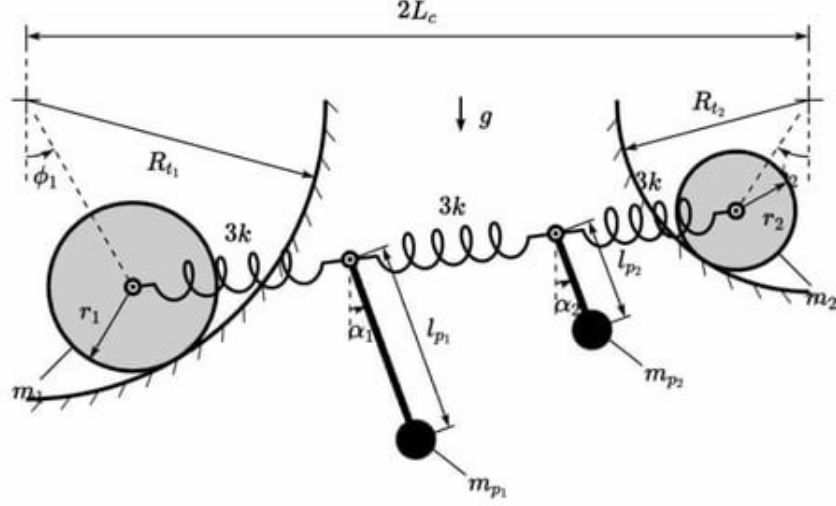


Figure 1: Schematic of dual rolling cylinders coupled to suspended pendulums via three springs.

1.2 System Description and Constants

The system consists of two rolling bodies on curved surfaces, two suspended pendulums with translating hinges, and three interconnecting springs, all oscillating in a two-dimensional vertical plane under gravity g :

- **Curved tracks:** Spaced symmetrically about a central origin with total reference span $2L_c$.
- **Left rolling body:** Mass M_1 , radius r_1 , track curvature radius R_1 , moment of inertia I_1 , angular position on the track ϕ_1 , and rotation angle θ_1 .
- **Right rolling body:** Mass M_2 , radius r_2 , track curvature radius R_2 , moment of inertia I_2 , angular position on the track ϕ_2 , and rotation angle θ_2 .
- **Pendulums:**
 - Left pendulum: Bob mass m_1 , massless rod of length l_1 , swing angle α_1 , and hinge translation coordinate x_{p1} .
 - Right pendulum: Bob mass m_2 , massless rod of length l_2 , swing angle α_2 , and hinge translation coordinate x_{p2} .
- **Coupling springs:** All three springs have stiffness $3k$ and identical unstretched natural length L_0 . The extensions of the leftmost, rightmost, and intermediate springs are denoted x_1 , x_2 , and x_3 , respectively.

1.3 Spring Extensions and Kinematic Relations

With the origin placed at the geometric center:

$$\text{Extension of leftmost spring } (x_1) = -L_c + (R_1 - r_1) \sin \phi_1 + x_{p_1} - L_0 \quad (1)$$

$$\text{Extension of rightmost spring } (x_2) = L_c - (R_2 - r_2) \sin \phi_2 + x_{p_2} - L_0 \quad (2)$$

For the middle spring, the geometric constraint over the span $2L_c$ gives:

$$2L_c = (L_0 + x_1) + (R_1 - r_1) \sin \phi_1 + (L_0 + x_2) + (R_2 - r_2) \sin \phi_2 + (L_0 + x_3) \quad (3)$$

Rearranging to solve for the extension x_3 :

$$x_3 = -(L_0 + x_1 + (R_1 - r_1) \sin \phi_1 + L_0 + x_2 - (R_2 - r_2) \sin \phi_2 - 2L_c + L_0) \quad (4)$$

1.4 Velocity Components and Pure Rolling Constraints

1. Left rolling cylinder center of mass:

$$v_x = \frac{d}{dt} (-L_c + (R_1 - r_1) \sin \phi_1) = (R_1 - r_1) \dot{\phi}_1 \cos \phi_1 \quad (5)$$

$$v_y = \frac{d}{dt} ((R_1 - r_1)(1 - \cos \phi_1)) = (R_1 - r_1) \dot{\phi}_1 \sin \phi_1 \quad (6)$$

2. Right rolling cylinder center of mass:

$$v'_x = \frac{d}{dt} (L_c - (R_2 - r_2) \sin \phi_2) = -(R_2 - r_2) \dot{\phi}_2 \cos \phi_2 \quad (7)$$

$$v'_y = \frac{d}{dt} ((R_2 - r_2)(1 - \cos \phi_2)) = (R_2 - r_2) \dot{\phi}_2 \sin \phi_2 \quad (8)$$

3. Left pendulum bob:

$$v_{2x} = \frac{d}{dt} (x_{p_1} + l_1 \sin \alpha_1) = \dot{x}_{p_1} + l_1 \dot{\alpha}_1 \cos \alpha_1 \quad (9)$$

$$v_{2y} = \frac{d}{dt} (l_1 \cos \alpha_1) = -l_1 \dot{\alpha}_1 \sin \alpha_1 \quad (10)$$

4. Right pendulum bob:

$$v'_{2x} = \frac{d}{dt} (x_{p_2} + l_2 \sin \alpha_2) = \dot{x}_{p_2} + l_2 \dot{\alpha}_2 \cos \alpha_2 \quad (11)$$

$$v'_{2y} = \frac{d}{dt} (l_2 \cos \alpha_2) = -l_2 \dot{\alpha}_2 \sin \alpha_2 \quad (12)$$

5. No-slip rolling constraints:

$$\dot{\theta}_1 = \left(\frac{R_1 - r_1}{r_1} \right) \dot{\phi}_1, \quad \dot{\theta}_2 = \left(\frac{R_2 - r_2}{r_2} \right) \dot{\phi}_2 \quad (13)$$

1.5 Total Kinetic and Potential Energy

Kinetic Energy (KE):

$$\begin{aligned}
KE = & \frac{1}{2} \left(M_1 + \frac{I_1}{r_1^2} \right) (R_1 - r_1)^2 \dot{\phi}_1^2 + \frac{1}{2} \left(M_2 + \frac{I_2}{r_2^2} \right) (R_2 - r_2)^2 \dot{\phi}_2^2 \\
& + \frac{1}{2} m_1 (\dot{x}_{p1}^2 + l_1^2 \dot{\alpha}_1^2 + 2l_1 \dot{x}_{p1} \dot{\alpha}_1 \cos \alpha_1) \\
& + \frac{1}{2} m_2 (\dot{x}_{p2}^2 + l_2^2 \dot{\alpha}_2^2 + 2l_2 \dot{x}_{p2} \dot{\alpha}_2 \cos \alpha_2)
\end{aligned} \tag{14}$$

where $\dot{x}_{p1} = (R_1 - r_1)\dot{\phi}_1 \cos \phi_1 + \dot{x}_1$ and $\dot{x}_{p2} = (R_2 - r_2)\dot{\phi}_2 \cos \phi_2 + \dot{x}_2$.

Potential Energy (PE):

$$\begin{aligned}
PE = & M_1 g (R_1 - r_1) (1 - \cos \phi_1) + M_2 g (R_2 - r_2) (1 - \cos \phi_2) \\
& + m_1 g l_1 (1 - \cos \alpha_1) + m_2 g l_2 (1 - \cos \alpha_2) \\
& + \frac{1}{2} (3k) x_1^2 + \frac{1}{2} (3k) x_2^2 + \frac{1}{2} (3k) x_3^2
\end{aligned} \tag{15}$$

1.6 Lagrange Equations of Motion

1. Degree of freedom α_1 :

$$l_1 \ddot{\alpha}_1 + \ddot{x}_{p1} \cos \alpha_1 + g \sin \alpha_1 = 0 \tag{16}$$

2. Degree of freedom α_2 :

$$l_2 \ddot{\alpha}_2 + \ddot{x}_{p2} \cos \alpha_2 + g \sin \alpha_2 = 0 \tag{17}$$

3. Degree of freedom x_1 :

$$m_1 \ddot{x}_{p1} + m_1 l_1 (\ddot{\alpha}_1 \cos \alpha_1 - \dot{\alpha}_1^2 \sin \alpha_1) + 3k(x_1 - x_3) = 0 \tag{18}$$

4. Degree of freedom x_2 :

$$m_2 \ddot{x}_{p2} + m_2 l_2 (\ddot{\alpha}_2 \cos \alpha_2 - \dot{\alpha}_2^2 \sin \alpha_2) + 3k(x_3 - x_2) = 0 \tag{19}$$

5. Degree of freedom ϕ_1 :

$$\left(M_1 + \frac{I_1}{r_1^2} \right) (R_1 - r_1) \ddot{\phi}_1 + M_1 g \sin \phi_1 - 3k x_1 \cos \phi_1 = 0 \tag{20}$$

6. Degree of freedom ϕ_2 :

$$\left(M_2 + \frac{I_2}{r_2^2} \right) (R_2 - r_2) \ddot{\phi}_2 + M_2 g \sin \phi_2 - 3k x_2 \cos \phi_2 = 0 \tag{21}$$

1.7 Explicit Equations of Motion for Each Degree of Freedom

1. Left rolling body ($\ddot{\phi}_1$):

$$\ddot{\phi}_1 = \frac{3k x_1 \cos \phi_1 - M_1 g \sin \phi_1}{\left(M_1 + \frac{I_1}{r_1^2} \right) (R_1 - r_1)} \tag{22}$$

2. Right rolling body ($\ddot{\phi}_2$):

$$\ddot{\phi}_2 = \frac{3kx_2 \cos \phi_2 - M_2 g \sin \phi_2}{\left(M_2 + \frac{I_2}{r_2^2}\right) (R_2 - r_2)} \quad (23)$$

3. Left pendulum bob ($\ddot{\alpha}_1$):

$$\ddot{\alpha}_1 = \frac{-g \sin \alpha_1 - \ddot{x}_{p1} \cos \alpha_1}{l_1} \quad (24)$$

4. Right pendulum bob ($\ddot{\alpha}_2$):

$$\ddot{\alpha}_2 = \frac{-g \sin \alpha_2 - \ddot{x}_{p2} \cos \alpha_2}{l_2} \quad (25)$$

5. Left hinge translation ($\ddot{x}_1$):

$$\begin{aligned} \ddot{x}_1 = & \frac{3k(x_3 - x_1)}{m_1} - (R_1 - r_1)\ddot{\phi}_1 \cos \phi_1 + (R_1 - r_1)\dot{\phi}_1^2 \sin \phi_1 \\ & - l_1 \ddot{\alpha}_1 \cos \alpha_1 + l_1 \dot{\alpha}_1^2 \sin \alpha_1 \end{aligned} \quad (26)$$

6. Right hinge translation ($\ddot{x}_2$):

$$\begin{aligned} \ddot{x}_2 = & \frac{3k(x_3 - x_2)}{m_2} - (R_2 - r_2)\ddot{\phi}_2 \cos \phi_2 + (R_2 - r_2)\dot{\phi}_2^2 \sin \phi_2 \\ & + l_2 \ddot{\alpha}_2 \cos \alpha_2 - l_2 \dot{\alpha}_2^2 \sin \alpha_2 \end{aligned} \quad (27)$$

2 Problem 2: Two Coupled Rotating Discs with Oblique Spring and Suspended Pendulum

2.1 Problem Diagram

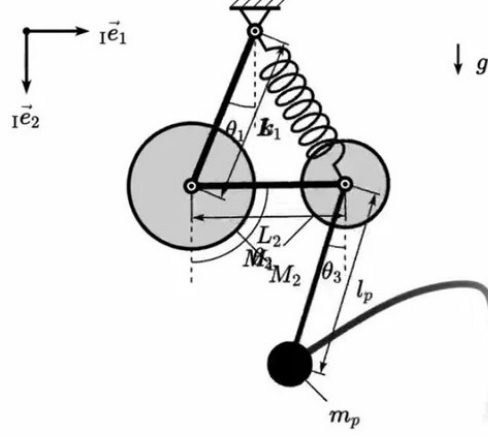


Figure 2: Schematic of two interconnected rotating discs with an oblique spring coupling to the fixed pivot and a suspended pendulum bob

2.2 System Description and Constants

The system consists of a top fixed pivot, two rigid links holding rotating bodies, a coupling spring to the fixed pivot, and a trailing pendulum bob:

- **First Link and First Disc:** Massless rod of length L_1 hinged to the fixed origin $(0, 0)$ oscillating at angle θ relative to the vertical downward axis. At its distal end is a disc of mass M_1 and moment of inertia I_1 .
- **Second Link and Second Disc:** Massless rod of length L_2 hinged to the center of the first disc, oriented at angle ϕ relative to the vertical axis. At its distal end is a second disc of mass M_2 and moment of inertia I_2 .
- **Pendulum Bob:** Massless rod of length l hinged to the center of the second disc, carrying a concentrated bob of mass m at swing angle α relative to the downward vertical.
- **Coupling Spring:** Linear extension spring of stiffness k and unstretched natural length L_0 , attached between the top fixed pivot $(0, 0)$ and the center of the second disc M_2 .

2.3 Position Coordinates and Velocity Components

Taking the top fixed pivot as origin $(0, 0)$ with x horizontal (positive rightward) and y vertical (positive downward):

1. Center of mass of first disc (M_1):

$$x_1 = -L_1 \sin \theta, \quad y_1 = L_1 \cos \theta \quad (28)$$

$$v_x = \frac{d}{dt} (-L_1 \sin \theta) = -L_1 \dot{\theta} \cos \theta \quad (29)$$

$$v_y = \frac{d}{dt} (L_1 \cos \theta) = -L_1 \dot{\theta} \sin \theta \quad (30)$$

2. Center of mass of second disc (M_2):

$$x_2 = L_2 \sin \phi - L_1 \sin \theta, \quad y_2 = L_1 \cos \theta + L_2 \cos \phi \quad (31)$$

$$v'_x = \frac{d}{dt} (L_2 \sin \phi - L_1 \sin \theta) = L_2 \dot{\phi} \cos \phi - L_1 \dot{\theta} \cos \theta \quad (32)$$

$$v'_y = \frac{d}{dt} (L_1 \cos \theta + L_2 \cos \phi) = -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \quad (33)$$

3. Position and velocity of pendulum bob (m):

$$x_p = L_2 \sin \phi - L_1 \sin \theta + l \sin \alpha \quad (34)$$

$$y_p = L_1 \cos \theta + L_2 \cos \phi + l \cos \alpha \quad (35)$$

$$v_{2x} = L_2 \dot{\phi} \cos \phi - L_1 \dot{\theta} \cos \theta + l \dot{\alpha} \cos \alpha \quad (36)$$

$$v_{2y} = -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi - l \dot{\alpha} \sin \alpha \quad (37)$$

2.4 Spring Geometry and Instantaneous Length

The instantaneous stretched length L_s of the spring connecting the fixed pivot $(0, 0)$ to disc M_2 is:

$$\begin{aligned} L_s &= \sqrt{(L_2 \sin \phi - L_1 \sin \theta)^2 + (L_1 \cos \theta + L_2 \cos \phi)^2} \\ &= \sqrt{L_1^2 + L_2^2 + 2L_1 L_2 \cos(\theta + \phi)} \end{aligned} \quad (38)$$

2.5 Total Kinetic and Potential Energy

Kinetic Energy (KE):

$$\begin{aligned} KE &= \frac{1}{2} (M_1 L_1^2 + I_1) \dot{\theta}^2 + \frac{1}{2} M_2 \left[L_1^2 \dot{\theta}^2 + L_2^2 \dot{\phi}^2 - 2L_1 L_2 \dot{\theta} \dot{\phi} \cos(\theta + \phi) \right] + \frac{1}{2} I_2 \dot{\phi}^2 \\ &\quad + \frac{1}{2} m \left[L_1^2 \dot{\theta}^2 + L_2^2 \dot{\phi}^2 + l^2 \dot{\alpha}^2 - 2L_1 L_2 \dot{\theta} \dot{\phi} \cos(\theta + \phi) \right. \\ &\quad \left. + 2L_2 l \dot{\phi} \dot{\alpha} \cos(\phi - \alpha) - 2L_1 l \dot{\theta} \dot{\alpha} \cos(\theta + \alpha) \right] \end{aligned} \quad (39)$$

Potential Energy (PE):

$$\begin{aligned} PE &= -(M_1 + M_2 + m)gL_1 \cos \theta - (M_2 + m)gL_2 \cos \phi - mgl \cos \alpha \\ &\quad + \frac{1}{2} k \left(\sqrt{L_1^2 + L_2^2 + 2L_1 L_2 \cos(\theta + \phi)} - L_0 \right)^2 \end{aligned} \quad (40)$$

2.6 Lagrange Equations of Motion

1. Degree of freedom θ :

$$\begin{aligned} & ((M_1 + M_2 + m)L_1^2 + I_1) \ddot{\theta} - (M_2 + m)L_1L_2\ddot{\phi}\cos(\theta + \phi) - mL_1l\ddot{\alpha}\cos(\theta + \alpha) \\ & + (M_2 + m)L_1L_2\dot{\phi}^2\sin(\theta + \phi) + mL_1l\dot{\alpha}^2\sin(\theta + \alpha) + (M_1 + M_2 + m)gL_1\sin\theta \\ & - k\left(1 - \frac{L_0}{L_s}\right)L_1L_2\sin(\theta + \phi) = 0 \end{aligned} \quad (41)$$

2. Degree of freedom ϕ :

$$\begin{aligned} & ((M_2 + m)L_2^2 + I_2) \ddot{\phi} - (M_2 + m)L_1L_2\ddot{\theta}\cos(\theta + \phi) + mL_2l\ddot{\alpha}\cos(\phi - \alpha) \\ & + (M_2 + m)L_1L_2\dot{\theta}^2\sin(\theta + \phi) + mL_2l\dot{\alpha}^2\sin(\phi - \alpha) + (M_2 + m)gL_2\sin\phi \\ & - k\left(1 - \frac{L_0}{L_s}\right)L_1L_2\sin(\theta + \phi) = 0 \end{aligned} \quad (42)$$

3. Degree of freedom α :

$$l\ddot{\alpha} + L_2\ddot{\phi}\cos(\phi - \alpha) - L_1\ddot{\theta}\cos(\theta + \alpha) - L_2\dot{\phi}^2\sin(\phi - \alpha) + L_1\dot{\theta}^2\sin(\theta + \alpha) + g\sin\alpha = 0 \quad (43)$$

2.7 Explicit Equations of Motion for Each Degree of Freedom

1. Rotation of first link ($\ddot{\theta}$):

$$\begin{aligned} \ddot{\theta} = & \frac{1}{(M_1 + M_2 + m)L_1^2 + I_1} \left[- (M_1 + M_2 + m)gL_1\sin\theta \right. \\ & + (M_2 + m)L_1L_2 \left(\ddot{\phi}\cos(\theta + \phi) - \dot{\phi}^2\sin(\theta + \phi) \right) \\ & + mL_1l \left(\ddot{\alpha}\cos(\theta + \alpha) - \dot{\alpha}^2\sin(\theta + \alpha) \right) \\ & \left. + k\left(1 - \frac{L_0}{L_s}\right)L_1L_2\sin(\theta + \phi) \right] \end{aligned} \quad (44)$$

2. Rotation of second link ($\ddot{\phi}$):

$$\begin{aligned} \ddot{\phi} = & \frac{1}{(M_2 + m)L_2^2 + I_2} \left[- (M_2 + m)gL_2\sin\phi \right. \\ & + (M_2 + m)L_1L_2 \left(\ddot{\theta}\cos(\theta + \phi) - \dot{\theta}^2\sin(\theta + \phi) \right) \\ & - mL_2l \left(\ddot{\alpha}\cos(\phi - \alpha) + \dot{\alpha}^2\sin(\phi - \alpha) \right) \\ & \left. + k\left(1 - \frac{L_0}{L_s}\right)L_1L_2\sin(\theta + \phi) \right] \end{aligned} \quad (45)$$

3. Oscillation of pendulum bob ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{-g\sin\alpha + L_1 \left(\ddot{\theta}\cos(\theta + \alpha) - \dot{\theta}^2\sin(\theta + \alpha) \right) - L_2 \left(\ddot{\phi}\cos(\phi - \alpha) - \dot{\phi}^2\sin(\phi - \alpha) \right)}{l} \quad (46)$$

3 Problem 3: Rolling Cylinder with Eccentric Pendulum on a Spring-Loaded Cart

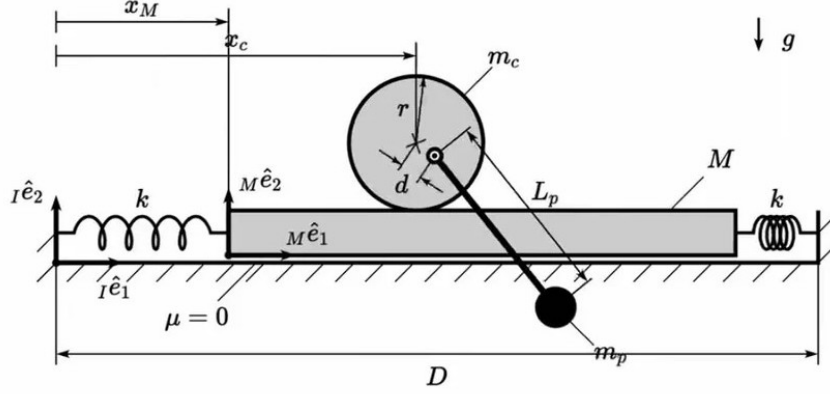


Figure 3: Schematic of a cart supported by two linear springs of stiffness k , carrying a pure-rolling cylinder with an eccentric hinge offset by distance d , from which a simple pendulum of length l hangs.

3.1 1. Generalized Coordinates and Parameters

The planar system has three degrees of freedom:

- $x(t)$: Horizontal displacement of the cart of mass M_2 .
- $\theta(t)$: Pure-rolling rotation angle of the cylinder of mass M_1 , radius r , and moment of inertia I .
- $\phi(t)$: Angular deflection of the pendulum bob of mass m and length l , measured clockwise from the downward vertical.
- d : Radial offset of the eccentric hinge from the cylinder center.
- k : Stiffness of each of the two identical horizontal springs flanking the cart.

3.2 2. Kinematics

The horizontal position of the rolling cylinder center relative to the origin is:

$$y = x + r\theta \implies V = \dot{y} = \dot{x} + r\dot{\theta} \quad (47)$$

The Cartesian position of the pendulum hinge is $(y - d \sin \theta, -d \cos \theta)$. Adding the pendulum bob displacements yields:

$$x_{\text{bob}} = y - d \sin \theta + l \sin \phi = x + r\theta - d \sin \theta + l \sin \phi \quad (48)$$

$$y_{\text{bob}} = -(d \cos \theta + l \cos \phi) \quad (49)$$

Differentiating with respect to time gives the velocity components of the bob:

$$v_x = \dot{x} + (r - d \cos \theta) \dot{\theta} + l \dot{\phi} \cos \phi \quad (50)$$

$$v_y = -d \dot{\theta} \sin \theta - l \dot{\phi} \sin \phi \quad (51)$$

3.3 3. Kinetic and Potential Energy

The total kinetic energy combines the translational energy of the cart, the planar rigid-body energy of the cylinder, and the particle kinetic energy of the bob:

$$\begin{aligned} KE &= \frac{1}{2} M_2 \dot{x}^2 + \frac{1}{2} M_1 (\dot{x} + r \dot{\theta})^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} m (v_x^2 + v_y^2) \\ &= \frac{1}{2} (M_1 + M_2 + m) \dot{x}^2 + \frac{1}{2} [M_1 r^2 + I + m (r^2 + d^2 - 2rd \cos \theta)] \dot{\theta}^2 + \frac{1}{2} m l^2 \dot{\phi}^2 \\ &\quad + [M_1 r + m(r - d \cos \theta)] \dot{x} \dot{\theta} + m l \dot{x} \dot{\phi} \cos \phi + m l [r \cos \phi - d \cos(\theta + \phi)] \dot{\theta} \dot{\phi} \end{aligned} \quad (52)$$

Taking the datum at the cart top surface, the total potential energy stored in the two springs (each stretched/compressed by x) and gravity acting on the bob is:

$$PE = \frac{1}{2} k x^2 + \frac{1}{2} k x^2 - mg(d \cos \theta + l \cos \phi) = k x^2 - mg(d \cos \theta + l \cos \phi) \quad (53)$$

3.4 4. Euler–Lagrange Equations

Applying the Euler–Lagrange formula $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$ for each coordinate:

For coordinate x :

$$(M_1 + M_2 + m) \ddot{x} + [M_1 r + m(r - d \cos \theta)] \ddot{\theta} + m l \ddot{\phi} \cos \phi + m d \dot{\theta}^2 \sin \theta - m l \dot{\phi}^2 \sin \phi + 2kx = 0 \quad (54)$$

For coordinate θ :

$$\begin{aligned} &[M_1 r^2 + I + m (r^2 + d^2 - 2rd \cos \theta)] \ddot{\theta} + [M_1 r + m(r - d \cos \theta)] \ddot{x} \\ &\quad + m l [r \cos \phi - d \cos(\theta + \phi)] \ddot{\phi} + m r d \dot{\theta}^2 \sin \theta - m l r \dot{\phi}^2 \sin \phi \\ &\quad + m l d \dot{\phi}^2 \sin(\theta + \phi) + m g d \sin \theta = 0 \end{aligned} \quad (55)$$

For coordinate ϕ :

$$l \ddot{\phi} + \ddot{x} \cos \phi + [r \cos \phi - d \cos(\theta + \phi)] \ddot{\theta} + d \dot{\theta}^2 \sin(\theta + \phi) + g \sin \phi = 0 \quad (56)$$

3.5 5. Explicit Equations of Motion

Isolating the second-order time derivatives for each degree of freedom:

Cart Translation ($\ddot{x}$):

$$\ddot{x} = \frac{-2kx - [M_1 r + m(r - d \cos \theta)] \ddot{\theta} - m l \ddot{\phi} \cos \phi - m d \dot{\theta}^2 \sin \theta + m l \dot{\phi}^2 \sin \phi}{M_1 + M_2 + m} \quad (57)$$

Rolling Cylinder Rotation ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{-mgd \sin \theta - [M_1 r + m(r - d \cos \theta)] \ddot{x} - ml[r \cos \phi - d \cos(\theta + \phi)] \ddot{\phi} - mrd\dot{\theta}^2 \sin \theta + mlr\dot{\phi}^2 \sin \phi - mld\dot{\phi}^2 \sin(\theta + \phi)}{M_1 r^2 + I + m(r^2 + d^2 - 2rd \cos \theta)} \quad (58)$$

Pendulum Oscillation ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{-g \sin \phi - \ddot{x} \cos \phi - [r \cos \phi - d \cos(\theta + \phi)] \ddot{\theta} - d\dot{\theta}^2 \sin(\theta + \phi)}{l} \quad (59)$$

4 Problem 4: Translating Bead on an Elastically Supported Tilt- ing Track

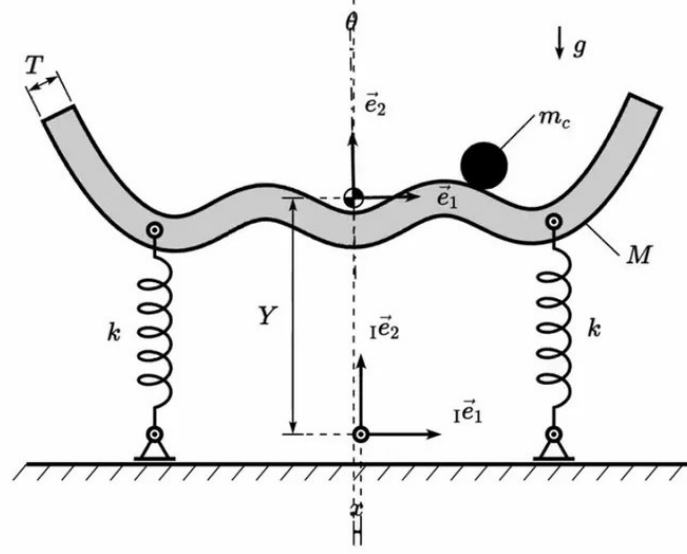


Figure 4: Curved track of mass M supported on two vertical springs of stiffness k , with a bead of mass m translating along the track surface.

Modeling Note on Track Geometry:

Some might be wondering why I didn't incorporate $f(d)$ or the track's explicit curve shape; the main reason is the absence of an explicit curve equation $y = f(x)$. Without that analytical profile, one cannot determine the varying local slope or radius of curvature as a function of d . Therefore, treating the track orientation along coordinate d as locally aligned with tilt angle θ is the only tractable formulation here.

4.1 1. Coordinate System and Parameters

The planar system has four degrees of freedom:

- $x(t)$: Horizontal displacement of the center of mass (CM) of the track.
- $y'(t)$: Vertical displacement of the CM of the track (taken positive downward along the direction of gravity).
- $\theta(t)$: Tilt / rotation angle of the track about its center of mass.
- $d(t)$: Position / arc distance travelled by the bead along the track.
- M, I : Mass and mass moment of inertia of the track about its CM.
- m : Mass of the translating bead.
- s : Track length between the two vertical spring attachment points.

- T : Transverse offset / thickness of the track.
- k : Stiffness of each vertical support spring.

4.2 2. Kinematics

The Cartesian velocities of the track center of mass are:

$$V_x = \frac{d(x)}{dt} = \dot{x}, \quad V_y = \frac{d(y')}{dt} = \dot{y}' \quad (60)$$

The position coordinates of the bead are expressed relative to the reference origin by accounting for track translation, rotation θ , and transverse offset T :

$$x_{\text{bead}} = x - \left(\frac{s}{2} - d\right) \cos \theta - T \sin \theta \quad (61)$$

$$y_{\text{bead}} = y' - \left(\frac{s}{2} - d\right) \sin \theta - T \cos \theta \quad (62)$$

Differentiating with respect to time yields the velocity components of the bead:

$$\begin{aligned} v'_x &= \frac{d}{dt} \left[x - \left(\frac{s}{2} - d\right) \cos \theta - T \sin \theta \right] \\ &= \dot{x} + \dot{d} \cos \theta + \left[\left(\frac{s}{2} - d\right) \sin \theta - T \cos \theta \right] \dot{\theta} \end{aligned} \quad (63)$$

$$\begin{aligned} v'_y &= \frac{d}{dt} \left[y' - \left(\frac{s}{2} - d\right) \sin \theta - T \cos \theta \right] \\ &= \dot{y}' + \dot{d} \sin \theta - \left[\left(\frac{s}{2} - d\right) \cos \theta - T \sin \theta \right] \dot{\theta} \end{aligned} \quad (64)$$

4.3 3. Kinetic and Potential Energy

The total kinetic energy combines the planar translation and rotation of the track with the particle kinetic energy of the bead:

$$\begin{aligned} KE &= \frac{1}{2} M (\dot{x}^2 + \dot{y}'^2) + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} m (v_x'^2 + v_y'^2) \\ &= \frac{1}{2} M (\dot{x}^2 + \dot{y}'^2) + \frac{1}{2} I \dot{\theta}^2 \\ &\quad + \frac{1}{2} m \left(\dot{x} + \dot{d} \cos \theta + \left[\left(\frac{s}{2} - d\right) \sin \theta - T \cos \theta \right] \dot{\theta} \right)^2 \\ &\quad + \frac{1}{2} m \left(\dot{y}' + \dot{d} \sin \theta - \left[\left(\frac{s}{2} - d\right) \cos \theta - T \sin \theta \right] \dot{\theta} \right)^2 \end{aligned} \quad (65)$$

The spring deflections are $z_1 = y' + \frac{s}{2} \sin \theta$ and $z_2 = y' - \frac{s}{2} \sin \theta$. With the datum set at the origin:

$$\begin{aligned} PE &= \frac{1}{2} k \left(y' + \frac{s}{2} \sin \theta \right)^2 + \frac{1}{2} k \left(y' - \frac{s}{2} \sin \theta \right)^2 - M g y' - m g y_{\text{bead}} \\ &= k y'^2 + \frac{1}{4} k s^2 \sin^2 \theta - (M + m) g y' + m g \left(\frac{s}{2} - d \right) \sin \theta + m g T \cos \theta \end{aligned} \quad (66)$$

4.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$ for each generalized coordinate:

For coordinate x :

$$(M+m)\ddot{x} + m\ddot{d}\cos\theta + m\left[\left(\frac{s}{2} - d\right)\sin\theta - T\cos\theta\right]\ddot{\theta} - 2m\dot{d}\dot{\theta}\sin\theta + m\left[\left(\frac{s}{2} - d\right)\cos\theta + T\sin\theta\right]\dot{\theta}^2 = 0 \quad (67)$$

For coordinate y' :

$$(M+m)\ddot{y}' + m\ddot{d}\sin\theta - m\left[\left(\frac{s}{2} - d\right)\cos\theta - T\sin\theta\right]\ddot{\theta} + 2m\dot{d}\dot{\theta}\cos\theta + m\left[\left(\frac{s}{2} - d\right)\sin\theta - T\cos\theta\right]\dot{\theta}^2 + 2ky' - (M+m)g = 0 \quad (68)$$

For coordinate d :

$$m\ddot{d} + m\ddot{x}\cos\theta + m\ddot{y}'\sin\theta - m\left(\frac{s}{2} - d\right)\dot{\theta}^2 - mg\sin\theta - mT\ddot{\theta} = 0 \quad (69)$$

For coordinate θ :

$$\left(I + m\left[\left(\frac{s}{2} - d\right)^2 + T^2\right]\right)\ddot{\theta} + m\left[\left(\frac{s}{2} - d\right)\sin\theta - T\cos\theta\right]\ddot{x} - m\left[\left(\frac{s}{2} - d\right)\cos\theta + T\sin\theta\right]\ddot{y}' - mT\ddot{d} - 2m\left(\frac{s}{2} - d\right)\dot{d}\dot{\theta} + \frac{1}{2}ks^2\sin\theta\cos\theta - mg\left[\left(\frac{s}{2} - d\right)\cos\theta - T\sin\theta\right] = 0 \quad (70)$$

4.5 5. Explicit Equations of Motion

Isolating the accelerations for each degree of freedom:

Track Horizontal Acceleration ($\ddot{x}$):

$$\ddot{x} = \frac{2m\dot{d}\dot{\theta}\sin\theta - m\left[\left(\frac{s}{2} - d\right)\cos\theta + T\sin\theta\right]\dot{\theta}^2 - m\ddot{d}\cos\theta - m\left[\left(\frac{s}{2} - d\right)\sin\theta - T\cos\theta\right]\ddot{\theta}}{M+m} \quad (71)$$

Track Vertical Acceleration ($\ddot{y}'$):

$$\ddot{y}' = \frac{(M+m)g - 2ky' - 2m\dot{d}\dot{\theta}\cos\theta - m\left[\left(\frac{s}{2} - d\right)\sin\theta - T\cos\theta\right]\dot{\theta}^2 - m\ddot{d}\sin\theta + m\left[\left(\frac{s}{2} - d\right)\cos\theta - T\sin\theta\right]\ddot{\theta}}{M+m} \quad (72)$$

Bead Acceleration Along Track ($\ddot{d}$):

$$\ddot{d} = \left(\frac{s}{2} - d\right)\dot{\theta}^2 + g\sin\theta - \ddot{x}\cos\theta - \ddot{y}'\sin\theta + T\ddot{\theta} \quad (73)$$

Track Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{2m\left(\frac{s}{2} - d\right)\dot{d}\dot{\theta} - \frac{1}{2}ks^2\sin\theta\cos\theta + mg\left[\left(\frac{s}{2} - d\right)\cos\theta - T\sin\theta\right] - m\left[\left(\frac{s}{2} - d\right)\sin\theta - T\cos\theta\right]\ddot{x} + m\left[\left(\frac{s}{2} - d\right)\cos\theta + T\sin\theta\right]\ddot{y}' + mT\ddot{d}}{I + m\left[\left(\frac{s}{2} - d\right)^2 + T^2\right]} \quad (74)$$

Net Track Acceleration:

$$a_{\text{track, net}} = \sqrt{\ddot{x}^2 + \ddot{y}^2} \quad (75)$$

5 Problem 5: Rolling Ring on a Parabolic Track with Nested Disc and Elastic Pendulum

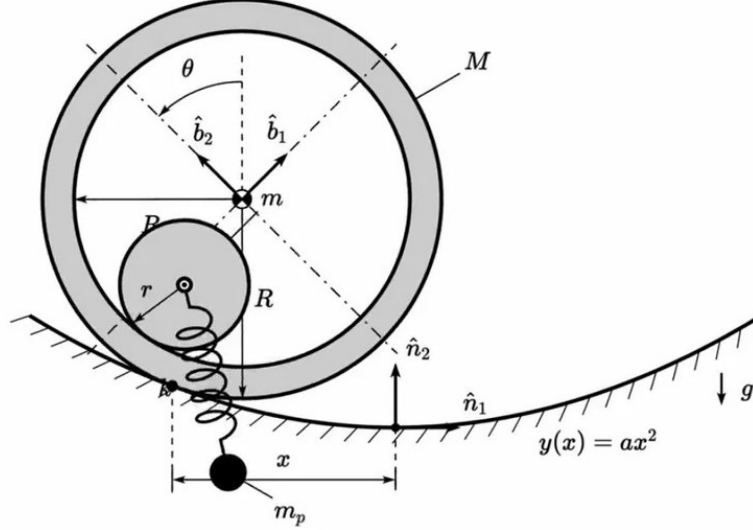


Figure 5: A ring of mass M , outer radius R_0 , and inner radius R_i rolls without slipping on a parabolic profile $y(x) = ax^2$. An inner cylinder of mass m and radius r rolls inside, from which a spring-suspended pendulum of mass m_p hangs.

5.1 1. Generalized Coordinates and Parameters

The planar system possesses four generalized degrees of freedom:

- $x(t)$: Horizontal position of the contact point between the outer ring and the parabolic substrate $y(x) = ax^2$.
- $\phi(t)$: Angular deflection of the center of mass of the internal rolling cylinder relative to the ring's center.
- $z(t)$: Dynamic extension of the pendulum spring measured from its nominal length l .
- $\alpha(t)$: Angular deflection of the suspended pendulum rod/spring from the vertical.
- M, I_M : Mass and central moment of inertia of the outer hollow ring (R_0 : outer radius, R_i : inner track radius).
- m, I_m, r : Mass, central moment of inertia, and radius of the inner rolling cylinder.
- m_p, k, l : Mass of the pendulum bob, stiffness of the spring, and unstretched/nominal pendulum length.
- $\theta(t)$: Pure-rolling rotation angle of the outer ring along the arc length $s(x)$ of the parabola.

5.2 2. Kinematics

The local inclination angle $\beta(x)$ of the parabolic track $y(x) = ax^2$ satisfies:

$$\tan \beta = 2ax \implies \sin \beta = \frac{2ax}{\sqrt{1+4a^2x^2}}, \quad \cos \beta = \frac{1}{\sqrt{1+4a^2x^2}} \quad (76)$$

Pure rolling of the outer ring along the differential arc length $ds = \sqrt{1+4a^2x^2} dx$ gives:

$$R_0 \dot{\theta} = \dot{s} = \sqrt{1+4a^2x^2} \dot{x} \implies \dot{\theta} = \frac{\sqrt{1+4a^2x^2}}{R_0} \dot{x} \quad (77)$$

The center of mass of the outer ring is located at $(x_M, y_M) = (x - R_0 \sin \beta, ax^2 + R_0 \cos \beta)$. Differentiating with respect to time yields the ring CM velocities:

$$\dot{x}_M = \left(1 - \frac{2aR_0}{(1+4a^2x^2)^{3/2}}\right) \dot{x} \quad (78)$$

$$\dot{y}_M = \left(2ax - \frac{4a^2xR_0}{(1+4a^2x^2)^{3/2}}\right) \dot{x} \quad (79)$$

Pure rolling of the inner disc of radius r inside the cavity of radius R_i establishes the spin angular velocity:

$$\omega_{\text{spin}} = \left(\frac{R_i - r}{r}\right) \dot{\phi} - \left(\frac{R_i}{r}\right) \dot{\theta} = \left(\frac{R_i - r}{r}\right) \dot{\phi} - \left(\frac{R_i \sqrt{1+4a^2x^2}}{rR_0}\right) \dot{x} \quad (80)$$

The position and velocity vectors of the inner disc center (x_m, y_m) and the pendulum bob (x_p, y_p) are:

$$x_m = x_M - (R_i - r) \sin \phi, \quad y_m = y_M - (R_i - r) \cos \phi \quad (81)$$

$$V_x = \dot{x}_M - (R_i - r) \dot{\phi} \cos \phi, \quad V_y = \dot{y}_M + (R_i - r) \dot{\phi} \sin \phi \quad (82)$$

$$x_p = x_m - (l + z) \sin \alpha, \quad y_p = y_m - (l + z) \cos \alpha \quad (83)$$

$$v_{px} = \dot{x}_M - (R_i - r) \dot{\phi} \cos \phi - \dot{z} \sin \alpha - (l + z) \dot{\alpha} \cos \alpha \quad (84)$$

$$v_{py} = \dot{y}_M + (R_i - r) \dot{\phi} \sin \phi - \dot{z} \cos \alpha + (l + z) \dot{\alpha} \sin \alpha \quad (85)$$

5.3 3. Kinetic and Potential Energy

The scalar kinetic energy combines the outer ring, inner cylinder, and pendulum bob:

$$\begin{aligned} KE &= \frac{1}{2}M(\dot{x}_M^2 + \dot{y}_M^2) + \frac{1}{2}I_M\dot{\theta}^2 + \frac{1}{2}m(V_x^2 + V_y^2) + \frac{1}{2}I_m\omega_{\text{spin}}^2 + \frac{1}{2}m_p(v_{px}^2 + v_{py}^2) \\ &= \frac{1}{2}(M + m + m_p)(\dot{x}_M^2 + \dot{y}_M^2) + \frac{1}{2}I_M\left(\frac{1+4a^2x^2}{R_0^2}\right)\dot{x}^2 + \frac{1}{2}(m + m_p)(R_i - r)^2\dot{\phi}^2 \\ &\quad + \frac{1}{2}I_m\left[\left(\frac{R_i - r}{r}\right)\dot{\phi} - \left(\frac{R_i\sqrt{1+4a^2x^2}}{rR_0}\right)\dot{x}\right]^2 + \frac{1}{2}m_p[\dot{z}^2 + (l + z)^2\dot{\alpha}^2] \\ &\quad - (m + m_p)(R_i - r)\dot{\phi}(\dot{x}_M \cos \phi - \dot{y}_M \sin \phi) - m_p\dot{z}(\dot{x}_M \sin \alpha + \dot{y}_M \cos \alpha) \\ &\quad - m_p(l + z)\dot{\alpha}(\dot{x}_M \cos \alpha - \dot{y}_M \sin \alpha) \end{aligned} \quad (86)$$

The total gravitational potential energy and elastic spring energy are:

$$\begin{aligned} PE &= Mg\left(ax^2 + \frac{R_0}{\sqrt{1+4a^2x^2}}\right) + mg\left(ax^2 + \frac{R_0}{\sqrt{1+4a^2x^2}} - (R_i - r) \cos \phi\right) \\ &\quad + m_pg\left(ax^2 + \frac{R_0}{\sqrt{1+4a^2x^2}} - (R_i - r) \cos \phi - (l + z) \cos \alpha\right) + \frac{1}{2}kz^2 \end{aligned} \quad (87)$$

5.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate z :

$$m_p \ddot{z} - m_p(l+z)\dot{\alpha}^2 + kz - m_p g \cos \alpha - m_p (\ddot{x}_M \sin \alpha + \ddot{y}_M \cos \alpha) = 0 \quad (88)$$

For coordinate α :

$$(l+z)\ddot{\alpha} + 2\dot{z}\dot{\alpha} + g \sin \alpha - (\ddot{x}_M \cos \alpha - \ddot{y}_M \sin \alpha) = 0 \quad (89)$$

For coordinate ϕ :

$$\begin{aligned} & \left[(m+m_p)(R_i-r)^2 + I_m \left(\frac{R_i-r}{r} \right)^2 \right] \ddot{\phi} \\ & - I_m \left(\frac{R_i-r}{r} \right) \left(\frac{R_i}{rR_0} \right) \left(\sqrt{1+4a^2x^2} \ddot{x} + \frac{4a^2x\dot{x}^2}{\sqrt{1+4a^2x^2}} \right) \\ & + (m+m_p)g(R_i-r) \sin \phi - (m+m_p)(R_i-r) (\ddot{x}_M \cos \phi - \ddot{y}_M \sin \phi) \\ & - m_p(R_i-r) \left[\ddot{z} \sin(\phi-\alpha) + (l+z)\ddot{\alpha} \cos(\phi-\alpha) \right. \\ & \quad \left. + 2\dot{z}\dot{\alpha} \cos(\phi-\alpha) - (l+z)\dot{\alpha}^2 \sin(\phi-\alpha) \right] = 0 \end{aligned} \quad (90)$$

For coordinate x : Let the effective coordinate mass $J_x(x)$ be defined as:

$$J_x(x) = (M+m+m_p) \left[\left(1 - \frac{2aR_0}{(1+4a^2x^2)^{3/2}} \right)^2 + \left(2ax - \frac{4a^2xR_0}{(1+4a^2x^2)^{3/2}} \right)^2 \right] + \left(I_M + I_m \frac{R_i^2}{r^2} \right) \frac{1+4a^2x^2}{R_0^2} \quad (91)$$

Retaining the differential operator $\frac{1}{2} \frac{\partial J_x(x)}{\partial x} \dot{x}^2$, the fourth equation of motion is:

$$\begin{aligned} & J_x(x)\ddot{x} + \frac{1}{2} \frac{\partial J_x(x)}{\partial x} \dot{x}^2 - I_m \left(\frac{R_i-r}{r} \right) \left(\frac{R_i\sqrt{1+4a^2x^2}}{rR_0} \right) \ddot{\phi} - I_m \left(\frac{R_i-r}{r} \right) \left(\frac{4a^2xR_i}{rR_0\sqrt{1+4a^2x^2}} \right) \dot{x}\dot{\phi} \\ & + (M+m+m_p)g \left(2ax - \frac{4a^2xR_0}{(1+4a^2x^2)^{3/2}} \right) \\ & - \left(1 - \frac{2aR_0}{(1+4a^2x^2)^{3/2}} \right) (m+m_p)(R_i-r) (\ddot{\phi} \cos \phi - \dot{\phi}^2 \sin \phi) \\ & + 2ax(m+m_p)(R_i-r) (\ddot{\phi} \sin \phi + \dot{\phi}^2 \cos \phi) \\ & - m_p \left[\ddot{z} \sin \alpha + 2\dot{z}\dot{\alpha} \cos \alpha + (l+z)\ddot{\alpha} \cos \alpha - (l+z)\dot{\alpha}^2 \sin \alpha \right] \\ & + 2ax m_p \left[\ddot{z} \cos \alpha - 2\dot{z}\dot{\alpha} \sin \alpha + (l+z)\ddot{\alpha} \sin \alpha - (l+z)\dot{\alpha}^2 \cos \alpha \right] = 0 \end{aligned} \quad (92)$$

5.5 5. Explicit Equations of Motion

Isolating each generalized second-order acceleration term:

Spring Radial Acceleration ($\ddot{z}$):

$$\ddot{z} = (l + z)\dot{\alpha}^2 - \frac{k}{m_p}z + g \cos \alpha + \ddot{x}_M \sin \alpha + \ddot{y}_M \cos \alpha \quad (93)$$

Pendulum Angular Acceleration ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{-g \sin \alpha - 2\dot{z}\dot{\alpha} + \ddot{x}_M \cos \alpha - \ddot{y}_M \sin \alpha}{l + z} \quad (94)$$

Inner Disc Angular Acceleration ($\ddot{\phi}$): Letting $J_\phi = (m + m_p)(R_i - r)^2 + I_m \left(\frac{R_i - r}{r} \right)^2$:

$$\begin{aligned} & I_m \left(\frac{R_i - r}{r} \right) \left(\frac{R_i}{r R_0} \right) \left(\sqrt{1 + 4a^2 x^2} \ddot{x} + \frac{4a^2 x \dot{x}^2}{\sqrt{1 + 4a^2 x^2}} \right) - (m + m_p)g(R_i - r) \sin \phi \\ & + (m + m_p)(R_i - r) (\ddot{x}_M \cos \phi - \ddot{y}_M \sin \phi) \\ & + m_p(R_i - r) \left[\ddot{z} \sin(\phi - \alpha) + (l + z)\ddot{\alpha} \cos(\phi - \alpha) + 2\dot{z}\dot{\alpha} \cos(\phi - \alpha) - (l + z)\dot{\alpha}^2 \sin(\phi - \alpha) \right] \\ \ddot{\phi} = & \frac{J_\phi}{J_\phi} \end{aligned} \quad (95)$$

Outer Ring Substrate Acceleration ($\ddot{x}$):

$$\begin{aligned} & -\frac{1}{2} \frac{\partial J_x(x)}{\partial x} \dot{x}^2 + I_m \left(\frac{R_i - r}{r} \right) \left(\frac{R_i \sqrt{1 + 4a^2 x^2}}{r R_0} \right) \ddot{\phi} + I_m \left(\frac{R_i - r}{r} \right) \left(\frac{4a^2 x R_i}{r R_0 \sqrt{1 + 4a^2 x^2}} \right) \dot{x} \dot{\phi} \\ & - (M + m + m_p)g \left(2ax - \frac{4a^2 x R_0}{(1 + 4a^2 x^2)^{3/2}} \right) \\ & + \left(1 - \frac{2a R_0}{(1 + 4a^2 x^2)^{3/2}} \right) (m + m_p)(R_i - r) (\ddot{\phi} \cos \phi - \dot{\phi}^2 \sin \phi) \\ & - 2ax(m + m_p)(R_i - r) (\ddot{\phi} \sin \phi + \dot{\phi}^2 \cos \phi) \\ & + m_p \left[\ddot{z} \sin \alpha + 2\dot{z}\dot{\alpha} \cos \alpha + (l + z)\ddot{\alpha} \cos \alpha - (l + z)\dot{\alpha}^2 \sin \alpha \right] \\ & - 2ax m_p \left[\ddot{z} \cos \alpha - 2\dot{z}\dot{\alpha} \sin \alpha + (l + z)\ddot{\alpha} \sin \alpha - (l + z)\dot{\alpha}^2 \cos \alpha \right] \\ \ddot{x} = & \frac{J_x(x)}{J_x(x)} \end{aligned} \quad (96)$$

6 Problem 6: Parallelogram Linked Discs with Eccentric Pendulum on a Parabolic Track

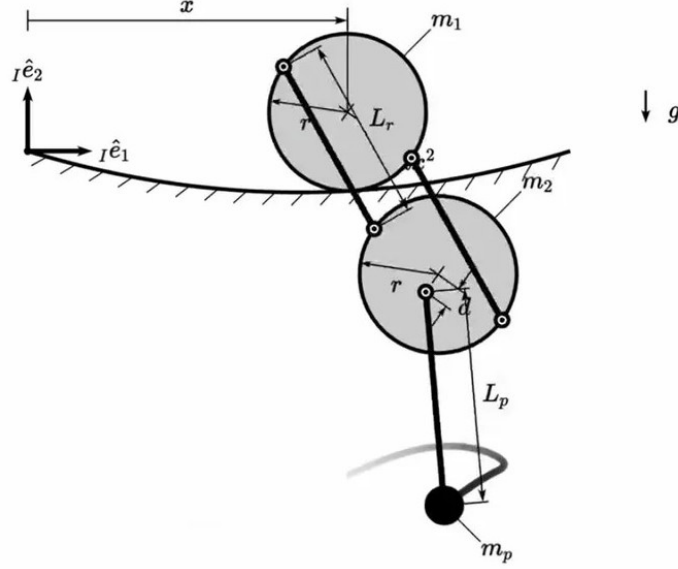


Figure 6: Two identical discs connected by a parallelogram linkage rolling on a parabolic substrate $y(x) = ax^2$, with an eccentric simple pendulum suspended from the lower disc.

6.1 1. Generalized Coordinates and Parameters

The planar system is described by three independent generalized coordinates:

- $x(t)$: Horizontal coordinate of the contact point between the upper rolling disc and the parabolic substrate $y(x) = ax^2$.
- $\phi(t)$: Angular orientation of the parallelogram linkage of length L connecting the two discs, measured clockwise from the downward vertical.
- $\alpha(t)$: Angular deflection of the pendulum rod of length l suspended from the eccentric pin, measured clockwise from the downward vertical.
- M_1, I_1, R_1 : Mass, central moment of inertia, and radius of the upper rolling disc.
- M_2, I_2, R_2 : Mass, central moment of inertia, and radius of the lower disc ($R_2 = R_1$).
- m : Mass of the pendulum bob suspended at the end of the rod.
- d : Eccentric radial offset of the pendulum pivot pin from the center of the lower disc.
- $\theta(t)$: Rolling rotation angle of both discs ($\dot{\theta}_1 = \dot{\theta}_2 = \dot{\theta}$ enforced by the parallelogram linkage).

6.2 2. Kinematics

The slope angle $\beta(x)$ of the parabolic substrate satisfies $\tan \beta = 2ax$, giving:

$$\sin \beta = \frac{2ax}{\sqrt{1 + 4a^2x^2}}, \quad \cos \beta = \frac{1}{\sqrt{1 + 4a^2x^2}} \quad (97)$$

Pure rolling along the arc length $ds = \sqrt{1 + 4a^2x^2} dx$ dictates:

$$\dot{\theta} = \frac{\sqrt{1 + 4a^2x^2}}{R_1} \dot{x} \implies \ddot{\theta} = \frac{\sqrt{1 + 4a^2x^2}}{R_1} \ddot{x} + \left(\frac{4a^2x}{R_1 \sqrt{1 + 4a^2x^2}} \right) \dot{x}^2 \quad (98)$$

Defining the geometry factor $f(x) = 1 - \frac{2aR_1}{(1+4a^2x^2)^{3/2}}$, the Cartesian position and velocity of the upper disc center (x_1, y_1) are:

$$x_1 = x - R_1 \sin \beta \implies \dot{x}_1 = f(x) \dot{x} \quad (99)$$

$$y_1 = ax^2 + R_1 \cos \beta \implies \dot{y}_1 = 2axf(x) \dot{x} \quad (100)$$

The Cartesian velocities of the lower disc center $(x_2, y_2) = (x_1 - L \sin \phi, y_1 - L \cos \phi)$ are:

$$v'_x = \dot{x}_1 - L \dot{\phi} \cos \phi, \quad v'_y = \dot{y}_1 + L \dot{\phi} \sin \phi \quad (101)$$

The pendulum bob is located at $(x_p, y_p) = (x_1 - L \sin \phi + d \cos \theta + l \sin \alpha, y_1 - L \cos \phi - d \sin \theta - l \cos \alpha)$, yielding:

$$v_{px} = \dot{x}_1 - L \dot{\phi} \cos \phi - d \dot{\theta} \sin \theta + l \dot{\alpha} \cos \alpha \quad (102)$$

$$v_{py} = \dot{y}_1 + L \dot{\phi} \sin \phi - d \dot{\theta} \cos \theta + l \dot{\alpha} \sin \alpha \quad (103)$$

6.3 3. Kinetic and Potential Energy

The total kinetic energy is formulated as:

$$\begin{aligned} KE &= \frac{1}{2} M_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} M_2 (v'^2_x + v'^2_y) + \frac{1}{2} I_2 \dot{\theta}^2 + \frac{1}{2} m (v_{px}^2 + v_{py}^2) \\ &= \frac{1}{2} (M_1 + M_2 + m) (1 + 4a^2x^2) [f(x)]^2 \dot{x}^2 + \frac{1}{2} (I_1 + I_2 + md^2) \left(\frac{1 + 4a^2x^2}{R_1^2} \right) \dot{x}^2 \\ &\quad + \frac{1}{2} (M_2 + m) L^2 \dot{\phi}^2 + \frac{1}{2} ml^2 \dot{\alpha}^2 - (M_2 + m) L \dot{\phi} (\dot{x}_1 \cos \phi - \dot{y}_1 \sin \phi) \\ &\quad - md \dot{\theta} (\dot{x}_1 \sin \theta + \dot{y}_1 \cos \theta) + ml \dot{\alpha} (\dot{x}_1 \cos \alpha + \dot{y}_1 \sin \alpha) \\ &\quad - mLd \dot{\phi} \dot{\theta} \sin(\phi - \theta) - mLl \dot{\phi} \dot{\alpha} \cos(\phi + \alpha) - mdl \dot{\theta} \dot{\alpha} \sin(\theta + \alpha) \end{aligned} \quad (104)$$

The gravitational potential energy relative to the origin datum is:

$$PE = (M_1 + M_2 + m)g \left(ax^2 + \frac{R_1}{\sqrt{1 + 4a^2x^2}} \right) - (M_2 + m)gL \cos \phi - mgd \sin \theta - mgl \cos \alpha \quad (105)$$

6.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate α :

$$\begin{aligned} l \ddot{\alpha} + g \sin \alpha + \ddot{x}_1 \cos \alpha + \ddot{y}_1 \sin \alpha - L \ddot{\phi} \cos(\phi + \alpha) \\ + L \dot{\phi}^2 \sin(\phi + \alpha) - d \ddot{\theta} \sin(\theta + \alpha) - d \dot{\theta}^2 \cos(\theta + \alpha) = 0 \end{aligned} \quad (106)$$

For coordinate ϕ :

$$\begin{aligned} (M_2 + m) L \ddot{\phi} + (M_2 + m) g \sin \phi - (M_2 + m) (\ddot{x}_1 \cos \phi - \ddot{y}_1 \sin \phi) \\ - md \left[\ddot{\theta} \sin(\phi - \theta) - \dot{\theta}^2 \cos(\phi - \theta) \right] - ml \left[\ddot{\alpha} \cos(\phi + \alpha) - \dot{\alpha}^2 \sin(\phi + \alpha) \right] = 0 \end{aligned} \quad (107)$$

For coordinate x :

$$\begin{aligned}
& (M_1 + M_2 + m)f(x) (\ddot{x}_1 + 2ax\ddot{y}_1 + 2agx) + (I_1 + I_2 + md^2) \left(\frac{\sqrt{1 + 4a^2x^2}}{R_1} \right) \ddot{\theta} \\
& - (M_2 + m)Lf(x) \left[\ddot{\phi} \cos \phi - \dot{\phi}^2 \sin \phi - 2ax(\ddot{\phi} \sin \phi + \dot{\phi}^2 \cos \phi) \right] \\
& - mdf(x) \left[\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta + 2ax(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) \right] - \frac{md\sqrt{1 + 4a^2x^2}}{R_1} (\ddot{x}_1 \sin \theta + \ddot{y}_1 \cos \theta + g \cos \theta) \\
& + mlf(x) [\ddot{\alpha} \cos \alpha - \dot{\alpha}^2 \sin \alpha + 2ax(\ddot{\alpha} \sin \alpha + \dot{\alpha}^2 \cos \alpha)] \\
& - \frac{mLd\sqrt{1 + 4a^2x^2}}{R_1} [\ddot{\phi} \sin(\phi - \theta) - \dot{\phi}^2 \cos(\phi - \theta)] \\
& - \frac{mdl\sqrt{1 + 4a^2x^2}}{R_1} [\ddot{\alpha} \sin(\theta + \alpha) + \dot{\alpha}^2 \cos(\theta + \alpha)] = 0
\end{aligned} \tag{108}$$

6.5 5. Explicit Equations of Motion

Isolating the second-order accelerations for each generalized coordinate:

Pendulum Angular Acceleration ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{-g \sin \alpha - (\ddot{x}_1 \cos \alpha + \ddot{y}_1 \sin \alpha) + L [\ddot{\phi} \cos(\phi + \alpha) - \dot{\phi}^2 \sin(\phi + \alpha)] + d [\ddot{\theta} \sin(\theta + \alpha) + \dot{\theta}^2 \cos(\theta + \alpha)]}{l} \tag{109}$$

Linkage Angular Acceleration ($\ddot{\phi}$):

$$\begin{aligned}
& -g \sin \phi + \frac{\ddot{x}_1 \cos \phi - \ddot{y}_1 \sin \phi}{L} + \frac{md}{(M_2 + m)L} [\ddot{\theta} \sin(\phi - \theta) - \dot{\theta}^2 \cos(\phi - \theta)] \\
& + \frac{ml}{(M_2 + m)L} [\ddot{\alpha} \cos(\phi + \alpha) - \dot{\alpha}^2 \sin(\phi + \alpha)] \\
\ddot{\phi} = & \frac{1}{1}
\end{aligned} \tag{110}$$

Discs Substrate Acceleration ($\ddot{x}$): Letting the denominator inertia factor be:

$$\begin{aligned}
\mathcal{D}_x = & (M_1 + M_2 + m)(1 + 4a^2x^2) \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right)^2 + (I_1 + I_2 + md^2) \left(\frac{1 + 4a^2x^2}{R_1^2} \right) \\
& - 2md \left(\frac{\sqrt{1 + 4a^2x^2}}{R_1} \right) \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right) (\sin \theta + 2ax \cos \theta)
\end{aligned} \tag{111}$$

The explicit equation for $\ddot{x}$ is:

$$\begin{aligned}
& - (M_1 + M_2 + m)g \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right) 2ax + mgd \left(\frac{\sqrt{1 + 4a^2x^2}}{R_1} \right) \cos \theta \\
& + \frac{mLd\sqrt{1 + 4a^2x^2}}{R_1} \left[\ddot{\phi} \sin(\phi - \theta) - \dot{\phi}^2 \cos(\phi - \theta) \right] \\
& + (M_2 + m)L \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right) \left[\ddot{\phi} \cos \phi - \dot{\phi}^2 \sin \phi - 2ax(\ddot{\phi} \sin \phi + \dot{\phi}^2 \cos \phi) \right] \\
& + \frac{mdl\sqrt{1 + 4a^2x^2}}{R_1} \left[\ddot{\alpha} \sin(\theta + \alpha) + \dot{\alpha}^2 \cos(\theta + \alpha) \right] \\
& - ml \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right) \left[\ddot{\alpha} \cos \alpha - \dot{\alpha}^2 \sin \alpha + 2ax(\ddot{\alpha} \sin \alpha + \dot{\alpha}^2 \cos \alpha) \right] \\
& + md \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right) \dot{\theta}^2 (\cos \theta - 2ax \sin \theta) \\
& - (M_1 + M_2 + m) \left(1 - \frac{2aR_1}{(1 + 4a^2x^2)^{3/2}} \right) \left[\frac{12a^2xR_1}{(1 + 4a^2x^2)^{5/2}} + 4a^2x \left(1 - \frac{2aR_1(1 - 8a^2x^2)}{(1 + 4a^2x^2)^{5/2}} \right) \right] \dot{x}^2 \\
& - (I_1 + I_2 + md^2) \left(\frac{4a^2x}{R_1^2} \right) \dot{x}^2
\end{aligned}$$

$$\ddot{x} = \frac{\mathcal{D}_x}{\mathcal{D}_x} \tag{112}$$

7 Problem 7: Two Pure-Rolling Discs with Interconnecting and Anchor Springs

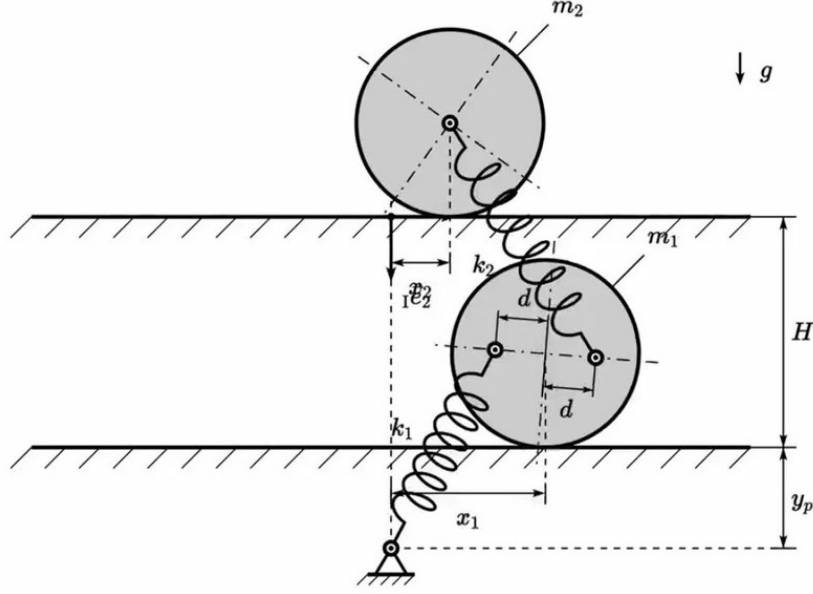


Figure 7: Two pure-rolling discs on parallel horizontal tracks separated by height H , connected by spring k_2 , with the lower disc anchored to a fixed ground pin by spring k_1 .

7.1 1. Generalized Coordinates and Parameters

The planar system has two generalized degrees of freedom:

- $\phi(t)$: Rolling rotation angle of the lower disc of mass m_1 , radius R_1 , and central moment of inertia I_1 .
- $\theta(t)$: Rolling rotation angle of the upper disc of mass m_2 , radius R_2 , and central moment of inertia I_2 .
- d : Radial offset of the two collinear eccentric pins from the center of the lower disc.
- k_1, L_{10} : Stiffness and unstretched length of the anchoring spring connected to the lower ground pin.
- k_2, L_{20} : Stiffness and unstretched length of the coupling spring between the two discs.
- H : Vertical distance between the two parallel rolling tracks.
- Y : Vertical depth of the fixed ground anchor below the lower track.

7.2 2. Kinematics and Spring Geometry

Under pure rolling without slipping on horizontal tracks, the translational displacements and velocities of the disc centers are:

$$x_1 = R_1\phi \implies V = \dot{x}_1 = R_1\dot{\phi} \quad (113)$$

$$x_2 = R_2\theta \implies V' = \dot{x}_2 = R_2\dot{\theta} \quad (114)$$

The fixed ground anchor is located at $(0, -Y)$. The lower anchor spring k_1 connects this anchor to the first eccentric pin on the lower disc, giving the geometric relation:

$$Y + R_1 - d \cos \phi = L_1 \cos \alpha_1 \quad (115)$$

$$R_1\phi - d \sin \phi = L_1 \sin \alpha_1 \quad (116)$$

The instantaneous extended length of spring 1 is:

$$L_1 = \sqrt{(Y + R_1 - d \cos \phi)^2 + (R_1\phi - d \sin \phi)^2} \quad (117)$$

The upper coupling spring k_2 spans from the diametrically opposite pin on the lower disc to the center of the upper disc at height $H + R_2$:

$$H + R_2 - (R_1 + d \cos \phi) = L_2 \cos \alpha_2 \quad (118)$$

$$R_2\theta - (R_1\phi + d \sin \phi) = L_2 \sin \alpha_2 \quad (119)$$

The instantaneous extended length of spring 2 is:

$$L_2 = \sqrt{(H + R_2 - R_1 - d \cos \phi)^2 + (R_2\theta - R_1\phi - d \sin \phi)^2} \quad (120)$$

7.3 3. Kinetic and Potential Energy

The total kinetic energy of the pure-rolling discs is:

$$\begin{aligned} KE &= \frac{1}{2}m_1V^2 + \frac{1}{2}I_1\dot{\phi}^2 + \frac{1}{2}m_2V'^2 + \frac{1}{2}I_2\dot{\theta}^2 \\ &= \frac{1}{2}(m_1R_1^2 + I_1)\dot{\phi}^2 + \frac{1}{2}(m_2R_2^2 + I_2)\dot{\theta}^2 \end{aligned} \quad (121)$$

Since both discs translate horizontally along flat parallel tracks, their gravitational potential energy remains constant. The total elastic potential energy stored in the two springs is:

$$PE = \frac{1}{2}k_1(L_1 - L_{10})^2 + \frac{1}{2}k_2(L_2 - L_{20})^2 \quad (122)$$

7.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate ϕ :

$$(m_1R_1^2 + I_1)\ddot{\phi} + k_1(L_1 - L_{10})\frac{\partial L_1}{\partial \phi} + k_2(L_2 - L_{20})\frac{\partial L_2}{\partial \phi} = 0 \quad (123)$$

Evaluating the geometric derivatives yields:

$$\frac{\partial L_1}{\partial \phi} = \frac{(R_1 - d \cos \phi)(R_1\phi - d \sin \phi) + d \sin \phi(Y + R_1 - d \cos \phi)}{L_1} \quad (124)$$

$$\frac{\partial L_2}{\partial \phi} = \frac{-(R_1 + d \cos \phi)(R_2\theta - R_1\phi - d \sin \phi) + d \sin \phi(H + R_2 - R_1 - d \cos \phi)}{L_2} \quad (125)$$

For coordinate θ :

$$(m_2 R_2^2 + I_2) \ddot{\theta} + k_2 (L_2 - L_{20}) \frac{\partial L_2}{\partial \theta} = 0 \quad (126)$$

where:

$$\frac{\partial L_2}{\partial \theta} = \frac{R_2 (R_2 \theta - R_1 \phi - d \sin \phi)}{L_2} \quad (127)$$

7.5 5. Explicit Equations of Motion

Isolating the angular acceleration of each disc:

Bottom Disc Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{k_1 \left(1 - \frac{L_{10}}{L_1}\right) \left[(R_1 - d \cos \phi) (R_1 \phi - d \sin \phi) + d \sin \phi (Y + R_1 - d \cos \phi) \right] - k_2 \left(1 - \frac{L_{20}}{L_2}\right) \left[(R_1 + d \cos \phi) (R_2 \theta - R_1 \phi - d \sin \phi) - d \sin \phi (H + R_2 - R_1 - d \cos \phi) \right]}{m_1 R_1^2 + I_1} \quad (128)$$

Upper Disc Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{-k_2 R_2 \left(1 - \frac{L_{20}}{L_2}\right) (R_2 \theta - R_1 \phi - d \sin \phi)}{m_2 R_2^2 + I_2} \quad (129)$$

8 Problem 8: Triple Inverted Rod System Coupled by Multi-Spring Network

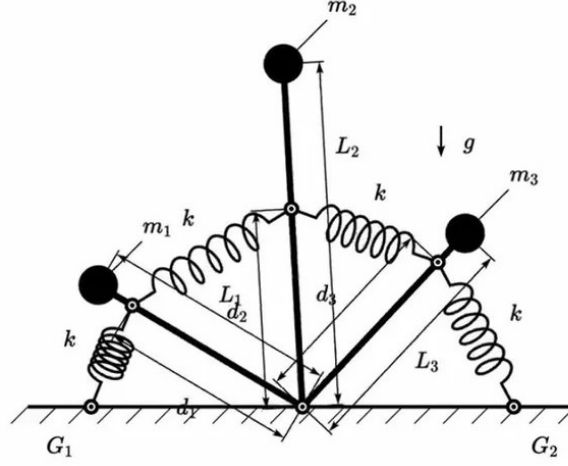


Figure 8: Three inverted rods of lengths L_1, L_2, L_3 pivoted at a common ground origin, coupled together and to ground anchors G_1, G_2 through four linear springs of stiffness k .

8.1 1. Generalized Coordinates and Parameters

The planar system has three independent angular degrees of freedom:

- $\theta_1(t)$: Angular orientation of rod 1, measured counter-clockwise from the horizontal left ground axis.
- $\theta_2(t)$: Angular orientation of rod 2, measured counter-clockwise from the upward vertical axis.
- $\theta_3(t)$: Angular orientation of rod 3, measured counter-clockwise from the horizontal right ground axis.
- m_1, m_2, m_3 : Masses of the bobs positioned at the ends of the rods.
- L_1, L_2, L_3 : Lengths of the three rods from the common pivot origin.
- d_1, d_2, d_3 : Distances along rods 1, 2, and 3 to their respective spring attachment points.
- G_1, G_2 : Distances from the central pivot to the left and right fixed ground anchors.
- k : Stiffness of each of the four linear springs.
- $l_{01}, l_{02}, l_{03}, l_{04}$: Unstretched lengths of springs 1, 2, 3, and 4.

8.2 2. Kinematics and Spring Geometry

The Cartesian coordinates of the four spring connection points yield the instantaneous stretched lengths:

$$x_1 = \sqrt{(G_1 - d_1 \cos \theta_1)^2 + (d_1 \sin \theta_1)^2} = \sqrt{G_1^2 + d_1^2 - 2G_1 d_1 \cos \theta_1} \quad (130)$$

$$x_2 = \sqrt{(d_1 \cos \theta_1 - d_2 \sin \theta_2)^2 + (d_2 \cos \theta_2 - d_1 \sin \theta_1)^2} = \sqrt{d_1^2 + d_2^2 - 2d_1 d_2 \sin(\theta_1 + \theta_2)} \quad (131)$$

$$x_3 = \sqrt{(d_2 \sin \theta_2 + d_3 \cos \theta_3)^2 + (d_3 \sin \theta_3 - d_2 \cos \theta_2)^2} = \sqrt{d_2^2 + d_3^2 - 2d_2 d_3 \sin(\theta_3 - \theta_2)} \quad (132)$$

$$x_4 = \sqrt{(G_2 - d_3 \cos \theta_3)^2 + (d_3 \sin \theta_3)^2} = \sqrt{G_2^2 + d_3^2 - 2G_2 d_3 \cos \theta_3} \quad (133)$$

The Cartesian position vectors and velocities of the three tip masses are:

$$\mathbf{r}_1 = \begin{pmatrix} -L_1 \cos \theta_1 \\ L_1 \sin \theta_1 \end{pmatrix} \Rightarrow \begin{aligned} V_{1x} &= L_1 \dot{\theta}_1 \sin \theta_1 \\ V_{1y} &= L_1 \dot{\theta}_1 \cos \theta_1 \end{aligned} \quad (134)$$

$$\mathbf{r}_2 = \begin{pmatrix} -L_2 \sin \theta_2 \\ L_2 \cos \theta_2 \end{pmatrix} \Rightarrow \begin{aligned} V_{2x} &= -L_2 \dot{\theta}_2 \cos \theta_2 \\ V_{2y} &= -L_2 \dot{\theta}_2 \sin \theta_2 \end{aligned} \quad (135)$$

$$\mathbf{r}_3 = \begin{pmatrix} L_3 \cos \theta_3 \\ L_3 \sin \theta_3 \end{pmatrix} \Rightarrow \begin{aligned} V_{3x} &= -L_3 \dot{\theta}_3 \sin \theta_3 \\ V_{3y} &= L_3 \dot{\theta}_3 \cos \theta_3 \end{aligned} \quad (136)$$

8.3 3. Kinetic and Potential Energy

The total kinetic energy of the system is given by:

$$KE = \frac{1}{2} m_1 L_1^2 \dot{\theta}_1^2 + \frac{1}{2} m_2 L_2^2 \dot{\theta}_2^2 + \frac{1}{2} m_3 L_3^2 \dot{\theta}_3^2 \quad (137)$$

Taking the horizontal ground plane as the reference datum, the total potential energy consists of the gravitational potentials and the elastic energy in all four springs:

$$\begin{aligned} PE &= m_1 g L_1 \sin \theta_1 + m_2 g L_2 \cos \theta_2 + m_3 g L_3 \sin \theta_3 \\ &\quad + \frac{1}{2} k (x_1 - l_{01})^2 + \frac{1}{2} k (x_2 - l_{02})^2 + \frac{1}{2} k (x_3 - l_{03})^2 + \frac{1}{2} k (x_4 - l_{04})^2 \end{aligned} \quad (138)$$

8.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} = 0$:

For coordinate θ_1 :

$$m_1 L_1^2 \ddot{\theta}_1 + m_1 g L_1 \cos \theta_1 + k \left(1 - \frac{l_{01}}{x_1} \right) G_1 d_1 \sin \theta_1 - k \left(1 - \frac{l_{02}}{x_2} \right) d_1 d_2 \cos(\theta_1 + \theta_2) = 0 \quad (139)$$

For coordinate θ_2 :

$$m_2 L_2^2 \ddot{\theta}_2 - m_2 g L_2 \sin \theta_2 - k \left(1 - \frac{l_{02}}{x_2} \right) d_1 d_2 \cos(\theta_1 + \theta_2) + k \left(1 - \frac{l_{03}}{x_3} \right) d_2 d_3 \cos(\theta_3 - \theta_2) = 0 \quad (140)$$

For coordinate θ_3 :

$$m_3 L_3^2 \ddot{\theta}_3 + m_3 g L_3 \cos \theta_3 - k \left(1 - \frac{l_{03}}{x_3} \right) d_2 d_3 \cos(\theta_3 - \theta_2) + k \left(1 - \frac{l_{04}}{x_4} \right) G_2 d_3 \sin \theta_3 = 0 \quad (141)$$

8.5 5. Explicit Equations of Motion

Isolating the angular acceleration of each inverted rod:

Rod 1 Angular Acceleration ($\ddot{\theta}_1$):

$$\ddot{\theta}_1 = \frac{- \left[m_1 g L_1 \cos \theta_1 + k \left(1 - \frac{l_{01}}{x_1} \right) G_1 d_1 \sin \theta_1 - k \left(1 - \frac{l_{02}}{x_2} \right) d_1 d_2 \cos(\theta_1 + \theta_2) \right]}{m_1 L_1^2} \quad (142)$$

Rod 2 Angular Acceleration ($\ddot{\theta}_2$):

$$\ddot{\theta}_2 = \frac{m_2 g L_2 \sin \theta_2 + k \left(1 - \frac{l_{02}}{x_2} \right) d_1 d_2 \cos(\theta_1 + \theta_2) - k \left(1 - \frac{l_{03}}{x_3} \right) d_2 d_3 \cos(\theta_3 - \theta_2)}{m_2 L_2^2} \quad (143)$$

Rod 3 Angular Acceleration ($\ddot{\theta}_3$):

$$\ddot{\theta}_3 = \frac{- \left[m_3 g L_3 \cos \theta_3 - k \left(1 - \frac{l_{03}}{x_3} \right) d_2 d_3 \cos(\theta_3 - \theta_2) + k \left(1 - \frac{l_{04}}{x_4} \right) G_2 d_3 \sin \theta_3 \right]}{m_3 L_3^2} \quad (144)$$

9 Problem 9: Rocking Frame with Spring-Loaded Slider and Suspended Pendulum

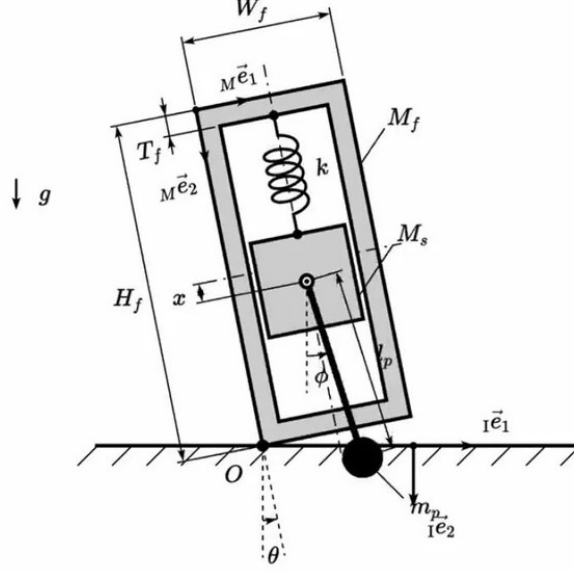


Figure 9: A rectangular frame of mass M_1 rocking on corner pivot O , enclosing a slider of mass M_2 restrained by spring k , and a simple pendulum of mass m suspended from the slider.

9.1 1. Generalized Coordinates and Parameters

The planar system has three coupled degrees of freedom:

- $\theta(t)$: Rocking angle of the rectangular frame about its contact corner pivot O , measured clockwise from the vertical.
- $x(t)$: Relative linear displacement of the slider along the guide axis of the frame, measured from the frame center.
- $\phi(t)$: Relative angular deflection of the pendulum rod from the frame orientation.
- M_1, I_1 : Mass and centroidal moment of inertia of the hollow rectangular frame.
- M_2, I_2 : Mass and moment of inertia of the internal slider.
- m, l : Mass of the pendulum bob and length of the pendulum rod.
- H, W, T : External height H_f , external width W_f , and top wall thickness T_f of the frame.
- k, L_0 : Stiffness and unstretched length of the internal linear spring.

9.2 2. Kinematics

The instantaneous extended length of the internal spring is:

$$L_{\text{spring}} = x + \frac{H}{2} - T \quad (145)$$

The Cartesian position coordinates of the frame center of mass relative to pivot O are:

$$x_1 = \frac{W}{2} \cos \theta + \frac{H}{2} \sin \theta, \quad y_1 = \frac{H}{2} \cos \theta - \frac{W}{2} \sin \theta \quad (146)$$

Differentiating with respect to time gives the frame CM velocity components:

$$V_x = \left(\frac{H}{2} \cos \theta - \frac{W}{2} \sin \theta \right) \dot{\theta} \quad (147)$$

$$V_y = - \left(\frac{H}{2} \sin \theta + \frac{W}{2} \cos \theta \right) \dot{\theta} \quad (148)$$

The Cartesian coordinates and velocities of the internal slider center are:

$$x_2 = \frac{W}{2} \cos \theta + \left(\frac{H}{2} - x \right) \sin \theta \implies V'_x = \left[\left(\frac{H}{2} - x \right) \cos \theta - \frac{W}{2} \sin \theta \right] \dot{\theta} - \dot{x} \sin \theta \quad (149)$$

$$y_2 = \left(\frac{H}{2} - x \right) \cos \theta - \frac{W}{2} \sin \theta \implies V'_y = - \left[\left(\frac{H}{2} - x \right) \sin \theta + \frac{W}{2} \cos \theta \right] \dot{\theta} - \dot{x} \cos \theta \quad (150)$$

The pendulum bob is located at:

$$x_p = x_2 + l \sin(\theta + \phi) \quad (151)$$

$$y_p = y_2 - l \cos(\theta + \phi) \quad (152)$$

Differentiating yields the bob velocity components:

$$v_x = \left[\left(\frac{H}{2} - x \right) \cos \theta - \frac{W}{2} \sin \theta \right] \dot{\theta} - \dot{x} \sin \theta + l(\dot{\theta} + \dot{\phi}) \cos(\theta + \phi) \quad (153)$$

$$v_y = - \left[\left(\frac{H}{2} - x \right) \sin \theta + \frac{W}{2} \cos \theta \right] \dot{\theta} - \dot{x} \cos \theta + l(\dot{\theta} + \dot{\phi}) \sin(\theta + \phi) \quad (154)$$

9.3 3. Kinetic and Potential Energy

Summing the kinetic contributions of the frame, slider, and pendulum bob:

$$\begin{aligned} KE &= \frac{1}{2} M_1 (V_x^2 + V_y^2) + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} M_2 (V_x'^2 + V_y'^2) + \frac{1}{2} I_2 \dot{\theta}^2 + \frac{1}{2} m (v_x^2 + v_y^2) \\ &= \frac{1}{2} M_1 \left[\left(\frac{W}{2} \right)^2 + \left(\frac{H}{2} \right)^2 \right] \dot{\theta}^2 + \frac{1}{2} I_1 \dot{\theta}^2 \\ &\quad + \frac{1}{2} M_2 \left[\left(\left(\frac{W}{2} \right)^2 + \left(\frac{H}{2} - x \right)^2 \right) \dot{\theta}^2 + \dot{x}^2 + W \dot{x} \dot{\theta} \right] + \frac{1}{2} I_2 \dot{\theta}^2 \\ &\quad + \frac{1}{2} m \left\{ V_x'^2 + V_y'^2 + l^2 (\dot{\theta} + \dot{\phi})^2 \right. \\ &\quad \left. + 2l(\dot{\theta} + \dot{\phi}) \left[\left(\frac{H}{2} - x \right) \dot{\theta} \cos(2\theta + \phi) - \frac{W}{2} \dot{\theta} \sin(2\theta + \phi) - \dot{x} \sin(2\theta + \phi) \right] \right\} \end{aligned} \quad (155)$$

Taking the ground surface as the reference datum, the total potential energy is:

$$\begin{aligned} PE &= g \left\{ \left[M_1 \frac{H}{2} + (M_2 + m) \left(\frac{H}{2} - x \right) \right] \cos \theta - (M_1 + M_2 + m) \frac{W}{2} \sin \theta - ml \cos(\theta + \phi) \right\} \\ &\quad + \frac{1}{2} k \left(x + \frac{H}{2} - T - L_0 \right)^2 \end{aligned} \quad (156)$$

9.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate x :

$$(M_2 + m)\ddot{x} + \frac{1}{2}(M_2 + m)W\ddot{\theta} - ml(\ddot{\theta} + \ddot{\phi})\sin(2\theta + \phi) - ml(\dot{\theta} + \dot{\phi})^2\cos(2\theta + \phi) \\ + (M_2 + m)\left(\frac{H}{2} - x\right)\dot{\theta}^2 + k\left(x + \frac{H}{2} - T - L_0\right) - (M_2 + m)g\cos\theta = 0 \quad (157)$$

For coordinate ϕ :

$$ml^2(\ddot{\theta} + \ddot{\phi}) + ml\ddot{\theta}\left[\left(\frac{H}{2} - x\right)\cos(2\theta + \phi) - \frac{W}{2}\sin(2\theta + \phi)\right] - ml\ddot{x}\sin(2\theta + \phi) \\ - ml\dot{x}\dot{\theta}\cos(2\theta + \phi) - ml\dot{\theta}^2\left[\left(\frac{H}{2} - x\right)\sin(2\theta + \phi) + \frac{W}{2}\cos(2\theta + \phi)\right] + mgl\sin(\theta + \phi) = 0 \quad (158)$$

For coordinate θ :

$$\left\{ I_1 + M_1 \left[\left(\frac{W}{2} \right)^2 + \left(\frac{H}{2} \right)^2 \right] + I_2 + (M_2 + m) \left[\left(\frac{W}{2} \right)^2 + \left(\frac{H}{2} - x \right)^2 \right] + ml^2 \right. \\ \left. + 2ml \left[\left(\frac{H}{2} - x \right) \cos(2\theta + \phi) - \frac{W}{2} \sin(2\theta + \phi) \right] \right\} \ddot{\theta} \\ + \left\{ ml^2 + ml \left[\left(\frac{H}{2} - x \right) \cos(2\theta + \phi) - \frac{W}{2} \sin(2\theta + \phi) \right] \right\} \ddot{\phi} \\ + \left[\frac{1}{2}(M_2 + m)W - ml\sin(2\theta + \phi) \right] \ddot{x} - 2(M_2 + m) \left(\frac{H}{2} - x \right) \dot{x}\dot{\theta} - 2ml\dot{x}(\dot{\theta} + \dot{\phi})\cos(2\theta + \phi) \\ - 2ml\dot{\theta}(\dot{\theta} + \dot{\phi}) \left[\left(\frac{H}{2} - x \right) \sin(2\theta + \phi) + \frac{W}{2} \cos(2\theta + \phi) \right] \\ - g \left\{ \left[M_1 \frac{H}{2} + (M_2 + m) \left(\frac{H}{2} - x \right) \right] \sin\theta + (M_1 + M_2 + m) \frac{W}{2} \cos\theta + ml\sin(\theta + \phi) \right\} = 0 \quad (159)$$

9.5 5. Explicit Equations of Motion

Isolating the accelerations for each generalized coordinate:

Slider Linear Acceleration ($\ddot{x}$):

$$(M_2 + m)g\cos\theta - k\left(x + \frac{H}{2} - T - L_0\right) - (M_2 + m)\left(\frac{H}{2} - x\right)\dot{\theta}^2 \\ - \frac{1}{2}(M_2 + m)W\ddot{\theta} + ml(\ddot{\theta} + \ddot{\phi})\sin(2\theta + \phi) + ml(\dot{\theta} + \dot{\phi})^2\cos(2\theta + \phi) \\ \ddot{x} = \frac{M_2 + m}{M_2 + m} \quad (160)$$

Pendulum Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = -\ddot{\theta} + \frac{-g \sin(\theta + \phi) + \ddot{x} \sin(2\theta + \phi) - \ddot{\theta} \left[\left(\frac{H}{2} - x \right) \cos(2\theta + \phi) - \frac{W}{2} \sin(2\theta + \phi) \right] + \dot{x} \dot{\theta} \cos(2\theta + \phi) + \dot{\theta}^2 \left[\left(\frac{H}{2} - x \right) \sin(2\theta + \phi) + \frac{W}{2} \cos(2\theta + \phi) \right]}{l} \quad (161)$$

Frame Rocking Angular Acceleration ($\ddot{\theta}$): Defining the effective rotational inertia about pivot O :

$$J_{\theta}(x, \theta, \phi) = I_1 + M_1 \left[\left(\frac{W}{2} \right)^2 + \left(\frac{H}{2} \right)^2 \right] + I_2 + (M_2 + m) \left[\left(\frac{W}{2} \right)^2 + \left(\frac{H}{2} - x \right)^2 \right] + ml^2 + 2ml \left[\left(\frac{H}{2} - x \right) \cos(2\theta + \phi) - \frac{W}{2} \sin(2\theta + \phi) \right] \quad (162)$$

The explicit angular acceleration is:

$$\ddot{\theta} = \frac{g \left\{ \left[M_1 \frac{H}{2} + (M_2 + m) \left(\frac{H}{2} - x \right) \right] \sin \theta + (M_1 + M_2 + m) \frac{W}{2} \cos \theta + ml \sin(\theta + \phi) \right\} - \left\{ ml^2 + ml \left[\left(\frac{H}{2} - x \right) \cos(2\theta + \phi) - \frac{W}{2} \sin(2\theta + \phi) \right] \right\} \ddot{\phi} - \left[\frac{1}{2} (M_2 + m) W - ml \sin(2\theta + \phi) \right] \ddot{x} + 2(M_2 + m) \left(\frac{H}{2} - x \right) \dot{x} \dot{\theta} + 2ml \dot{x} (\dot{\theta} + \dot{\phi}) \cos(2\theta + \phi) + 2ml \dot{\theta} (\dot{\theta} + \dot{\phi}) \left[\left(\frac{H}{2} - x \right) \sin(2\theta + \phi) + \frac{W}{2} \cos(2\theta + \phi) \right]}{J_{\theta}(x, \theta, \phi)} \quad (163)$$

10 Problem 10: Pure-Rolling Disc with Triple Suspended Pendulums

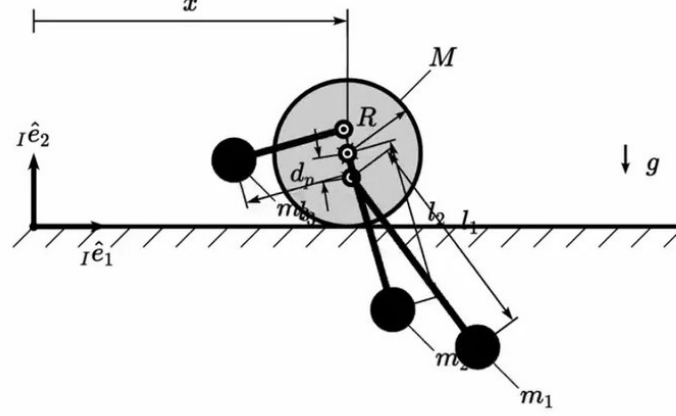


Figure 10: A cylinder of mass M and radius R rolling without slipping on a horizontal plane, carrying three simple pendulums: one pivoted at the center, and two pivoted at symmetric eccentric offsets d .

10.1 1. Generalized Coordinates and Parameters

Under the pure rolling condition $x = R\theta$, the planar system has four degrees of freedom:

- $\theta(t)$: Pure-rolling rotation angle of the disc of mass M , radius R , and central moment of inertia I .
- $\phi_1(t)$: Angular deflection of the bottom eccentric pendulum bob of mass m_1 and length l_1 .
- $\phi_2(t)$: Angular deflection of the central pendulum bob of mass m_2 and length l_2 .
- $\phi_3(t)$: Angular deflection of the top eccentric pendulum bob of mass m_3 and length l_3 .
- d : Radial offset of the top and bottom pendulum pivots from the center of the disc.

10.2 2. Kinematics

With pure rolling enforced ($x = R\theta, \dot{x} = R\dot{\theta}$), the positions and velocities of the three pendulum bobs are:

$$\mathbf{r}_1 = \begin{pmatrix} R\theta - d \sin \theta + l_1 \sin \phi_1 \\ R - d \cos \theta - l_1 \cos \phi_1 \end{pmatrix} \Rightarrow \begin{aligned} v_{1x} &= (R - d \cos \theta)\dot{\theta} + l_1\dot{\phi}_1 \cos \phi_1 \\ v_{1y} &= d\dot{\theta} \sin \theta + l_1\dot{\phi}_1 \sin \phi_1 \end{aligned} \quad (164)$$

$$\mathbf{r}_2 = \begin{pmatrix} R\theta + l_2 \sin \phi_2 \\ R - l_2 \cos \phi_2 \end{pmatrix} \Rightarrow \begin{aligned} v_{2x} &= R\dot{\theta} + l_2\dot{\phi}_2 \cos \phi_2 \\ v_{2y} &= l_2\dot{\phi}_2 \sin \phi_2 \end{aligned} \quad (165)$$

$$\mathbf{r}_3 = \begin{pmatrix} R\theta + d \sin \theta + l_3 \sin \phi_3 \\ R + d \cos \theta - l_3 \cos \phi_3 \end{pmatrix} \Rightarrow \begin{aligned} v_{3x} &= (R + d \cos \theta)\dot{\theta} + l_3\dot{\phi}_3 \cos \phi_3 \\ v_{3y} &= -d\dot{\theta} \sin \theta + l_3\dot{\phi}_3 \sin \phi_3 \end{aligned} \quad (166)$$

10.3 3. Kinetic and Potential Energy

The total kinetic energy of the rolling disc and three suspended bobs is:

$$\begin{aligned} KE &= \frac{1}{2}(MR^2 + I)\dot{\theta}^2 + \frac{1}{2}m_1(v_{1x}^2 + v_{1y}^2) + \frac{1}{2}m_2(v_{2x}^2 + v_{2y}^2) + \frac{1}{2}m_3(v_{3x}^2 + v_{3y}^2) \\ &= \frac{1}{2}(MR^2 + I)\dot{\theta}^2 + \frac{1}{2}m_1 \left[(R^2 + d^2 - 2Rd \cos \theta)\dot{\theta}^2 + l_1^2\dot{\phi}_1^2 + 2l_1\dot{\theta}\dot{\phi}_1(R \cos \phi_1 - d \cos(\theta + \phi_1)) \right] \\ &\quad + \frac{1}{2}m_2 \left[R^2\dot{\theta}^2 + l_2^2\dot{\phi}_2^2 + 2Rl_2\dot{\theta}\dot{\phi}_2 \cos \phi_2 \right] \\ &\quad + \frac{1}{2}m_3 \left[(R^2 + d^2 + 2Rd \cos \theta)\dot{\theta}^2 + l_3^2\dot{\phi}_3^2 + 2l_3\dot{\theta}\dot{\phi}_3(R \cos \phi_3 + d \cos(\theta + \phi_3)) \right] \end{aligned} \quad (167)$$

Taking the horizontal rolling plane as the potential energy datum:

$$PE = (m_3 - m_1)gd \cos \theta - g(m_1 l_1 \cos \phi_1 + m_2 l_2 \cos \phi_2 + m_3 l_3 \cos \phi_3) \quad (168)$$

10.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate ϕ_1 :

$$l_1\ddot{\phi}_1 + \ddot{\theta} [R \cos \phi_1 - d \cos(\theta + \phi_1)] + d\dot{\theta}^2 \sin(\theta + \phi_1) + g \sin \phi_1 = 0 \quad (169)$$

For coordinate ϕ_2 :

$$l_2\ddot{\phi}_2 + R\ddot{\theta} \cos \phi_2 + g \sin \phi_2 = 0 \quad (170)$$

For coordinate ϕ_3 :

$$l_3\ddot{\phi}_3 + \ddot{\theta} [R \cos \phi_3 + d \cos(\theta + \phi_3)] - d\dot{\theta}^2 \sin(\theta + \phi_3) + g \sin \phi_3 = 0 \quad (171)$$

For coordinate θ :

$$\begin{aligned}
& [(M + m_1 + m_2 + m_3)R^2 + I + (m_1 + m_3)d^2 + 2(m_3 - m_1)Rd \cos \theta] \ddot{\theta} \\
& - (m_3 - m_1)Rd\dot{\theta}^2 \sin \theta + m_2 R l_2 \left(\ddot{\phi}_2 \cos \phi_2 - \dot{\phi}_2^2 \sin \phi_2 \right) \\
& + m_1 l_1 \ddot{\phi}_1 [R \cos \phi_1 - d \cos(\theta + \phi_1)] - m_1 l_1 \dot{\phi}_1^2 [R \sin \phi_1 - d \sin(\theta + \phi_1)] \\
& + m_3 l_3 \ddot{\phi}_3 [R \cos \phi_3 + d \cos(\theta + \phi_3)] - m_3 l_3 \dot{\phi}_3^2 [R \sin \phi_3 + d \sin(\theta + \phi_3)] \\
& - (m_3 - m_1)gd \sin \theta = 0
\end{aligned} \tag{172}$$

10.5 5. Decoupled Explicit Equations of Motion

To obtain explicit, closed-form equations of motion without matrix inversion, define the geometric projections:

$$A_1 = R \cos \phi_1 - d \cos(\theta + \phi_1), \quad B_1 = R \sin \phi_1 - d \sin(\theta + \phi_1) \tag{173}$$

$$A_2 = R \cos \phi_2, \quad B_2 = R \sin \phi_2 \tag{174}$$

$$A_3 = R \cos \phi_3 + d \cos(\theta + \phi_3), \quad B_3 = R \sin \phi_3 + d \sin(\theta + \phi_3) \tag{175}$$

along with the baseline rotational inertia:

$$J(\theta) = (M + m_1 + m_2 + m_3)R^2 + I + (m_1 + m_3)d^2 + 2(m_3 - m_1)Rd \cos \theta \tag{176}$$

Eliminating the accelerations $\ddot{\phi}_1, \ddot{\phi}_2, \ddot{\phi}_3$ from the disc equation yields the effective inertia $\mathcal{D}(\theta, \phi_1, \phi_2, \phi_3)$ and generalized forcing $Y(\theta, \phi, \dot{\phi})$:

$$\mathcal{D}(\theta, \phi_1, \phi_2, \phi_3) = J(\theta) - m_1 A_1^2 - m_2 A_2^2 - m_3 A_3^2 \tag{177}$$

$$\begin{aligned}
Y(\theta, \phi, \dot{\phi}) = & (m_3 - m_1)Rd\dot{\theta}^2 \sin \theta + (m_3 - m_1)gd \sin \theta \\
& + m_1 l_1 \dot{\phi}_1^2 B_1 + m_2 R l_2 \dot{\phi}_2^2 \sin \phi_2 + m_3 l_3 \dot{\phi}_3^2 B_3 \\
& + m_1 A_1 \left(d\dot{\theta}^2 \sin(\theta + \phi_1) + g \sin \phi_1 \right) + m_2 A_2 (g \sin \phi_2) \\
& + m_3 A_3 \left(-d\dot{\theta}^2 \sin(\theta + \phi_3) + g \sin \phi_3 \right)
\end{aligned} \tag{178}$$

Disc Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{Y(\theta, \phi, \dot{\phi})}{\mathcal{D}(\theta, \phi_1, \phi_2, \phi_3)} \tag{179}$$

Bottom Pendulum Angular Acceleration ($\ddot{\phi}_1$):

$$\ddot{\phi}_1 = \frac{-A_1 \ddot{\theta} - d\dot{\theta}^2 \sin(\theta + \phi_1) - g \sin \phi_1}{l_1} \tag{180}$$

Middle Pendulum Angular Acceleration ($\ddot{\phi}_2$):

$$\ddot{\phi}_2 = \frac{-R\ddot{\theta} \cos \phi_2 - g \sin \phi_2}{l_2} \tag{181}$$

Top Pendulum Angular Acceleration ($\ddot{\phi}_3$):

$$\ddot{\phi}_3 = \frac{-A_3 \ddot{\theta} + d\dot{\theta}^2 \sin(\theta + \phi_3) - g \sin \phi_3}{l_3} \tag{182}$$

11 Problem 11: Regular Octagonal Rocker on Parabolic Track with Elastic Pendulum

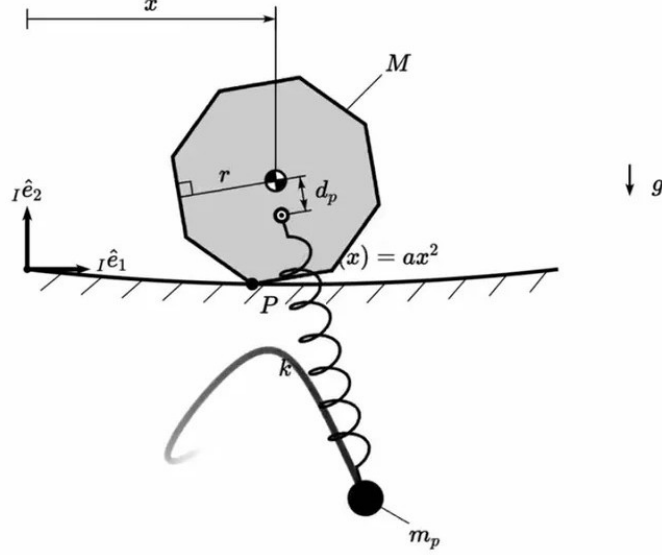


Figure 11: A regular octagonal cylinder of mass M , inradius r , and eccentric pin offset d rocking/rolling on a parabolic substrate $y(x) = ax^2$, supporting an elastic spring-pendulum of bob mass m .

11.1 1. Generalized Coordinates and Parameters

The planar system is formulated with three independent generalized coordinates:

- $\theta(t)$: Rocking/rotation angle of the regular octagonal body.
- $l(t)$: Instantaneous length of the elastic pendulum spring.
- $\phi(t)$: Angular deflection of the elastic pendulum bob from the downward vertical.
- $x(t)$: Horizontal coordinate of the contact point on the parabolic foundation $y(x) = ax^2$.
- M, I : Mass and centroidal moment of inertia of the regular octagonal body.
- r : Apothem (inradius) of the regular octagon; for a regular 8-gon, $\tan(\pi/8) = \sqrt{2} - 1 \approx 0.414$.
- d : Eccentric downward offset of the suspension pin from the octagon center of mass.
- m, k, l_0 : Mass of the pendulum bob, spring stiffness, and unstretched spring length.

11.2 2. Kinematics

The local slope angle $\beta(x)$ of the parabolic substrate $y(x) = ax^2$ satisfies:

$$\tan \beta = 2ax \implies \sin \beta = \frac{2ax}{\sqrt{1 + 4a^2x^2}}, \quad \cos \beta = \frac{1}{\sqrt{1 + 4a^2x^2}} \quad (183)$$

The center of mass coordinates (x_M, y_M) of the octagonal cylinder are:

$$x_M = x - r \sin \beta, \quad y_M = ax^2 + r \cos \beta \quad (184)$$

Differentiating with respect to time yields the velocity components:

$$\dot{x}_M = \left(1 - \frac{2ar}{(1 + 4a^2x^2)^{3/2}}\right) \dot{x} \quad (185)$$

$$\dot{y}_M = 2ax \left(1 - \frac{2ar}{(1 + 4a^2x^2)^{3/2}}\right) \dot{x} = 2ax\dot{x}_M \quad (186)$$

The instantaneous effective normal lever arm h_{eff} of the faceted octagonal face is:

$$h_{\text{eff}} = r [\cos(\theta - \beta) + 0.414 \sin(\theta - \beta)] \quad (187)$$

Enforcing the non-slip contact rolling constraint along the arc length relates $\dot{x}$ and $\dot{\theta}$:

$$\dot{x} \sqrt{1 + 4a^2x^2} = \dot{\theta} h_{\text{eff}} \implies \dot{\theta} = \frac{\sqrt{1 + 4a^2x^2}}{h_{\text{eff}}} \dot{x} \quad (188)$$

The position of the pendulum bob (x_p, y_p) relative to the fixed origin is:

$$x_p = x_M - d \sin \theta - l \sin \phi \quad (189)$$

$$y_p = y_M - d \cos \theta - l \cos \phi \quad (190)$$

Differentiating gives the Cartesian velocity components of the bob:

$$v_{px} = \dot{x}_M - d\dot{\theta} \cos \theta - \dot{l} \sin \phi - l\dot{\phi} \cos \phi \quad (191)$$

$$v_{py} = \dot{y}_M + d\dot{\theta} \sin \theta - \dot{l} \cos \phi + l\dot{\phi} \sin \phi \quad (192)$$

11.3 3. Kinetic and Potential Energy

The total scalar kinetic energy of the system is:

$$\begin{aligned} KE &= \frac{1}{2}M(\dot{x}_M^2 + \dot{y}_M^2) + \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}m(v_{px}^2 + v_{py}^2) \\ &= \frac{1}{2}(M + m)(\dot{x}_M^2 + \dot{y}_M^2) + \frac{1}{2}(I + md^2)\dot{\theta}^2 + \frac{1}{2}m(\dot{l}^2 + l^2\dot{\phi}^2) \\ &\quad - md\dot{\theta}(\dot{x}_M \cos \theta - \dot{y}_M \sin \theta) - m\dot{l}(\dot{x}_M \sin \phi + \dot{y}_M \cos \phi) \\ &\quad - ml\dot{\phi}(\dot{x}_M \cos \phi - \dot{y}_M \sin \phi) + mdl\dot{\theta}\dot{\phi} \cos(\phi - \theta) + mdl\dot{\theta} \sin(\phi - \theta) \end{aligned} \quad (193)$$

Taking the reference datum at the base of the parabola, the total potential energy is:

$$PE = (M + m)g \left(ax^2 + \frac{r}{\sqrt{1 + 4a^2x^2}} \right) - mgd \cos \theta - mgl \cos \phi + \frac{1}{2}k(l - l_0)^2 \quad (194)$$

11.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For spring length l :

$$\ddot{l} - l\dot{\phi}^2 + d\ddot{\theta} \sin(\phi - \theta) - d\dot{\theta}^2 \cos(\phi - \theta) - \ddot{x}_M \sin \phi - (\ddot{y}_M + g) \cos \phi + \frac{k}{m}(l - l_0) = 0 \quad (195)$$

For pendulum angle ϕ :

$$l\ddot{\phi} + 2\dot{l}\dot{\phi} + d\ddot{\theta}\cos(\phi - \theta) + d\dot{\theta}^2\sin(\phi - \theta) - \ddot{x}_M\cos\phi + (\ddot{y}_M + g)\sin\phi = 0 \quad (196)$$

For octagon angle θ :

$$\begin{aligned} & (I + md^2 + (M + m)h_{\text{eff}}^2)\ddot{\theta} + md\left[\ddot{l}\sin(\phi - \theta) + l\ddot{\phi}\cos(\phi - \theta) + 2\dot{l}\dot{\phi}\cos(\phi - \theta) - l\dot{\phi}^2\sin(\phi - \theta)\right] \\ & - mh_{\text{eff}}\left[\ddot{l}\sin(\phi - \beta) + l\ddot{\phi}\cos(\phi - \beta) + 2\dot{l}\dot{\phi}\cos(\phi - \beta) - l\dot{\phi}^2\sin(\phi - \beta)\right] \\ & + (M + m)g(h_{\text{eff}}\sin\beta - d\sin\theta) + (M + m)h_{\text{eff}}\dot{h}_{\text{eff}}\dot{\theta}^2 = 0 \end{aligned} \quad (197)$$

11.5 5. Explicit Equations of Motion

Isolating the second-order accelerations for each generalized coordinate:

Octagon Angular Acceleration ($\ddot{\theta}$):

$$\begin{aligned} & - (M + m)g(h_{\text{eff}}\sin\beta - d\sin\theta) - (M + m)h_{\text{eff}}\dot{h}_{\text{eff}}\dot{\theta}^2 \\ & - md\left(2\dot{l}\dot{\phi}\cos(\phi - \theta) - l\dot{\phi}^2\sin(\phi - \theta)\right) + mh_{\text{eff}}\left(2\dot{l}\dot{\phi}\cos(\phi - \beta) - l\dot{\phi}^2\sin(\phi - \beta)\right) \\ & - [m(d\sin(\phi - \theta) - h_{\text{eff}}\sin(\phi + \beta))] \left(l\dot{\phi}^2 + d\dot{\theta}^2\cos(\phi - \theta) + g\cos\phi - \frac{k}{m}(l - l_0) + h_{\text{eff}}\dot{\theta}\sin(\phi + \beta)\right) \\ & - [m(d\cos(\phi - \theta) - h_{\text{eff}}\cos(\phi + \beta))] \left(-2\dot{l}\dot{\phi} - d\dot{\theta}^2\sin(\phi - \theta) - g\sin\phi + h_{\text{eff}}\dot{\theta}\cos(\phi + \beta)\right) \\ \ddot{\theta} = & \frac{}{I + Mh_{\text{eff}}^2 + 2mdh_{\text{eff}}\cos(\theta + \beta)} \end{aligned} \quad (198)$$

Spring Length Acceleration ($\ddot{l}$):

$$\ddot{l} = l\dot{\phi}^2 + d\dot{\theta}^2\cos(\phi - \theta) + g\cos\phi - \frac{k}{m}(l - l_0) + h_{\text{eff}}\dot{\theta}\sin(\phi + \beta) - [d\sin(\phi - \theta) - h_{\text{eff}}\sin(\phi + \beta)]\ddot{\theta} \quad (199)$$

Bob Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{-2\dot{l}\dot{\phi} - d\dot{\theta}^2\sin(\phi - \theta) - g\sin\phi + h_{\text{eff}}\dot{\theta}\cos(\phi + \beta) - [d\cos(\phi - \theta) - h_{\text{eff}}\cos(\phi + \beta)]\ddot{\theta}}{l} \quad (200)$$

12 Problem 12: Rocking Semicircular Shell with Nested Rolling Cylinder and T-Bar Pendulum

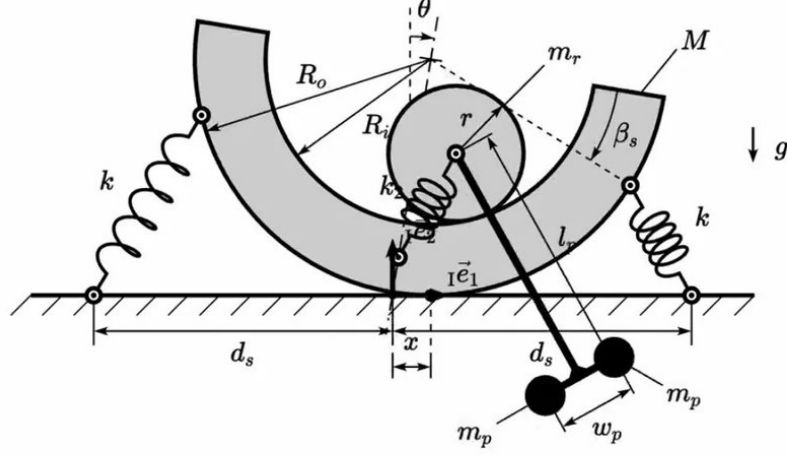


Figure 12: A thick semicircular cylindrical shell of mass M rolling on a horizontal surface restrained by two flanking ground springs, containing a pure-rolling cylinder of mass m_r coupled to the inner wall via spring k_2 , which suspends a rigid T-bar pendulum carrying two tip masses m_p .

12.1 1. Generalized Coordinates and Parameters

Under pure rolling of the outer shell ($x = R_o\theta$), the planar system has three generalized coordinates:

- $\theta(t)$: Rolling tilt angle of the outer semicircular shell of mass M , outer radius R_o , inner radius R_i , and central moment of inertia I_M .
- $\phi(t)$: Relative angular position of the inner rolling cylinder center inside the shell cavity.
- $\alpha(t)$: Angular deflection of the suspended T-bar pendulum rod from the downward vertical.
- m_r, I_m, r : Mass, central moment of inertia, and radius of the internal rolling cylinder.
- m_p : Mass of each of the two symmetric tip bobs mounted on the crossbar of width w_p at stem length L .
- β_s : Angular span defining the mounting anchors of the two external restraint springs on the shell.
- d_s : Horizontal distance from the central contact origin to each fixed ground anchor.
- k_1, k_3 : Stiffness of the left and right ground-anchoring linear springs (l_{01}, l_{03} : unstretched lengths).
- k_2, R_s : Stiffness of the internal coupling spring (l_{02} : unstretched length) and radial anchor position on the shell.

12.2 2. Kinematics and Spring Geometry

Enforcing pure rolling of the internal cylinder along the inner cavity boundary yields its spin angular velocity:

$$(\dot{\theta} + \dot{\phi})(R_i - r) + r\omega_{\text{spin}} = R_i\dot{\theta} \implies \omega_{\text{spin}} = \dot{\theta} - \left(\frac{R_i - r}{r}\right)\dot{\phi} \quad (201)$$

The instantaneous lengths of the left external spring l_1 , right external spring l_3 , and internal cavity spring l_2 are:

$$l_1 = \sqrt{(R_0\theta - R_0\sin(\beta_s + \theta) + d_s)^2 + (R_0 - R_0\cos(\beta_s + \theta))^2} \quad (202)$$

$$l_2 = \sqrt{(R_i - r)^2 + R_s^2 + 2R_s(R_i - r)\cos\phi} \quad (203)$$

$$l_3 = \sqrt{(R_0\theta + R_0\sin(\beta_s - \theta) - d_s)^2 + (R_0 - R_0\cos(\beta_s - \theta))^2} \quad (204)$$

Their partial derivatives with respect to coordinates θ and ϕ are:

$$\frac{\partial l_1}{\partial \theta} = \frac{[R_0\theta - R_0\sin(\beta_s + \theta) + d_s] R_0 (1 - \cos(\beta_s + \theta)) - [R_0 - R_0\cos(\beta_s + \theta)] R_0 \sin(\beta_s + \theta)}{l_1} \quad (205)$$

$$\frac{\partial l_2}{\partial \phi} = \frac{R_s(R_i - r)\sin\phi}{l_2} \quad (206)$$

$$\frac{\partial l_3}{\partial \theta} = \frac{[R_0\theta + R_0\sin(\beta_s - \theta) - d_s] R_0 (1 + \cos(\beta_s - \theta)) - [R_0 - R_0\cos(\beta_s - \theta)] R_0 \sin(\beta_s - \theta)}{l_3} \quad (207)$$

The Cartesian position coordinates of the internal cylinder center (x_c, y_c) and the two pendulum bobs $(x_{p\pm}, y_{p\pm})$ are:

$$x_c = R_0\theta + (R_i - r)\sin(\theta + \phi), \quad y_c = R_0 - (R_i - r)\cos(\theta + \phi) \quad (208)$$

$$x_{p\pm} = x_c + L\sin\alpha \mp \frac{w_p}{2}\cos\alpha, \quad y_{p\pm} = y_c - L\cos\alpha \mp \frac{w_p}{2}\sin\alpha \quad (209)$$

Differentiating with respect to time yields the velocity vectors:

$$v_{cx} = R_0\dot{\theta} + (R_i - r)(\dot{\theta} + \dot{\phi})\cos(\theta + \phi) \quad (210)$$

$$v_{cy} = (R_i - r)(\dot{\theta} + \dot{\phi})\sin(\theta + \phi) \quad (211)$$

$$v_{px\pm} = v_{cx} + L\dot{\alpha}\cos\alpha \pm \frac{w_p}{2}\dot{\alpha}\sin\alpha \quad (212)$$

$$v_{py\pm} = v_{cy} + L\dot{\alpha}\sin\alpha \mp \frac{w_p}{2}\dot{\alpha}\cos\alpha \quad (213)$$

12.3 3. Kinetic and Potential Energy

Summing the kinetic energies of the rolling shell, inner rolling cylinder, and both pendulum bobs (noting cross terms with $\pm w_p/2$ cancel identically):

$$\begin{aligned} KE &= \frac{1}{2}MR_0^2\dot{\theta}^2 + \frac{1}{2}I_M\dot{\theta}^2 + \frac{1}{2}m_r(v_{cx}^2 + v_{cy}^2) + \frac{1}{2}I_m\omega_{\text{spin}}^2 + \frac{1}{2}m_p(v_{px+}^2 + v_{py+}^2 + v_{px-}^2 + v_{py-}^2) \\ &= \frac{1}{2}(MR_0^2 + I_M)\dot{\theta}^2 + \frac{1}{2}m_r\left[R_0^2\dot{\theta}^2 + (R_i - r)^2(\dot{\theta} + \dot{\phi})^2 + 2R_0(R_i - r)\dot{\theta}(\dot{\theta} + \dot{\phi})\cos(\theta + \phi)\right] \\ &\quad + \frac{1}{2}I_m\left[\dot{\theta} - \left(\frac{R_i - r}{r}\right)\dot{\phi}\right]^2 + m_p\left[R_0^2\dot{\theta}^2 + (R_i - r)^2(\dot{\theta} + \dot{\phi})^2 + 2R_0(R_i - r)\dot{\theta}(\dot{\theta} + \dot{\phi})\cos(\theta + \phi)\right] \\ &\quad + m_p\left(L^2 + \frac{w_p^2}{4}\right)\dot{\alpha}^2 + 2m_pL\dot{\alpha}\left[R_0\dot{\theta}\cos\alpha + (R_i - r)(\dot{\theta} + \dot{\phi})\cos(\theta + \phi - \alpha)\right] \end{aligned} \quad (214)$$

Taking the flat ground surface as the reference datum, the total potential energy stored in gravity and the three linear springs is:

$$PE = (m_r + 2m_p)g [R_0 - (R_i - r) \cos(\theta + \phi)] - 2m_p g L \cos \alpha + \frac{1}{2}k_1(l_1 - l_{01})^2 + \frac{1}{2}k_2(l_2 - l_{02})^2 + \frac{1}{2}k_3(l_3 - l_{03})^2 \quad (215)$$

12.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate α :

$$2m_p \left(L^2 + \frac{w_p^2}{4} \right) \ddot{\alpha} + 2m_p L R_0 \ddot{\theta} \cos \alpha + 2m_p L (R_i - r) (\ddot{\theta} + \ddot{\phi}) \cos(\theta + \phi - \alpha) - 2m_p L (R_i - r) (\dot{\theta} + \dot{\phi})^2 \sin(\theta + \phi - \alpha) + 2m_p g L \sin \alpha = 0 \quad (216)$$

For coordinate ϕ :

$$\begin{aligned} & \left[(m_r + 2m_p)(R_i - r)^2 + \left(\frac{R_i - r}{r} \right)^2 I_m \right] \ddot{\phi} \\ & + \left[(m_r + 2m_p)(R_i - r)^2 + (m_r + 2m_p)R_0(R_i - r) \cos(\theta + \phi) - \left(\frac{R_i - r}{r} \right) I_m \right] \ddot{\theta} \\ & + 2m_p L (R_i - r) \ddot{\alpha} \cos(\theta + \phi - \alpha) + 2m_p L (R_i - r) \dot{\alpha}^2 \sin(\theta + \phi - \alpha) \\ & + (m_r + 2m_p)g(R_i - r) \sin(\theta + \phi) + k_2(l_2 - l_{02}) \frac{\partial l_2}{\partial \phi} = 0 \end{aligned} \quad (217)$$

For coordinate θ :

$$\begin{aligned} & [MR_0^2 + I_M + (m_r + 2m_p) (R_0^2 + (R_i - r)^2 + 2R_0(R_i - r) \cos(\theta + \phi)) + I_m] \ddot{\theta} \\ & + \left[(m_r + 2m_p) ((R_i - r)^2 + R_0(R_i - r) \cos(\theta + \phi)) - \left(\frac{R_i - r}{r} \right) I_m \right] \ddot{\phi} \\ & + 2m_p L [R_0 \cos \alpha + (R_i - r) \cos(\theta + \phi - \alpha)] \ddot{\alpha} \\ & - (m_r + 2m_p)R_0(R_i - r) (\dot{\theta} + \dot{\phi})^2 \sin(\theta + \phi) - 2m_p L R_0 \dot{\alpha}^2 \sin \alpha \\ & + 2m_p L (R_i - r) \dot{\alpha}^2 \sin(\theta + \phi - \alpha) + (m_r + 2m_p)g(R_i - r) \sin(\theta + \phi) \\ & + k_1(l_1 - l_{01}) \frac{\partial l_1}{\partial \theta} + k_3(l_3 - l_{03}) \frac{\partial l_3}{\partial \theta} = 0 \end{aligned} \quad (218)$$

12.5 5. Explicit Decoupled Equations of Motion

To obtain explicit scalar acceleration forms, define the symmetric inertia coefficients:

$$M_\theta = MR_0^2 + I_M + (m_r + 2m_p) [R_0^2 + (R_i - r)^2 + 2R_0(R_i - r) \cos(\theta + \phi)] + I_m \quad (219)$$

$$M_\phi = (m_r + 2m_p)(R_i - r)^2 + \left(\frac{R_i - r}{r}\right)^2 I_m \quad (220)$$

$$M_\alpha = 2m_p \left(L^2 + \frac{w_p^2}{4} \right) \quad (221)$$

$$M_{\theta\phi} = (m_r + 2m_p) [(R_i - r)^2 + R_0(R_i - r) \cos(\theta + \phi)] - \left(\frac{R_i - r}{r}\right) I_m \quad (222)$$

$$M_{\theta\alpha} = 2m_p L [R_0 \cos \alpha + (R_i - r) \cos(\theta + \phi - \alpha)] \quad (223)$$

$$M_{\phi\alpha} = 2m_p L (R_i - r) \cos(\theta + \phi - \alpha) \quad (224)$$

and generalized forcing terms:

$$\begin{aligned} Q_1 = & (m_r + 2m_p) R_0 (R_i - r) (\dot{\theta} + \dot{\phi})^2 \sin(\theta + \phi) + 2m_p L R_0 \dot{\alpha}^2 \sin \alpha \\ & - 2m_p L (R_i - r) \dot{\alpha}^2 \sin(\theta + \phi - \alpha) - (m_r + 2m_p) g (R_i - r) \sin(\theta + \phi) \\ & - k_1 (l_1 - l_{01}) \frac{\partial l_1}{\partial \theta} - k_3 (l_3 - l_{03}) \frac{\partial l_3}{\partial \theta} \end{aligned} \quad (225)$$

$$Q_2 = -2m_p L (R_i - r) \dot{\alpha}^2 \sin(\theta + \phi - \alpha) - (m_r + 2m_p) g (R_i - r) \sin(\theta + \phi) - k_2 (l_2 - l_{02}) \frac{\partial l_2}{\partial \phi} \quad (226)$$

$$Q_3 = 2m_p L (R_i - r) (\dot{\theta} + \dot{\phi})^2 \sin(\theta + \phi - \alpha) - 2m_p g L \sin \alpha \quad (227)$$

The system determinant is:

$$Y = M_\theta (M_\phi M_\alpha - M_{\phi\alpha}^2) - M_{\theta\phi} (M_{\theta\phi} M_\alpha - M_{\phi\alpha} M_{\theta\alpha}) + M_{\theta\alpha} (M_{\theta\phi} M_{\phi\alpha} - M_\phi M_{\theta\alpha}) \quad (228)$$

Shell Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{Q_1 (M_\phi M_\alpha - M_{\phi\alpha}^2) - M_{\theta\phi} (Q_2 M_\alpha - M_{\phi\alpha} Q_3) + M_{\theta\alpha} (Q_2 M_{\phi\alpha} - M_\phi Q_3)}{Y} \quad (229)$$

Nested Cylinder Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{M_\theta (Q_2 M_\alpha - M_{\phi\alpha} Q_3) - Q_1 (M_{\theta\phi} M_\alpha - M_{\phi\alpha} M_{\theta\alpha}) + M_{\theta\alpha} (M_{\theta\phi} Q_3 - Q_2 M_{\theta\alpha})}{Y} \quad (230)$$

T-Bar Pendulum Angular Acceleration ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{M_\theta (M_\phi Q_3 - Q_2 M_{\phi\alpha}) - M_{\theta\phi} (M_{\theta\phi} Q_3 - Q_2 M_{\theta\alpha}) + Q_1 (M_{\theta\phi} M_{\phi\alpha} - M_\phi M_{\theta\alpha})}{Y} \quad (231)$$

13 Problem 13: Pure-Rolling Cylinder on a Vertical Wall with Offset Spring and Suspended Pendulum

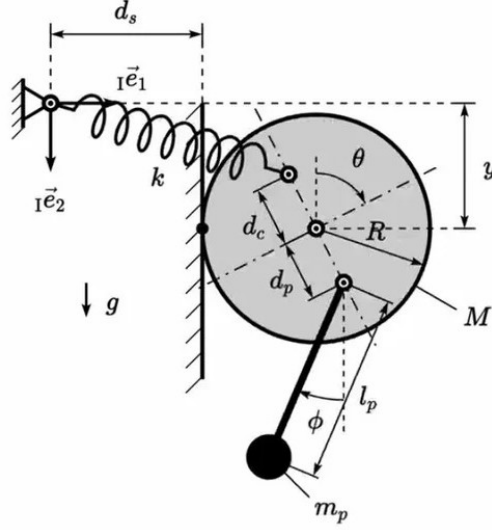


Figure 13: A cylinder of mass M and radius R rolling vertically without slipping along a wall, restrained by a linear spring k connected to an eccentric pin d_c , and carrying a simple pendulum of mass m and length l attached to an eccentric pivot d_p .

13.1 1. Generalized Coordinates and Parameters

Under pure rolling along the vertical wall ($y = R\theta$), the planar system has two degrees of freedom:

- $\theta(t)$: Rolling rotation angle of the cylinder of mass M , radius R , and central moment of inertia I .
- $\phi(t)$: Angular deflection of the pendulum rod from the downward vertical.
- m, l : Mass of the pendulum bob and length of the pendulum rod.
- d_c : Eccentric distance of the spring attachment pin from the cylinder center.
- d_p : Eccentric distance of the pendulum pivot pin from the cylinder center.
- d_s : Horizontal distance between the fixed spring anchor and the vertical wall.
- k, L_0 : Stiffness and unstretched length of the linear anchor spring.
- g : Acceleration due to gravity acting downwards.

13.2 2. Kinematics and Spring Geometry

Taking the fixed spring anchor as the origin $(0, 0)$, the Cartesian coordinates of the cylinder center are $(d_s + R, R\theta)$. The spring connects to the pin at:

$$x_{\text{spring}} = d_s + R - d_c \cos \theta \quad (232)$$

$$y_{\text{spring}} = R\theta - d_c \sin \theta \quad (233)$$

The instantaneous extended length of the spring L_s and its derivative with respect to θ are:

$$L_s = \sqrt{(d_s + R - d_c \cos \theta)^2 + (R\theta - d_c \sin \theta)^2} \quad (234)$$

$$\frac{\partial L_s}{\partial \theta} = \frac{(d_s + R - d_c \cos \theta)d_c \sin \theta + (R\theta - d_c \sin \theta)(R - d_c \cos \theta)}{L_s} \quad (235)$$

The pendulum bob is located at:

$$x_{\text{bob}} = d_s + R - d_p \cos \theta + l \sin \phi \quad (236)$$

$$y_{\text{bob}} = R\theta + d_p \sin \theta + l \cos \phi \quad (237)$$

Differentiating with respect to time yields the velocity components of the bob:

$$v_x = -d_p \dot{\theta} \sin \theta + l \dot{\phi} \cos \phi \quad (238)$$

$$v_y = (R + d_p \cos \theta) \dot{\theta} - l \dot{\phi} \sin \phi \quad (239)$$

13.3 3. Kinetic and Potential Energy

The total kinetic energy of the rolling cylinder and suspended bob is:

$$\begin{aligned} KE &= \frac{1}{2}(MR^2 + I)\dot{\theta}^2 + \frac{1}{2}m(v_x^2 + v_y^2) \\ &= \frac{1}{2}[MR^2 + I + m(R^2 + d_p^2 + 2Rd_p \cos \theta)]\dot{\theta}^2 + \frac{1}{2}ml^2\dot{\phi}^2 \\ &\quad - ml\dot{\theta}\dot{\phi}[d_p \sin(\theta + \phi) + R \sin \phi] \end{aligned} \quad (240)$$

Taking the horizontal line passing through the anchor origin as the potential energy datum:

$$PE = -(M + m)gR\theta - mgd_p \sin \theta - mgl \cos \phi + \frac{1}{2}k(L_s - L_0)^2 \quad (241)$$

13.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For coordinate ϕ :

$$ml^2\ddot{\phi} - ml[d_p \sin(\theta + \phi) + R \sin \phi]\ddot{\theta} - mld_p\dot{\theta}^2 \cos(\theta + \phi) + mgl \sin \phi = 0 \quad (242)$$

For coordinate θ :

$$\begin{aligned} &[MR^2 + I + m(R^2 + d_p^2 + 2Rd_p \cos \theta)]\ddot{\theta} - ml[d_p \sin(\theta + \phi) + R \sin \phi]\ddot{\phi} \\ &\quad - mRd_p\dot{\theta}^2 \sin \theta - ml\dot{\phi}^2[d_p \cos(\theta + \phi) + R \cos \phi] - (M + m)gR - mgd_p \cos \theta \\ &\quad + k(L_s - L_0) \left[\frac{(d_s + R - d_c \cos \theta)d_c \sin \theta + (R\theta - d_c \sin \theta)(R - d_c \cos \theta)}{L_s} \right] = 0 \end{aligned} \quad (243)$$

13.5 5. Explicit Decoupled Equations of Motion

To represent the solution in closed form, define the inertia factors:

$$A = MR^2 + I + m(R^2 + d_p^2 + 2Rd_p \cos \theta) \quad (244)$$

$$B = d_p \sin(\theta + \phi) + R \sin \phi \quad (245)$$

$$Y = A - mB^2 \quad (246)$$

along with the generalized forcing terms:

$$Q_1 = mRd_p\dot{\theta}^2 \sin \theta + ml\dot{\phi}^2 [d_p \cos(\theta + \phi) + R \cos \phi] + (M + m)gR + mgd_p \cos \theta - k(L_s - L_0) \left[\frac{(d_s + R - d_c \cos \theta)d_c \sin \theta + (R\theta - d_c \sin \theta)(R - d_c \cos \theta)}{L_s} \right] \quad (247)$$

$$Q_2 = d_p\dot{\theta}^2 \cos(\theta + \phi) - g \sin \phi \quad (248)$$

Cylinder Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{Q_1 + mBQ_2}{Y} \quad (249)$$

Pendulum Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{BQ_1 + AQ_2}{lY} \quad (250)$$

14 Problem 14: Parallel-Link Suspended Frame with Spring-Loaded Slider and Pendulum

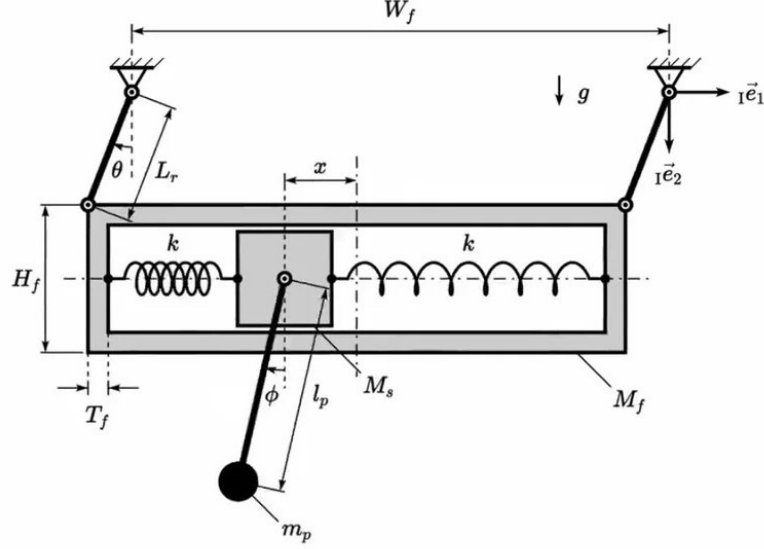


Figure 14: A rectangular frame of mass M suspended by parallel links of length L at tilt angle θ , containing an internal slider of mass m restrained by two symmetric springs of stiffness k , and suspending a simple pendulum of bob mass m' and length l .

14.1 1. Generalized Coordinates and Parameters

The planar system has three coupled degrees of freedom:

- $\theta(t)$: Angular deflection of the parallel suspension links of length L from the downward vertical.
- $x(t)$: Relative linear displacement of the slider inside the frame from its neutral central position.
- $\phi(t)$: Angular deflection of the suspended pendulum rod from the downward vertical.
- M : Mass of the outer rectangular frame.
- m : Mass of the internal translating slider.
- m' : Mass of the pendulum bob suspended from the slider.
- L, l : Length of the frame suspension links and length of the pendulum rod, respectively.
- k : Stiffness of each of the two horizontal centering springs (effective total stiffness $2k$).
- W, H : Width and height of the rectangular frame.

14.2 2. Kinematics

Taking the midpoint between the top ground pivots as the reference origin, the Cartesian position and velocity of the frame center are:

$$x_M = \frac{W}{2} + L \sin \theta \implies V_x = L\dot{\theta} \cos \theta \quad (251)$$

$$y_M = \frac{H}{2} + L \cos \theta \implies V_y = -L\dot{\theta} \sin \theta \quad (252)$$

The Cartesian position and velocity components of the internal slider are:

$$x_m = x + \frac{W}{2} + L \sin \theta \implies v'_x = \dot{x} + L\dot{\theta} \cos \theta \quad (253)$$

$$y_m = \frac{H}{2} + L \cos \theta \implies v'_y = -L\dot{\theta} \sin \theta \quad (254)$$

The Cartesian position and velocity components of the pendulum bob are:

$$x_p = x + \frac{W}{2} + L \sin \theta + l \sin \phi \implies v_x = \dot{x} + L\dot{\theta} \cos \theta + l\dot{\phi} \cos \phi \quad (255)$$

$$y_p = \frac{H}{2} + L \cos \theta + l \cos \phi \implies v_y = -L\dot{\theta} \sin \theta - l\dot{\phi} \sin \phi \quad (256)$$

14.3 3. Kinetic and Potential Energy

The total kinetic energy combines the translational motion of the frame (guided without rotation by the parallelogram suspension), the slider, and the pendulum bob:

$$\begin{aligned} KE &= \frac{1}{2}M(V_x^2 + V_y^2) + \frac{1}{2}m(v_x'^2 + v_y'^2) + \frac{1}{2}m'(v_x^2 + v_y^2) \\ &= \frac{1}{2}ML^2\dot{\theta}^2 + \frac{1}{2}m(\dot{x}^2 + L^2\dot{\theta}^2 + 2L\dot{x}\dot{\theta} \cos \theta) \\ &\quad + \frac{1}{2}m'[\dot{x}^2 + L^2\dot{\theta}^2 + l^2\dot{\phi}^2 + 2l\dot{x}\dot{\phi} \cos \phi + 2L\dot{x}\dot{\theta} \cos \theta + 2Ll\dot{\theta}\dot{\phi} \cos(\theta - \phi)] \end{aligned} \quad (257)$$

Taking the pivot origin as the potential energy datum, the total potential energy stored in gravity and the two springs is:

$$PE = -(M + m + m')gL \cos \theta - m'gl \cos \phi + kx^2 \quad (258)$$

14.4 4. Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For slider displacement x :

$$(m + m')\ddot{x} + (m + m')L\ddot{\theta} \cos \theta + m'l\ddot{\phi} \cos \phi - (m + m')L\dot{\theta}^2 \sin \theta - m'l\dot{\phi}^2 \sin \phi + 2kx = 0 \quad (259)$$

For frame link angle θ :

$$\begin{aligned} &(M + m + m')L^2\ddot{\theta} + (m + m')L\ddot{x} \cos \theta + m'Ll\ddot{\phi} \cos(\theta - \phi) \\ &\quad + m'Ll\dot{\phi}^2 \sin(\theta - \phi) + (M + m + m')gL \sin \theta = 0 \end{aligned} \quad (260)$$

For pendulum angle ϕ :

$$m'l^2\ddot{\phi} + m'l\ddot{x} \cos \phi + m' Ll\ddot{\theta} \cos(\theta - \phi) - m' Ll\dot{\theta}^2 \sin(\theta - \phi) + m' gl \sin \phi = 0 \quad (261)$$

14.5 5. Explicit Decoupled Equations of Motion

To express the system in closed explicit form, define the mass and coupling terms:

$$m_1 = m + m', \quad m_2 = M + m + m' \quad (262)$$

$$m_x = m_1, \quad m_\theta = m_2 L^2, \quad m_\phi = m' l^2 \quad (263)$$

$$M_1 = m_1 L \cos \theta, \quad M_2 = m' l \cos \phi, \quad M_3 = m' Ll \cos(\theta - \phi) \quad (264)$$

along with the generalized forcing terms:

$$Q_1 = m_1 L \dot{\theta}^2 \sin \theta + m' l \dot{\phi}^2 \sin \phi - 2kx \quad (265)$$

$$Q_2 = -m' Ll \dot{\phi}^2 \sin(\theta - \phi) - m_2 g L \sin \theta \quad (266)$$

$$Q_3 = m' Ll \dot{\theta}^2 \sin(\theta - \phi) - m' gl \sin \phi \quad (267)$$

The inertia determinant Y is:

$$Y = m_x (m_\theta m_\phi - M_3^2) - M_1 (M_1 m_\phi - M_2 M_3) + M_2 (M_1 M_3 - M_2 m_\theta) \quad (268)$$

Slider Linear Acceleration ($\ddot{x}$):

$$\ddot{x} = \frac{Q_1 (m_\theta m_\phi - M_3^2) + Q_2 (M_2 M_3 - M_1 m_\phi) + Q_3 (M_1 M_3 - M_2 m_\theta)}{Y} \quad (269)$$

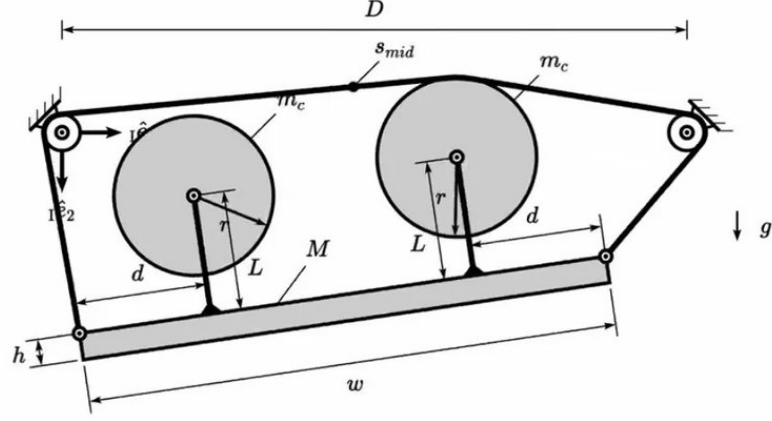
Suspended Frame Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{Q_1 (M_2 M_3 - M_1 m_\phi) + Q_2 (m_x m_\phi - M_2^2) + Q_3 (M_1 M_2 - m_x M_3)}{Y} \quad (270)$$

Pendulum Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{Q_1 (M_1 M_3 - M_2 m_\theta) + Q_2 (M_1 M_2 - m_x M_3) + Q_3 (m_x m_\theta - M_1^2)}{Y} \quad (271)$$

15 Problem 15: Suspended Platform with Dual Cylinders and Cable Constraint



$$M\ddot{q} - F_{\text{ext}} = J^T \lambda, \quad 0 \leq \lambda \perp \Phi(q) \geq 0$$

$$\begin{bmatrix} M & -J^T \\ J & 0 \end{bmatrix} \begin{bmatrix} \dot{q}^+ \\ \Lambda \end{bmatrix} = \begin{bmatrix} M\dot{q}^- \\ 0 \end{bmatrix}$$

$$q = [x, y, \theta, \phi_1, \phi_2]^T, \quad M = \text{diag}(m_{\text{tot}}, m_{\text{tot}}, I_{\text{base}}, I_c, I_c), \quad J = \nabla \Phi(q)$$

Figure 15: A rigid platform of mass M and width w carrying two cylinders of mass m atop posts of height L , suspended by a continuous cable routed across two fixed pulleys separated by span D .

15.1 Coordinate System and Parameters

The planar platform system is parameterized by:

- $x(t)$: Vertical displacement of the bottom-left corner of the bar (taken positive downward).
- $y(t)$: Horizontal displacement of the bottom-left corner of the bar.
- $\alpha(t)$: Tilt angle of the rigid platform from the horizontal.
- $\phi_1(t), \phi_2(t)$: Rotation angles of the left and right cylinders about their respective axes.
- M, I_M : Mass and centroidal moment of inertia of the main platform bar of width w .
- m, I_m : Mass and moment of inertia of each of the two identical cylinders.
- L : Height of the rigid vertical stalks supporting the cylinders.
- $d, w - d$: Horizontal positions along the bar where the two stalks are mounted.
- D : Horizontal separation between the two top fixed pulleys.
- λ : Lagrange multiplier representing the scalar magnitude of the string tension.
- s_1, s_2, s_3, s_4, s_5 : Lengths of the five distinct cable segments.

15.2 Kinematics

Taking the top-left fixed pulley as the coordinate origin $(0, 0)$, the Cartesian coordinates and velocities of the platform center of mass are:

$$x_{\text{CM}} = x + \frac{w}{2} \sin \alpha \implies v_y = \dot{x} + \frac{w}{2} \dot{\alpha} \cos \alpha \quad (272)$$

$$y_{\text{CM}} = y + \frac{w}{2} \cos \alpha \implies v_x = \dot{y} - \frac{w}{2} \dot{\alpha} \sin \alpha \quad (273)$$

The coordinates and velocities of the left cylinder center (x_L, y_L) and right cylinder center (x_R, y_R) are:

$$x_L = x + d \sin \alpha - L \cos \alpha \implies V_y = \dot{x} + d \dot{\alpha} \cos \alpha + L \dot{\alpha} \sin \alpha \quad (274)$$

$$y_L = y + d \cos \alpha + L \sin \alpha \implies V_x = \dot{y} - d \dot{\alpha} \sin \alpha + L \dot{\alpha} \cos \alpha \quad (275)$$

$$x_R = x + (w - d) \sin \alpha - L \cos \alpha \implies V'_y = \dot{x} + (w - d) \dot{\alpha} \cos \alpha + L \dot{\alpha} \sin \alpha \quad (276)$$

$$y_R = y + (w - d) \cos \alpha + L \sin \alpha \implies V'_x = \dot{y} - (w - d) \dot{\alpha} \sin \alpha + L \dot{\alpha} \cos \alpha \quad (277)$$

15.3 Kinetic and Potential Energy

Summing the kinetic energies of the platform, both cylinders, and their axle rotations:

$$\begin{aligned} KE = & \frac{1}{2}(M + 2m) (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} \left[I_M + \frac{1}{4} M w^2 + m (d^2 + (w - d)^2 + 2L^2) \right] \dot{\alpha}^2 \\ & + \dot{\alpha} \left[\dot{x} \left(\frac{1}{2}(M + 2m)w \cos \alpha + 2mL \sin \alpha \right) + \dot{y} \left(-\frac{1}{2}(M + 2m)w \sin \alpha + 2mL \cos \alpha \right) \right] \\ & + \frac{1}{2} I_m (\dot{\phi}_1^2 + \dot{\phi}_2^2) \end{aligned} \quad (278)$$

Taking the top pulley line as the reference datum for gravity:

$$\begin{aligned} PE = & -Mg \left(x + \frac{w}{2} \sin \alpha \right) - mg (x + d \sin \alpha - L \cos \alpha) - mg (x + (w - d) \sin \alpha - L \cos \alpha) \\ = & -(M + 2m)gx - \frac{1}{2}(M + 2m)gw \sin \alpha + 2mgL \cos \alpha \end{aligned} \quad (279)$$

15.4 Case 1: Unconstrained System Formulation

Treating the system without the kinematic cable constraint, the standard Euler–Lagrange equations yield:

Vertical Coordinate (x) :

$$\begin{aligned} (M + 2m)\ddot{x} + \left(\frac{1}{2}(M + 2m)w \cos \alpha + 2mL \sin \alpha \right) \ddot{\alpha} \\ + \left(-\frac{1}{2}(M + 2m)w \sin \alpha + 2mL \cos \alpha \right) \dot{\alpha}^2 - (M + 2m)g = 0 \end{aligned} \quad (280)$$

Horizontal Coordinate (y) :

$$\begin{aligned} (M + 2m)\ddot{y} + \left(-\frac{1}{2}(M + 2m)w \sin \alpha + 2mL \cos \alpha \right) \ddot{\alpha} \\ - \left(\frac{1}{2}(M + 2m)w \cos \alpha + 2mL \sin \alpha \right) \dot{\alpha}^2 = 0 \end{aligned} \quad (281)$$

Platform Tilt Angle (α):

$$I_1 \ddot{\alpha} + \left(\frac{1}{2}(M + 2m)w \cos \alpha + 2mL \sin \alpha \right) \ddot{x} + \left(-\frac{1}{2}(M + 2m)w \sin \alpha + 2mL \cos \alpha \right) \ddot{y} - g \left(\frac{1}{2}(M + 2m)w \cos \alpha + 2mL \sin \alpha \right) = 0 \quad (282)$$

Unconstrained Equations of Motion: Direct algebraic substitution yields:

$$\ddot{\alpha} = 0 \quad (283)$$

$$\ddot{x} = g + \left(\frac{w}{2} \sin \alpha - \frac{2mL}{M + 2m} \cos \alpha \right) \dot{\alpha}^2 \quad (284)$$

$$\ddot{y} = \left(\frac{w}{2} \cos \alpha + \frac{2mL}{M + 2m} \sin \alpha \right) \dot{\alpha}^2 \quad (285)$$

$$\ddot{\phi}_1 = \ddot{\phi}_2 = 0 \quad (286)$$

15.5 Case 2: Constrained System via Lagrange Multipliers

The total cable length is decomposed into five segments:

$$L_{\text{string}} = s_1 + s_2 + s_3 + s_4 + s_5 \quad (287)$$

where:

$$s_1 = \sqrt{x^2 + y^2} \quad (288)$$

$$s_2 = \sqrt{x_L^2 + y_L^2} = \sqrt{(x + d \sin \alpha - L \cos \alpha)^2 + (y + d \cos \alpha + L \sin \alpha)^2} \quad (289)$$

$$s_3 = w - 2d \quad (290)$$

$$s_4 = \sqrt{x_R^2 + (D - y_R)^2} = \sqrt{(x + (w - d) \sin \alpha - L \cos \alpha)^2 + (D - y - (w - d) \cos \alpha - L \sin \alpha)^2} \quad (291)$$

$$s_5 = \sqrt{(x + w \sin \alpha)^2 + (D - y - w \cos \alpha)^2} \quad (292)$$

Constraint Gradient Components:

$$\frac{\partial L_{\text{string}}}{\partial x} = \frac{x}{s_1} + \frac{x_L}{s_2} + \frac{x_R}{s_4} + \frac{x + w \sin \alpha}{s_5} \quad (293)$$

$$\frac{\partial L_{\text{string}}}{\partial y} = \frac{y}{s_1} + \frac{y_L}{s_2} - \frac{D - y_R}{s_4} - \frac{D - y - w \cos \alpha}{s_5} \quad (294)$$

$$\begin{aligned} \frac{\partial L_{\text{string}}}{\partial \alpha} &= \frac{y_L(-d \sin \alpha + L \cos \alpha) + x_L(d \cos \alpha + L \sin \alpha)}{s_2} \\ &+ \frac{-(D - y_R)(-(w - d) \sin \alpha + L \cos \alpha) + x_R((w - d) \cos \alpha + L \sin \alpha)}{s_4} \\ &+ \frac{-(D - y - w \cos \alpha)(-w \sin \alpha) + (x + w \sin \alpha)(w \cos \alpha)}{s_5} \end{aligned} \quad (295)$$

15.6 Modified Equations of Motion

With tension multiplier λ , the constrained equations are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = \lambda \frac{\partial L_{\text{string}}}{\partial x}, \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}} \right) - \frac{\partial L}{\partial y} = \lambda \frac{\partial L_{\text{string}}}{\partial y}, \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\alpha}} \right) - \frac{\partial L}{\partial \alpha} = \lambda \frac{\partial L_{\text{string}}}{\partial \alpha} \quad (296)$$

Defining:

$$M_{\text{tot}} = M + 2m \quad (297)$$

$$F_{x\alpha} = \frac{1}{2}M_{\text{tot}}w \cos \alpha + 2mL \sin \alpha \quad (298)$$

$$F_{y\alpha} = -\frac{1}{2}M_{\text{tot}}w \sin \alpha + 2mL \cos \alpha \quad (299)$$

$$\bar{I}_1 = I_m + \frac{1}{4}Mw^2 + m(d^2 + (w-d)^2 + 2L^2) - \frac{F_{x\alpha}^2 + F_{y\alpha}^2}{M_{\text{tot}}} \quad (300)$$

Angular Acceleration of Platform ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{\lambda}{\bar{I}_1} \left(\frac{\partial L_{\text{string}}}{\partial \alpha} - \frac{F_{x\alpha}}{M_{\text{tot}}} \frac{\partial L_{\text{string}}}{\partial x} - \frac{F_{y\alpha}}{M_{\text{tot}}} \frac{\partial L_{\text{string}}}{\partial y} \right) \quad (301)$$

Vertical Acceleration of Platform ($\ddot{x}$):

$$\ddot{x} = g + \left(\frac{w}{2} \sin \alpha - \frac{2mL}{M_{\text{tot}}} \cos \alpha \right) \dot{\alpha}^2 - \frac{F_{x\alpha}}{M_{\text{tot}}} \ddot{\alpha} + \frac{\lambda}{M_{\text{tot}}} \frac{\partial L_{\text{string}}}{\partial x} \quad (302)$$

Horizontal Acceleration of Platform ($\ddot{y}$):

$$\ddot{y} = \left(\frac{w}{2} \cos \alpha + \frac{2mL}{M_{\text{tot}}} \sin \alpha \right) \dot{\alpha}^2 - \frac{F_{y\alpha}}{M_{\text{tot}}} \ddot{\alpha} + \frac{\lambda}{M_{\text{tot}}} \frac{\partial L_{\text{string}}}{\partial y} \quad (303)$$

16 Problem 16: Seven-Body Suspended Multi-Link Viscoelastic Network under Harmonic Forcing

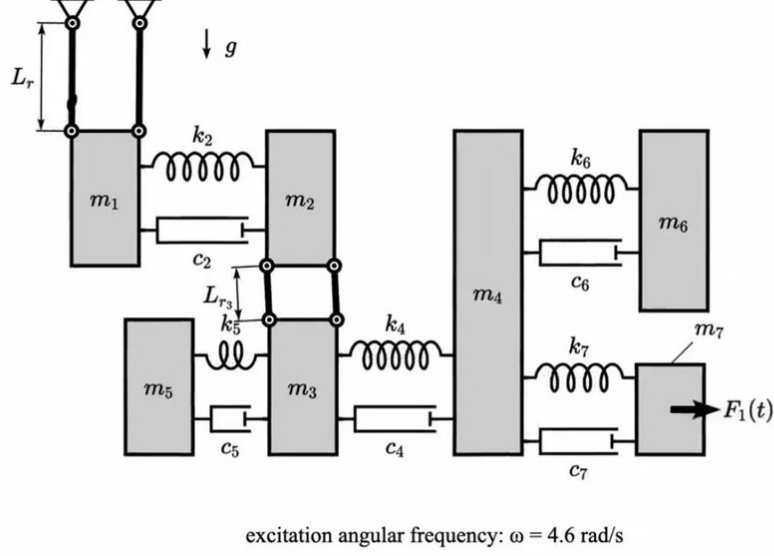


Figure 16: Seven-body mechanical network featuring dual parallel-link suspensions, viscoelastic spring-damper couplings, and dynamic external harmonic forcing $F_1(t)$.

16.1 Generalized Coordinates and System Parameters

The planar system consists of 7 degrees of freedom:

- $\theta(t)$: Angular deflection of the upper parallel suspension links of length L_1 (L_r) from the downward vertical.
- $\phi(t)$: Angular deflection of the intermediate parallel suspension links of length L_2 (L_{r3}) from the downward vertical.
- $k'_1(t)$: Relative horizontal extension between body m_1 and body m_2 .
- $k'_2(t)$: Relative horizontal extension between body m_6 (or m_5 in diagram) and body m_3 .
- $k'_3(t)$: Relative horizontal extension between body m_3 and body m_4 .
- $k'_4(t)$: Relative horizontal extension between body m_4 and body m_5 (or m_6 in diagram).
- $k'_5(t)$: Relative horizontal extension between body m_4 and body m_7 .
- $m_1, m_2, m_3, m_4, m_5, m_6, m_7$: Masses of the seven translating bodies.
- k_j, c_j, k'_{j0} : Stiffness, viscous damping coefficient, and unstretched length for each spring-dashpot stage ($j = 1, \dots, 5$).
- $F_1(t)$: External horizontal dynamic force applied directly to body m_7 with excitation angular frequency $\omega = 4.6 \text{ rad/s}$.

16.2 Kinematic Chain and Viscoelastic Internal Forces

Define the net viscoelastic forces for each spring-damper pair:

$$F'_j = k_j(k'_j - k'_{j0}) + c_j \dot{k}'_j \quad (j = 1, 2, 3, 4, 5) \quad (304)$$

The horizontal kinematic chain accelerations propagating through the stages are:

$$\ddot{x}_1 = L_1 \ddot{\theta} \cos \theta - L_1 \dot{\theta}^2 \sin \theta \quad (305)$$

$$\ddot{x}_2 = \ddot{x}_1 + \ddot{k}'_1 \quad (306)$$

$$\ddot{x}_3 = \ddot{x}_2 + L_2 \ddot{\phi} \cos \phi - L_2 \dot{\phi}^2 \sin \phi \quad (307)$$

$$\ddot{x}_4 = \ddot{x}_3 + \ddot{k}'_3 \quad (308)$$

Let the total mass and lower sub-assembly mass aggregates be defined as:

$$M_{\text{tot}} = \sum_{i=1}^7 m_i = m_1 + m_2 + m_3 + m_4 + m_5 + m_6 + m_7 \quad (309)$$

$$M_x = \sum_{i=3}^7 m_i = m_3 + m_4 + m_5 + m_6 + m_7 \quad (310)$$

16.3 Kinetic Energy, Potential Energy, and Dissipation

The Cartesian velocities of each mass are:

$$\mathbf{v}_1 = \begin{pmatrix} L_1 \dot{\theta} \cos \theta \\ -L_1 \dot{\theta} \sin \theta \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} L_1 \dot{\theta} \cos \theta + \dot{k}'_1 \\ -L_1 \dot{\theta} \sin \theta \end{pmatrix} \quad (311)$$

$$\mathbf{v}_3 = \begin{pmatrix} v_{3x} \\ -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \end{pmatrix} = \begin{pmatrix} L_1 \dot{\theta} \cos \theta + \dot{k}'_1 + L_2 \dot{\phi} \cos \phi \\ -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \end{pmatrix} \quad (312)$$

$$\mathbf{v}_6 = \begin{pmatrix} v_{3x} - \dot{k}'_2 \\ -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} v_{3x} + \dot{k}'_3 \\ -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \end{pmatrix} \quad (313)$$

$$\mathbf{v}_5 = \begin{pmatrix} v_{3x} + \dot{k}'_3 + \dot{k}'_4 \\ -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \end{pmatrix}, \quad \mathbf{v}_7 = \begin{pmatrix} v_{3x} + \dot{k}'_3 + \dot{k}'_5 \\ -L_1 \dot{\theta} \sin \theta - L_2 \dot{\phi} \sin \phi \end{pmatrix} \quad (314)$$

Summing the kinetic energies:

$$\begin{aligned} KE = & \frac{1}{2} m_1 L_1^2 \dot{\theta}^2 + \frac{1}{2} m_2 \left[L_1^2 \dot{\theta}^2 + 2 L_1 \dot{k}'_1 \dot{\theta} \cos \theta + (\dot{k}'_1)^2 \right] \\ & + \frac{1}{2} M_x \left[L_1^2 \dot{\theta}^2 \sin^2 \theta + L_2^2 \dot{\phi}^2 \sin^2 \phi + 2 L_1 L_2 \dot{\theta} \dot{\phi} \sin \theta \sin \phi \right] \\ & + \frac{1}{2} m_3 v_{3x}^2 + \frac{1}{2} m_6 (v_{3x} - \dot{k}'_2)^2 + \frac{1}{2} m_4 (v_{3x} + \dot{k}'_3)^2 \\ & + \frac{1}{2} m_5 (v_{3x} + \dot{k}'_3 + \dot{k}'_4)^2 + \frac{1}{2} m_7 (v_{3x} + \dot{k}'_3 + \dot{k}'_5)^2 \end{aligned} \quad (315)$$

The total gravitational and elastic potential energy is:

$$PE = -M_{\text{tot}} g L_1 \cos \theta - M_x g L_2 \cos \phi + \frac{1}{2} \sum_{j=1}^5 k_j (k'_j - k'_{j0})^2 \quad (316)$$

The Rayleigh dissipation function for the linear dampers is:

$$\mathcal{D} = \frac{1}{2} \sum_{j=1}^5 c_j (\dot{k}'_j)^2 \quad (317)$$

16.4 Euler–Lagrange Equations with Dissipation and External Work

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} + \frac{\partial \mathcal{D}}{\partial \dot{q}_j} = Q_j$ with generalized force components $Q_j = F_1(t) \frac{\partial x_T}{\partial q_j}$:

For coordinate θ :

$$\begin{aligned} M_{\text{tot}} L_1^2 \ddot{\theta} + M_x L_1 L_2 \left[\ddot{\phi} \cos(\theta - \phi) + \dot{\phi}^2 \sin(\theta - \phi) \right] \\ + L_1 \cos \theta \left[(m_2 + M_x) \ddot{k}'_1 - m_6 \ddot{k}'_2 + (m_4 + m_5 + m_7) \ddot{k}'_3 + m_5 \ddot{k}'_4 + m_7 \ddot{k}'_5 \right] \\ + M_{\text{tot}} g L_1 \sin \theta - F_1(t) L_1 \cos \theta = 0 \end{aligned} \quad (318)$$

For coordinate ϕ :

$$\begin{aligned} M_x L_2^2 \ddot{\phi} + M_x L_1 L_2 \left[\ddot{\theta} \cos(\theta - \phi) - \dot{\theta}^2 \sin(\theta - \phi) \right] \\ + L_2 \cos \phi \left[M_x \ddot{k}'_1 - m_6 \ddot{k}'_2 + (m_4 + m_5 + m_7) \ddot{k}'_3 + m_5 \ddot{k}'_4 + m_7 \ddot{k}'_5 \right] \\ + M_x g L_2 \sin \phi - F_1(t) L_2 \cos \phi = 0 \end{aligned} \quad (319)$$

For relative coordinate k'_1 :

$$\begin{aligned} (m_2 + M_x) \ddot{k}'_1 + (m_2 + M_x) \ddot{x}_1 + M_x L_2 \left(\ddot{\phi} \cos \phi - \dot{\phi}^2 \sin \phi \right) - m_6 \ddot{k}'_2 \\ + (m_4 + m_5 + m_7) \ddot{k}'_3 + m_5 \ddot{k}'_4 + m_7 \ddot{k}'_5 + c_1 \dot{k}'_1 + k_1 (k'_1 - k'_{10}) - F_1(t) = 0 \end{aligned} \quad (320)$$

For relative coordinate k'_2 :

$$m_6 \ddot{k}'_2 - m_6 \ddot{x}_3 + c_2 \dot{k}'_2 + k_2 (k'_2 - k'_{20}) = 0 \quad (321)$$

For relative coordinate k'_3 :

$$(m_4 + m_5 + m_7) \ddot{k}'_3 + (m_4 + m_5 + m_7) \ddot{x}_3 + m_5 \ddot{k}'_4 + m_7 \ddot{k}'_5 + c_3 \dot{k}'_3 + k_3 (k'_3 - k'_{30}) - F_1(t) = 0 \quad (322)$$

For relative coordinate k'_4 :

$$m_5 \ddot{k}'_4 + m_5 (\ddot{x}_3 + \ddot{k}'_3) + c_4 \dot{k}'_4 + k_4 (k'_4 - k'_{40}) = 0 \quad (323)$$

For relative coordinate k'_5 :

$$m_7 \ddot{k}'_5 + m_7 (\ddot{x}_3 + \ddot{k}'_3) + c_5 \dot{k}'_5 + k_5 (k'_5 - k'_{50}) - F_1(t) = 0 \quad (324)$$

16.5 Explicit Equations of Motion

Isolating the second-order accelerations for each degree of freedom:

Top Link Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{F_1(t) \cos \theta - M_{\text{tot}} g \sin \theta - M_x L_2 \left[\ddot{\phi} \cos(\theta - \phi) + \dot{\phi}^2 \sin(\theta - \phi) \right] - \cos \theta \left[(m_2 + M_x) \ddot{k}'_1 - m_6 \ddot{k}'_2 + (m_4 + m_5 + m_7) \ddot{k}'_3 + m_5 \ddot{k}'_4 + m_7 \ddot{k}'_5 \right]}{M_{\text{tot}} L_1} \quad (325)$$

Intermediate Link Angular Acceleration ($\ddot{\phi}$):

$$\ddot{\phi} = \frac{F_1(t) \cos \phi - M_x g \sin \phi - M_x L_1 \left[\ddot{\theta} \cos(\theta - \phi) - \dot{\theta}^2 \sin(\theta - \phi) \right] - \cos \phi \left[M_x \ddot{k}'_1 - m_6 \ddot{k}'_2 + (m_4 + m_5 + m_7) \ddot{k}'_3 + m_5 \ddot{k}'_4 + m_7 \ddot{k}'_5 \right]}{M_x L_2} \quad (326)$$

Spring 1 Extension Acceleration ($\ddot{k}'_1$):

$$\ddot{k}'_1 = -\ddot{x}_1 + \frac{F'_3 - F'_2 - F'_1 - M_x L_2 \left(\ddot{\phi} \cos \phi - \dot{\phi}^2 \sin \phi \right)}{m_2} \quad (327)$$

Spring 2 Extension Acceleration ($\ddot{k}'_2$):

$$\ddot{k}'_2 = \ddot{x}_3 - \frac{F'_2}{m_6} \quad (328)$$

Spring 3 Extension Acceleration ($\ddot{k}'_3$):

$$\ddot{k}'_3 = -\ddot{x}_3 + \frac{F'_4 + F'_5 - F'_3}{m_4} \quad (329)$$

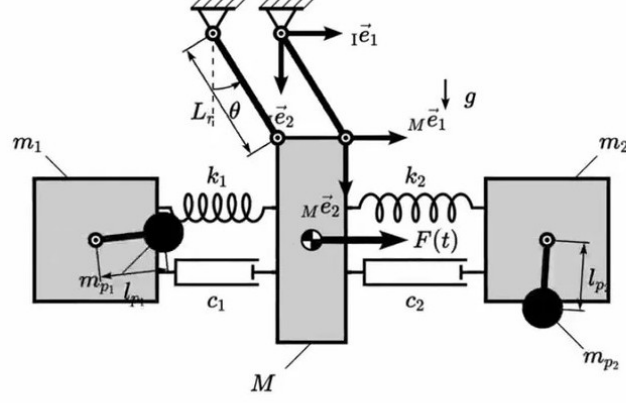
Spring 4 Extension Acceleration ($\ddot{k}'_4$):

$$\ddot{k}'_4 = -\ddot{x}_4 - \frac{F'_4}{m_5} \quad (330)$$

Spring 5 Extension Acceleration ($\ddot{k}'_5$):

$$\ddot{k}'_5 = -\ddot{x}_4 + \frac{F_1(t) - F'_5}{m_7} \quad (331)$$

17 Problem 17: Parallel-Suspended Central Block with Flanking Viscoelastic Blocks and Dual Pendulums



excitation angular frequency: $\omega = 1.4 \text{ rad/s}$

Figure 17: A central block of mass M suspended by parallel links of length L , coupled to flanking blocks M_1 and M_2 via viscoelastic spring-damper pairs (k_1, c_1) and (k_2, c_2) , each carrying a simple pendulum of length l_1, l_2 and bob mass m_1, m_2 .

17.1 Generalized Coordinates and Parameters

The planar system has five degrees of freedom:

- $\theta(t)$: Angular deflection of the central block parallel suspension links of length L from the downward vertical.
- $k'_1(t)$: Relative horizontal extension between the left block M_1 and the central block M .
- $k'_2(t)$: Relative horizontal extension between the right block M_2 and the central block M .
- $\phi_1(t)$: Angular deflection of the left pendulum of length l_1 and bob mass m_1 from the downward vertical.
- $\phi_2(t)$: Angular deflection of the right pendulum of length l_2 and bob mass m_2 from the downward vertical.
- M, M_1, M_2 : Masses of the central suspended block, left slider block, and right slider block, respectively.
- m_1, m_2 : Masses of the left and right pendulum bobs.
- $k_1, k_2, k'_{10}, k'_{20}$: Stiffness values and unstretched lengths of the left and right coupling springs.
- c_1, c_2 : Viscous damping coefficients of the left and right dashpots.
- $M_{\text{tot}} = M + M_1 + M_2 + m_1 + m_2$: Total mass of the system.
- $M'_1 = M_1 + m_1$: Total mass of the left sub-assembly.
- $M'_2 = M_2 + m_2$: Total mass of the right sub-assembly.

17.2 Kinematics

Taking the fixed suspension pivot as the origin, the Cartesian velocities of the three rigid blocks are:

$$\text{Left Block } M_1 : \quad v_{1x} = L\dot{\theta} \cos \theta - \dot{k}'_1, \quad v_{1y} = -L\dot{\theta} \sin \theta \quad (332)$$

$$\text{Central Block } M : \quad v_{2x} = L\dot{\theta} \cos \theta, \quad v_{2y} = -L\dot{\theta} \sin \theta \quad (333)$$

$$\text{Right Block } M_2 : \quad v_{3x} = L\dot{\theta} \cos \theta + \dot{k}'_2, \quad v_{3y} = -L\dot{\theta} \sin \theta \quad (334)$$

The Cartesian velocities of the two suspended pendulum bobs are:

$$\text{Left Bob } m_1 : \quad v_{p1x} = L\dot{\theta} \cos \theta - \dot{k}'_1 + l_1\dot{\phi}_1 \cos \phi_1, \quad v_{p1y} = -L\dot{\theta} \sin \theta - l_1\dot{\phi}_1 \sin \phi_1 \quad (335)$$

$$\text{Right Bob } m_2 : \quad v_{p2x} = L\dot{\theta} \cos \theta + \dot{k}'_2 + l_2\dot{\phi}_2 \cos \phi_2, \quad v_{p2y} = -L\dot{\theta} \sin \theta - l_2\dot{\phi}_2 \sin \phi_2 \quad (336)$$

17.3 Kinetic Energy, Potential Energy, and Dissipation

Summing the kinetic energies of all five bodies:

$$\begin{aligned} KE = & \frac{1}{2}M_{\text{tot}}L^2\dot{\theta}^2 + \frac{1}{2}M'_1(\dot{k}'_1)^2 + \frac{1}{2}M'_2(\dot{k}'_2)^2 + \frac{1}{2}m_1l_1^2\dot{\phi}_1^2 + \frac{1}{2}m_2l_2^2\dot{\phi}_2^2 \\ & - M'_1L\dot{k}'_1\dot{\theta} \cos \theta + M'_2L\dot{k}'_2\dot{\theta} \cos \theta + m_1Ll_1\dot{\theta}\dot{\phi}_1 \cos(\theta - \phi_1) + m_2Ll_2\dot{\theta}\dot{\phi}_2 \cos(\theta - \phi_2) \\ & - m_1l_1\dot{k}'_1\dot{\phi}_1 \cos \phi_1 + m_2l_2\dot{k}'_2\dot{\phi}_2 \cos \phi_2 \end{aligned} \quad (337)$$

Taking the suspension pivot line as the reference datum for gravity:

$$PE = -M_{\text{tot}}gL \cos \theta - m_1gl_1 \cos \phi_1 - m_2gl_2 \cos \phi_2 + \frac{1}{2}k_1(k'_1 - k'_{10})^2 + \frac{1}{2}k_2(k'_2 - k'_{20})^2 \quad (338)$$

The Rayleigh dissipation function for the two viscous dashpots is:

$$\mathcal{D} = \frac{1}{2}c_1(\dot{k}'_1)^2 + \frac{1}{2}c_2(\dot{k}'_2)^2 \quad (339)$$

17.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \frac{\partial \mathcal{D}}{\partial \dot{q}_i} = 0$:

For suspension link angle θ :

$$\begin{aligned} M_{\text{tot}}L^2\ddot{\theta} - M'_1L\ddot{k}'_1 \cos \theta + M'_2L\ddot{k}'_2 \cos \theta + m_1Ll_1 \left[\ddot{\phi}_1 \cos(\theta - \phi_1) + \dot{\phi}_1^2 \sin(\theta - \phi_1) \right] \\ + m_2Ll_2 \left[\ddot{\phi}_2 \cos(\theta - \phi_2) + \dot{\phi}_2^2 \sin(\theta - \phi_2) \right] + M_{\text{tot}}gL \sin \theta = 0 \end{aligned} \quad (340)$$

For left spring coordinate k'_1 :

$$M'_1\ddot{k}'_1 - M'_1L\ddot{\theta} \cos \theta + M'_1L\dot{\theta}^2 \sin \theta - m_1l_1\ddot{\phi}_1 \cos \phi_1 + m_1l_1\dot{\phi}_1^2 \sin \phi_1 + c_1\dot{k}'_1 + k_1(k'_1 - k'_{10}) = 0 \quad (341)$$

For right spring coordinate k'_2 :

$$M'_2\ddot{k}'_2 + M'_2L\ddot{\theta} \cos \theta - M'_2L\dot{\theta}^2 \sin \theta + m_2l_2\ddot{\phi}_2 \cos \phi_2 - m_2l_2\dot{\phi}_2^2 \sin \phi_2 + c_2\dot{k}'_2 + k_2(k'_2 - k'_{20}) = 0 \quad (342)$$

For left pendulum angle ϕ_1 :

$$l_1 \ddot{\phi}_1 + L \ddot{\theta} \cos(\theta - \phi_1) - \ddot{k}'_1 \cos \phi_1 - L \dot{\theta}^2 \sin(\theta - \phi_1) + g \sin \phi_1 = 0 \quad (343)$$

For right pendulum angle ϕ_2 :

$$l_2 \ddot{\phi}_2 + L \ddot{\theta} \cos(\theta - \phi_2) + \ddot{k}'_2 \cos \phi_2 - L \dot{\theta}^2 \sin(\theta - \phi_2) + g \sin \phi_2 = 0 \quad (344)$$

17.5 Explicit Equations of Motion

Isolating the second-order accelerations for each degree of freedom:

Suspension Link Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{M'_1 \ddot{k}'_1 \cos \theta - M'_2 \ddot{k}'_2 \cos \theta - m_1 l_1 \left[\ddot{\phi}_1 \cos(\theta - \phi_1) + \dot{\phi}_1^2 \sin(\theta - \phi_1) \right] - m_2 l_2 \left[\ddot{\phi}_2 \cos(\theta - \phi_2) + \dot{\phi}_2^2 \sin(\theta - \phi_2) \right] - M_{\text{tot}} g \sin \theta}{M_{\text{tot}} L} \quad (345)$$

Left Spring Acceleration ($\ddot{k}'_1$):

$$\ddot{k}'_1 = L \ddot{\theta} \cos \theta - L \dot{\theta}^2 \sin \theta + \frac{m_1 l_1}{M'_1} \left(\ddot{\phi}_1 \cos \phi_1 - \dot{\phi}_1^2 \sin \phi_1 \right) - \frac{c_1 \dot{k}'_1}{M'_1} - \frac{k_1(k'_1 - k'_{10})}{M'_1} \quad (346)$$

Right Spring Acceleration ($\ddot{k}'_2$):

$$\ddot{k}'_2 = -L \ddot{\theta} \cos \theta + L \dot{\theta}^2 \sin \theta - \frac{m_2 l_2}{M'_2} \left(\ddot{\phi}_2 \cos \phi_2 - \dot{\phi}_2^2 \sin \phi_2 \right) - \frac{c_2 \dot{k}'_2}{M'_2} - \frac{k_2(k'_2 - k'_{20})}{M'_2} \quad (347)$$

Left Pendulum Angular Acceleration ($\ddot{\phi}_1$):

$$\ddot{\phi}_1 = \frac{\ddot{k}'_1 \cos \phi_1 - L \ddot{\theta} \cos(\theta - \phi_1) + L \dot{\theta}^2 \sin(\theta - \phi_1) - g \sin \phi_1}{l_1} \quad (348)$$

Right Pendulum Angular Acceleration ($\ddot{\phi}_2$):

$$\ddot{\phi}_2 = \frac{-\ddot{k}'_2 \cos \phi_2 - L \ddot{\theta} \cos(\theta - \phi_2) + L \dot{\theta}^2 \sin(\theta - \phi_2) - g \sin \phi_2}{l_2} \quad (349)$$

18 Problem 18: Parabolic Track with Nested Cylinder and Link-Coupled External Disc

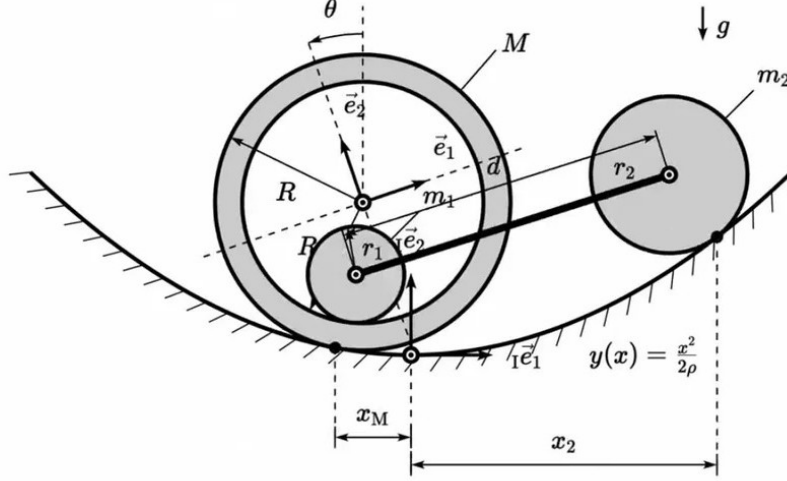


Figure 18: A hollow cylinder of mass M with an internal rolling disc of mass m_1 on a parabolic track $y(x) = \frac{x^2}{2\rho}$, coupled via a rigid link of length d to an external rolling disc of mass m_2 .

18.1 Generalized Coordinates and Parameters

The constrained planar multi-body system is described by three coordinates subject to one holonomic constraint:

- $x(t)$: Horizontal substrate contact coordinate of the large hollow cylinder.
- $\phi(t)$: Relative angular position of the inner cylinder inside the hollow cavity.
- $x'(t)$: Horizontal substrate contact coordinate of the external rolling cylinder (m_2).
- M, I_M : Mass and central moment of inertia of the hollow cylinder (R_0 : outer radius, R_i : inner cavity radius).
- m_1, I_1, r : Mass, moment of inertia, and radius of the internal rolling cylinder.
- m_2, I_2, R' : Mass, moment of inertia, and radius of the external rolling cylinder.
- ρ : Characteristic focal parameter of the parabolic substrate $y(x) = \frac{x^2}{2\rho}$.
- d : Fixed length of the rigid massless link connecting the center of m_1 to the center of m_2 .
- λ : Lagrange multiplier corresponding to the rigid link constraint force.

18.2 Kinematics and Constraint Geometry

The local slope angles $\beta(x)$ and $\gamma(x')$ along the parabolic track $y = \frac{x^2}{2\rho}$ satisfy:

$$\tan \beta = \frac{x}{\rho} \implies \sin \beta = \frac{x}{\sqrt{x^2 + \rho^2}}, \quad \cos \beta = \frac{\rho}{\sqrt{x^2 + \rho^2}} \quad (350)$$

$$\tan \gamma = \frac{x'}{\rho} \implies \sin \gamma = \frac{x'}{\sqrt{x'^2 + \rho^2}}, \quad \cos \gamma = \frac{\rho}{\sqrt{x'^2 + \rho^2}} \quad (351)$$

Defining the geometric scale factors:

$$F(x) = 1 - \frac{R_0 \rho^2}{(x^2 + \rho^2)^{3/2}}, \quad G(x') = 1 - \frac{R' \rho^2}{(x'^2 + \rho^2)^{3/2}} \quad (352)$$

The pure rolling conditions along the parabolic arc length relate the spin angles to the contact translations:

$$\dot{\theta} = \frac{\sqrt{x^2 + \rho^2}}{\rho R_0} \dot{x} \implies \ddot{\theta} = \frac{\sqrt{x^2 + \rho^2}}{\rho R_0} \ddot{x} + \frac{x \dot{x}^2}{\rho R_0 \sqrt{x^2 + \rho^2}} \quad (353)$$

$$\dot{\alpha} = \frac{\sqrt{x'^2 + \rho^2}}{\rho R'} \dot{x}' \implies \ddot{\alpha} = \frac{\sqrt{x'^2 + \rho^2}}{\rho R'} \ddot{x}' + \frac{x' \dot{x}'^2}{\rho R' \sqrt{x'^2 + \rho^2}} \quad (354)$$

The spin angular velocity of the inner cylinder rolling without slipping inside the cavity is:

$$\omega_{\text{spin}} = \frac{R_i}{r} \dot{\theta} - \left(\frac{R_i - r}{r} \right) \dot{\phi} \quad (355)$$

The Cartesian coordinates of the inner cylinder center (x_1, y_1) and external disc center (x_2, y_2) are:

$$x_1 = x - \frac{R_0 x}{\sqrt{x^2 + \rho^2}} - (R_i - r) \sin(\theta + \phi), \quad y_1 = \frac{x^2}{2\rho} + \frac{R_0 \rho}{\sqrt{x^2 + \rho^2}} - (R_i - r) \cos(\theta + \phi) \quad (356)$$

$$x_2 = x' - \frac{R' x'}{\sqrt{x'^2 + \rho^2}}, \quad y_2 = \frac{x'^2}{2\rho} + \frac{R' \rho}{\sqrt{x'^2 + \rho^2}} \quad (357)$$

The holonomic constraint enforcing constant link length d is:

$$\Phi(x', \phi, x) = (x_2 - x_1)^2 + (y_2 - y_1)^2 - d^2 = 0 \quad (358)$$

Defining $\Delta x = x_2 - x_1$ and $\Delta y = y_2 - y_1$, the constraint gradients are:

$$\frac{\partial \Phi}{\partial x'} = 2G(x') \left(\Delta x + \frac{x'}{\rho} \Delta y \right) \quad (359)$$

$$\frac{\partial \Phi}{\partial \phi} = 2(R_i - r) [\Delta x \cos(\theta + \phi) - \Delta y \sin(\theta + \phi)] \quad (360)$$

$$\frac{\partial \Phi}{\partial x} = -2F(x) \left(\Delta x + \frac{x}{\rho} \Delta y \right) + \frac{2(R_i - r) \sqrt{x^2 + \rho^2}}{\rho R_0} [\Delta x \cos(\theta + \phi) - \Delta y \sin(\theta + \phi)] \quad (361)$$

18.3 Kinetic and Potential Energy Formulations

The system kinetic energy is the sum of translational and rotational energies:

$$\begin{aligned}
KE = & \frac{1}{2}M \left(\frac{x^2 + \rho^2}{\rho^2} \right) [F(x)]^2 \dot{x}^2 + \frac{1}{2}I_M \left(\frac{x^2 + \rho^2}{\rho^2 R_0^2} \right) \dot{x}^2 \\
& + \frac{1}{2}m_1 \left\{ \left(\frac{x^2 + \rho^2}{\rho^2} \right) [F(x)]^2 \dot{x}^2 + (R_i - r)^2 (\dot{\theta} + \dot{\phi})^2 \right. \\
& \quad \left. - 2(R_i - r)F(x)\dot{x}(\dot{\theta} + \dot{\phi}) \left[\cos(\theta + \phi) - \frac{x}{\rho} \sin(\theta + \phi) \right] \right\} \\
& + \frac{1}{2}I_1 \left[\frac{R_i \sqrt{x^2 + \rho^2}}{\rho r R_0} \dot{x} - \left(\frac{R_i - r}{r} \right) \dot{\phi} \right]^2 \\
& + \frac{1}{2}m_2 \left(\frac{x'^2 + \rho^2}{\rho^2} \right) [G(x')]^2 \dot{x}'^2 + \frac{1}{2}I_2 \left(\frac{x'^2 + \rho^2}{\rho^2 R'^2} \right) \dot{x}'^2
\end{aligned} \tag{362}$$

The gravitational potential energy relative to the vertex datum is:

$$PE = (M + m_1)g \left(\frac{x^2}{2\rho} + \frac{R_0 \rho}{\sqrt{x^2 + \rho^2}} \right) - m_1 g (R_i - r) \cos(\theta + \phi) + m_2 g \left(\frac{x'^2}{2\rho} + \frac{R' \rho}{\sqrt{x'^2 + \rho^2}} \right) \tag{363}$$

18.4 Euler–Lagrange Equations with Constraint Multipliers

The constrained Lagrange equations are $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \lambda \frac{1}{2} \frac{\partial \Phi}{\partial q_i}$:

For external cylinder coordinate x' :

$$M_{x'x'} \ddot{x}' + \frac{1}{2} \frac{\partial M_{x'x'}}{\partial x'} \dot{x}'^2 + m_2 g G(x') \frac{x'}{\rho} = \lambda G(x') \left(\Delta x + \frac{x'}{\rho} \Delta y \right) \tag{364}$$

where:

$$M_{x'x'}(x') = m_2 \left(\frac{x'^2 + \rho^2}{\rho^2} \right) [G(x')]^2 + \frac{I_2(x'^2 + \rho^2)}{\rho^2 R'^2} \tag{365}$$

$$\frac{\partial M_{x'x'}}{\partial x'} = \frac{2x'}{\rho^2} \left(m_2 [G(x')]^2 + \frac{I_2}{R'^2} \right) + \frac{6m_2 R' x'}{\sqrt{x'^2 + \rho^2}} G(x') \tag{366}$$

For cavity relative angle ϕ :

$$\begin{aligned}
M_{\phi\phi} \ddot{\phi} + M_{\phi x} \ddot{x} - m_1 (R_i - r) F(x) \dot{x}^2 \left[\frac{1}{\rho} \sin(\theta + \phi) + \frac{x}{\rho^2} \cos(\theta + \phi) \right] \\
+ m_1 g (R_i - r) \sin(\theta + \phi) = \lambda (R_i - r) [\Delta x \cos(\theta + \phi) - \Delta y \sin(\theta + \phi)]
\end{aligned} \tag{367}$$

where:

$$M_{\phi\phi} = m_1 (R_i - r)^2 + I_1 \left(\frac{R_i - r}{r} \right)^2 \tag{368}$$

$$M_{\phi x} = m_1 (R_i - r)^2 \frac{\sqrt{x^2 + \rho^2}}{\rho R_0} - I_1 \frac{R_i (R_i - r) \sqrt{x^2 + \rho^2}}{\rho r^2 R_0} - m_1 (R_i - r) F(x) \left[\cos(\theta + \phi) - \frac{x}{\rho} \sin(\theta + \phi) \right] \tag{369}$$

For outer ring substrate coordinate x :

$$M_{xx}\ddot{x} + M_{x\phi}\ddot{\phi} + H_x(x, \phi, \dot{x}, \dot{\phi}) = \lambda \frac{1}{2} \frac{\partial \Phi}{\partial x} \quad (370)$$

where:

$$\begin{aligned} M_{xx} = & (M + m_1) \left(\frac{x^2 + \rho^2}{\rho^2} \right) [F(x)]^2 + \left(I_M + m_1(R_i - r)^2 + I_1 \frac{R_i^2}{r^2} \right) \frac{x^2 + \rho^2}{\rho^2 R_0^2} \\ & - 2m_1(R_i - r)F(x) \frac{\sqrt{x^2 + \rho^2}}{\rho R_0} \left[\cos(\theta + \phi) - \frac{x}{\rho} \sin(\theta + \phi) \right] \end{aligned} \quad (371)$$

and the Coriolis, centrifugal, and gravitational term H_x is:

$$\begin{aligned} H_x = & \frac{1}{2} \frac{\partial M_{xx}}{\partial x} \dot{x}^2 + (M + m_1)gF(x) \frac{x}{\rho} + m_1g(R_i - r) \frac{\sqrt{x^2 + \rho^2}}{\rho R_0} \sin(\theta + \phi) \\ & + m_1(R_i - r)F(x) \dot{\phi}^2 \left[\sin(\theta + \phi) + \frac{x}{\rho} \cos(\theta + \phi) \right] \end{aligned} \quad (372)$$

18.5 Explicit Decoupled Equations of Motion

Isolating the second-order accelerations as explicit functions of the coordinates, velocities, and tension multiplier λ :

External Disc Substrate Acceleration ($\ddot{x}'$):

$$\ddot{x}' = \frac{\lambda G(x') \left(\Delta x + \frac{x'}{\rho} \Delta y \right) - m_2 g G(x') \frac{x'}{\rho} - \frac{1}{2} \frac{\partial M_{x'x'}}{\partial x'} \dot{x}'^2}{M_{x'x'}(x')} \quad (373)$$

Cavity Disc and Outer Ring Accelerations ($\ddot{\phi}, \ddot{x}$): Letting the generalized forces be:

$$\begin{aligned} Q_\phi = & \lambda(R_i - r) [\Delta x \cos(\theta + \phi) - \Delta y \sin(\theta + \phi)] - m_1 g(R_i - r) \sin(\theta + \phi) \\ & + m_1(R_i - r)F(x) \dot{x}^2 \left[\frac{1}{\rho} \sin(\theta + \phi) + \frac{x}{\rho^2} \cos(\theta + \phi) \right] \end{aligned} \quad (374)$$

$$Q_x = \lambda \frac{1}{2} \frac{\partial \Phi}{\partial x} - H_x(x, \phi, \dot{x}, \dot{\phi}) \quad (375)$$

and the determinant of the 2×2 coupled inertia block be:

$$\Delta_M = M_{xx}M_{\phi\phi} - M_{x\phi}^2 \quad (376)$$

The decoupled explicit accelerations are:

$$\ddot{x} = \frac{M_{\phi\phi}Q_x - M_{x\phi}Q_\phi}{\Delta_M} \quad (377)$$

$$\ddot{\phi} = \frac{M_{xx}Q_\phi - M_{x\phi}Q_x}{\Delta_M} \quad (378)$$

Lagrange Multiplier Determination: The algebraic value of λ is uniquely obtained by taking the second time derivative of the constraint $\ddot{\Phi} = 0$:

$$\frac{\partial \Phi}{\partial x'} \ddot{x}' + \frac{\partial \Phi}{\partial \phi} \ddot{\phi} + \frac{\partial \Phi}{\partial x} \ddot{x} + \frac{d}{dt} \left(\frac{\partial \Phi}{\partial x'} \right) \dot{x}' + \frac{d}{dt} \left(\frac{\partial \Phi}{\partial \phi} \right) \dot{\phi} + \frac{d}{dt} \left(\frac{\partial \Phi}{\partial x} \right) \dot{x} = 0 \quad (379)$$

Substituting the explicit expressions for $\ddot{x}'$, $\ddot{\phi}$, and $\ddot{x}$ yields a linear scalar equation for λ .

19 Problem 19: Elastically Supported Guiding Bar with Freely Moving Discs

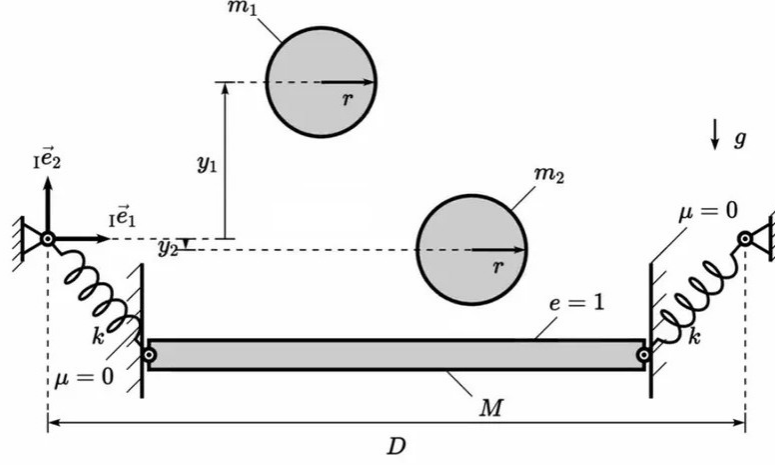


Figure 19: A horizontal bar of mass M translating along frictionless vertical guides, supported by two inclined springs of stiffness k , beneath two falling/bouncing discs of masses m_1 and m_2 .

19.1 Generalized Coordinates and Parameters

The planar system is parameterized by three translational degrees of freedom:

- $x(t)$: Vertical displacement of the horizontal bar of mass M (taken positive downward).
- $y_1(t)$: Relative vertical position of disc 1 of mass m_1 and radius r with respect to the bar.
- $y_2(t)$: Relative vertical position of disc 2 of mass m_2 and radius r with respect to the bar.
- M : Mass of the rigid horizontal guiding bar.
- m_1, m_2 : Masses of the two discs.
- k, L_0 : Stiffness and unstretched length of each of the two symmetrical inclined springs.
- b : Horizontal distance from the vertical guide wall to each fixed spring anchor.
- h : Vertical distance from the bar resting datum to the fixed spring anchors.
- g : Acceleration due to gravity acting downwards.

19.2 Kinematics and Spring Geometry

The absolute vertical downward velocity of the bar is:

$$V = \frac{d(x)}{dt} = \dot{x} \quad (380)$$

The relative vertical velocities of the two discs with respect to the moving bar are:

$$v'_1 = \frac{d(y_1 - x)}{dt} = \dot{y}_1 - \dot{x} \quad (381)$$

$$v'_2 = \frac{d(y_2 - x)}{dt} = \dot{y}_2 - \dot{x} \quad (382)$$

Each of the two symmetrical inclined springs undergoes a geometric stretch. The instantaneous extended length of each spring is:

$$L_s = \sqrt{b^2 + (h + x)^2} \quad (383)$$

19.3 Kinetic and Potential Energy

The total kinetic energy of the system is the sum of the translational energies of the bar and the two discs:

$$\begin{aligned} KE &= \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m_1(\dot{y}_1 - \dot{x})^2 + \frac{1}{2}m_2(\dot{y}_2 - \dot{x})^2 \\ &= \frac{1}{2}(M + m_1 + m_2)\dot{x}^2 - m_1\dot{x}\dot{y}_1 - m_2\dot{x}\dot{y}_2 + \frac{1}{2}m_1\dot{y}_1^2 + \frac{1}{2}m_2\dot{y}_2^2 \end{aligned} \quad (384)$$

Taking the fixed anchor level as the gravitational datum, the total potential energy stored in gravity and the two springs ($2 \times \frac{1}{2}k(L_s - L_0)^2$) is:

$$PE = -(M + m_1 + m_2)gx + m_1gy_1 + m_2gy_2 + k \left(\sqrt{b^2 + (h + x)^2} - L_0 \right)^2 \quad (385)$$

19.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

For relative coordinate y_1 :

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_1} \right) - \frac{\partial L}{\partial y_1} = m_1(\ddot{y}_1 - \ddot{x}) + m_1g = 0 \implies \ddot{y}_1 - \ddot{x} + g = 0 \quad (386)$$

For relative coordinate y_2 :

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_2} \right) - \frac{\partial L}{\partial y_2} = m_2(\ddot{y}_2 - \ddot{x}) + m_2g = 0 \implies \ddot{y}_2 - \ddot{x} + g = 0 \quad (387)$$

For bar coordinate x :

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} &= M\ddot{x} + m_1(\ddot{x} - \ddot{y}_1) + m_2(\ddot{x} - \ddot{y}_2) - (M + m_1 + m_2)g \\ &\quad + 2k \left(1 - \frac{L_0}{\sqrt{b^2 + (h + x)^2}} \right) (h + x) = 0 \end{aligned} \quad (388)$$

19.5 Explicit Equations of Motion

Substituting the relations $\ddot{x} - \ddot{y}_1 = g$ and $\ddot{x} - \ddot{y}_2 = g$ from the first two equations into the third equation directly cancels the disc gravitational reaction terms, decoupling the system:

Bar Vertical Acceleration ($\ddot{x}$):

$$\ddot{x} = g - \frac{2k}{M} \left(1 - \frac{L_0}{\sqrt{b^2 + (h+x)^2}} \right) (h+x) \quad (389)$$

Disc 1 Relative Acceleration ($\ddot{y}_1$):

$$\ddot{y}_1 = \ddot{x} - g = -\frac{2k}{M} \left(1 - \frac{L_0}{\sqrt{b^2 + (h+x)^2}} \right) (h+x) \quad (390)$$

Disc 2 Relative Acceleration ($\ddot{y}_2$):

$$\ddot{y}_2 = \ddot{x} - g = -\frac{2k}{M} \left(1 - \frac{L_0}{\sqrt{b^2 + (h+x)^2}} \right) (h+x) \quad (391)$$

20 Problem 20: Parallel-Suspended E-Frame with Dual Spring-Sliders and Pendulums

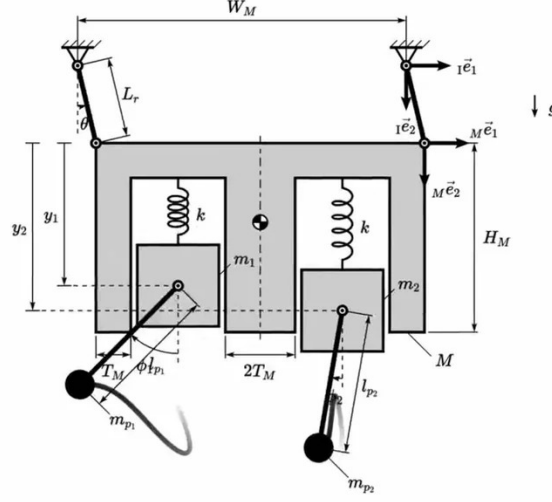


Figure 20: An E-shaped rigid frame of mass M suspended by parallel links of length L , carrying two internal spring-loaded sliders of masses m_1, m_2 on vertical guides, each suspending a simple pendulum of bob mass m_{p1}, m_{p2} and length l_1, l_2 .

20.1 Generalized Coordinates and Parameters

The planar multi-body system possesses five independent degrees of freedom:

- $\theta(t)$: Angular deflection of the parallel suspension links of length L from the downward vertical.
- $k'_1(t), k'_2(t)$: Relative vertical displacements/spring extensions of the left (m_1) and right (m_2) sliders along their respective frame guides.
- $\phi_1(t), \phi_2(t)$: Angular deflections of the left and right pendulums from the downward vertical.
- M : Mass of the outer E-shaped suspended frame.
- m_1, m_2 : Masses of the left and right translating sliders.
- m_{p1}, m_{p2} : Masses of the left and right pendulum bobs.
- l_1, l_2 : Lengths of the left and right pendulum rods.
- k, l_0 : Stiffness and unstretched length of each vertical slider spring.
- $M_{\text{tot}} = M + m_1 + m_2 + m_{p1} + m_{p2}$: Total mass of the system.

20.2 Kinematics

Taking the fixed suspension pivot as the reference origin, the Cartesian velocity components of the outer frame are:

$$V_x = L\dot{\theta} \cos \theta, \quad V_y = -L\dot{\theta} \sin \theta \quad (392)$$

The Cartesian velocity components of the two translating sliders ($i \in \{1, 2\}$) are:

$$v'_{ix} = L\dot{\theta} \cos \theta, \quad v'_{iy} = \dot{k}'_i - L\dot{\theta} \sin \theta \quad (393)$$

The Cartesian velocity components of the two suspended pendulum bobs ($i \in \{1, 2\}$) are:

$$v_{ix} = L\dot{\theta} \cos \theta + l_i \dot{\phi}_i \cos \phi_i \quad (394)$$

$$v_{iy} = \dot{k}'_i - L\dot{\theta} \sin \theta - l_i \dot{\phi}_i \sin \phi_i \quad (395)$$

20.3 Kinetic and Potential Energy

Summing the kinetic energies of the frame, the two sliders, and both pendulum bobs:

$$\begin{aligned} KE = & \frac{1}{2}ML^2\dot{\theta}^2 + \sum_{i=1}^2 \frac{1}{2}m_i \left(L^2\dot{\theta}^2 + (\dot{k}'_i)^2 - 2L\dot{k}'_i\dot{\theta} \sin \theta \right) \\ & + \sum_{i=1}^2 \frac{1}{2}m_{pi} \left[L^2\dot{\theta}^2 + (\dot{k}'_i)^2 + l_i^2\dot{\phi}_i^2 - 2L\dot{k}'_i\dot{\theta} \sin \theta \right. \\ & \quad \left. - 2l_i\dot{k}'_i\dot{\phi}_i \sin \phi_i - 2Ll_i\dot{\theta}\dot{\phi}_i \cos(\theta + \phi_i) \right] \end{aligned} \quad (396)$$

Taking the top suspension pivot level as the potential energy datum:

$$PE = \frac{1}{2}k(k'_1 - l_0)^2 + \frac{1}{2}k(k'_2 - l_0)^2 - g \left[M_{\text{tot}}L \cos \theta + \sum_{i=1}^2 (m_i + m_{pi})k'_i + \sum_{i=1}^2 m_{pi}l_i \cos \phi_i \right] \quad (397)$$

20.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$:

For frame link angle θ :

$$\begin{aligned} M_{\text{tot}}L^2\ddot{\theta} - L \sin \theta \sum_{i=1}^2 (m_i + m_{pi})\ddot{k}'_i \\ - L \sum_{i=1}^2 m_{pi}l_i \left[\ddot{\phi}_i \cos(\theta + \phi_i) - \dot{\phi}_i^2 \sin(\theta + \phi_i) \right] + M_{\text{tot}}gL \sin \theta = 0 \end{aligned} \quad (398)$$

For slider displacements k'_i ($i \in \{1, 2\}$):

$$(m_i + m_{pi})\ddot{k}'_i - (m_i + m_{pi})L \left(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta \right) - m_{pi}l_i \left(\ddot{\phi}_i \sin \phi_i + \dot{\phi}_i^2 \cos \phi_i \right) + k(k'_i - l_0) - (m_i + m_{pi})g = 0 \quad (399)$$

For pendulum angles ϕ_i ($i \in \{1, 2\}$):

$$l_i\ddot{\phi}_i - \ddot{k}'_i \sin \phi_i - L\ddot{\theta} \cos(\theta + \phi_i) + L\dot{\theta}^2 \sin(\theta + \phi_i) + g \sin \phi_i = 0 \quad (400)$$

20.5 Explicit Equations of Motion

Isolating the second-order accelerations for each generalized coordinate:

Frame Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = -\frac{g}{L} \sin \theta + \frac{1}{M_{\text{tot}} L} \sum_{i=1}^2 \left[(m_i + m_{pi}) \ddot{k}_i' \sin \theta + m_{pi} l_i \left(\ddot{\phi}_i \cos(\theta + \phi_i) - \dot{\phi}_i^2 \sin(\theta + \phi_i) \right) \right] \quad (401)$$

Slider Accelerations ($\ddot{k}_i'$ for $i \in \{1, 2\}$):

$$\ddot{k}_i' = g - \frac{k}{m_i + m_{pi}} (k_i' - l_0) + L \left(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta \right) + \frac{m_{pi} l_i}{m_i + m_{pi}} \left(\ddot{\phi}_i \sin \phi_i + \dot{\phi}_i^2 \cos \phi_i \right) \quad (402)$$

Pendulum Angular Accelerations ($\ddot{\phi}_i$ for $i \in \{1, 2\}$):

$$\ddot{\phi}_i = -\frac{g}{l_i} \sin \phi_i + \frac{1}{l_i} \left[\ddot{k}_i' \sin \phi_i + L \ddot{\theta} \cos(\theta + \phi_i) - L \dot{\theta}^2 \sin(\theta + \phi_i) \right] \quad (403)$$

21 Problem 21: Coupled Wheels with Connecting Rod, Spring-Slider, and T-Bar Pendulum

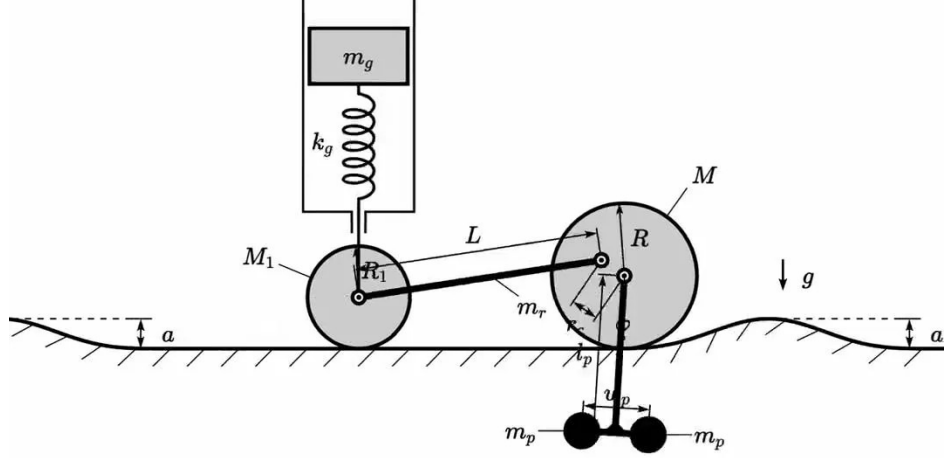


Figure 21: Two pure-rolling wheels linked by an intermediate rod of length L : the left wheel carries a vertically guided spring-mass block m_g , while the right wheel carries an eccentric pin at radius r_c and suspends a T-bar pendulum with two tip masses m_p .

21.1 Coordinate Reduction and Kinematic Loop Closure

The system begins with five degrees of freedom: the wheel rotation angles ϕ and θ , the connecting rod orientation angle α , the T-bar pendulum angle γ , and the spring-block relative extension k'_1 . Enforcing the closed kinematic vector loop of the rigid connecting rod of length L yields two algebraic constraints, eliminating α and ϕ as independent coordinates and reducing the system to three active degrees of freedom: $\{\theta, \gamma, k'_1\}$.

Geometric Loop Relations: Taking the vertical elevation difference $a + R_2 - R_1$ and horizontal separation across the wheel centers:

$$\sin \alpha = \frac{(a + R_2 - R_1) + r_c \cos \theta}{L} \quad (404)$$

$$L \cos \alpha = R_2 \theta - R_1 \phi - r_c \sin \theta \quad (405)$$

Differentiating these geometric constraints with respect to time eliminates $\dot{\alpha}$ and $\dot{\phi}$ in terms of $\dot{\theta}$:

$$\dot{\alpha} = - \left(\frac{r_c \sin \theta}{L \cos \alpha} \right) \dot{\theta} \quad (406)$$

$$\dot{\phi} = \left(\frac{R_2 - r_c(\cos \theta + \tan \alpha \sin \theta)}{R_1} \right) \dot{\theta} \quad (407)$$

21.2 Kinematics

Taking the contact point of the left wheel as reference:

$$\text{Left Wheel Center: } V_x = R_1 \dot{\phi}, \quad V_y = 0 \quad (408)$$

$$\text{Spring Block } m_g : v_x = R_1 \dot{\phi}, \quad v_y = \dot{k}'_1 \quad (409)$$

$$\text{Connecting Rod Center: } v''_x = R_1 \dot{\phi} - \frac{L}{2} \dot{\alpha} \sin \alpha, \quad v''_y = \frac{L}{2} \dot{\alpha} \cos \alpha \quad (410)$$

$$\text{Right Wheel Center: } v'_x = R_2 \dot{\theta}, \quad v'_y = 0 \quad (411)$$

For the two symmetric pendulum bobs of mass m_p mounted on the crossbar of width w at distance l :

$$v'_{1x} = R_2 \dot{\theta} + l \dot{\gamma} \cos \gamma + \frac{w}{2} \dot{\gamma} \sin \gamma, \quad v'_{1y} = l \dot{\gamma} \sin \gamma - \frac{w}{2} \dot{\gamma} \cos \gamma \quad (412)$$

$$v'_{2x} = R_2 \dot{\theta} + l \dot{\gamma} \cos \gamma - \frac{w}{2} \dot{\gamma} \sin \gamma, \quad v'_{2y} = l \dot{\gamma} \sin \gamma + \frac{w}{2} \dot{\gamma} \cos \gamma \quad (413)$$

21.3 Kinetic and Potential Energy

Summing the kinetic energies of the wheels, slider block, rod, and both pendulum bobs:

$$\begin{aligned} KE = & \frac{1}{2}(M_1 R_1^2 + I_{M1}) \dot{\phi}^2 + \frac{1}{2} m_g \left(R_1^2 \dot{\phi}^2 + (\dot{k}'_1)^2 \right) + \frac{1}{2} (M R_2^2 + I_M) \dot{\theta}^2 + m_p R_2^2 \dot{\theta}^2 \\ & + m_p \left(l^2 + \frac{w^2}{4} \right) \dot{\gamma}^2 + 2 m_p R_2 l \cos \gamma \dot{\theta} \dot{\gamma} + \frac{1}{2} m_r R_1^2 \dot{\phi}^2 - \frac{1}{2} m_r R_1 L \dot{\phi} \dot{\alpha} \sin \alpha \\ & + \frac{1}{8} m_r L^2 \dot{\alpha}^2 + \frac{1}{2} I_r \dot{\alpha}^2 \end{aligned} \quad (414)$$

Substituting $\dot{\phi}$ and $\dot{\alpha}$ as functions of $\dot{\theta}$ condenses the coupled rotational terms into an effective inertia $M_\theta(\theta)$:

$$\begin{aligned} M_\theta(\theta) = & (M_1 R_1^2 + I_{M1} + m_g R_1^2 + m_r R_1^2) \left(\frac{R_2 - r_c (\cos \theta + \tan \alpha \sin \theta)}{R_1} \right)^2 + (M R_2^2 + I_M + 2 m_p R_2^2) \\ & - m_r R_1 L \sin \alpha \left(\frac{R_2 - r_c (\cos \theta + \tan \alpha \sin \theta)}{R_1} \right) \left(-\frac{r_c \sin \theta}{L \cos \alpha} \right) \\ & + \left(I_r + \frac{1}{4} m_r L^2 \right) \left(\frac{r_c^2 \sin^2 \theta}{L^2 \cos^2 \alpha} \right) \end{aligned} \quad (415)$$

Taking the reference datum for gravity:

$$PE = m_g g k'_1 + \frac{1}{2} k (k'_1 - L_0)^2 + \frac{1}{2} m_r g r_c \cos \theta - 2 m_p g l \cos \gamma \quad (416)$$

21.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$ for the generalized coordinates $\{k'_1, \gamma, \theta\}$:

For spring-block displacement k'_1 :

$$m_g \ddot{k}'_1 + k (k'_1 - L_0) + m_g g = 0 \quad (417)$$

For T-bar pendulum angle γ :

$$2m_p \left(l^2 + \frac{w^2}{4} \right) \ddot{\gamma} + 2m_p R_2 l \cos \gamma \ddot{\theta} + 2m_p g l \sin \gamma = 0 \quad (418)$$

For coupled wheel-rod angle θ :

$$M_\theta \ddot{\theta} + \frac{1}{2} \left(\frac{dM_\theta}{d\theta} \right) \dot{\theta}^2 + 2m_p R_2 l \cos \gamma \ddot{\gamma} - 2m_p R_2 l \sin \gamma \dot{\gamma}^2 - \frac{1}{2} m_r g r_c \sin \theta = 0 \quad (419)$$

21.5 Explicit Equations of Motion

Decoupling the accelerations in terms of state coordinates and velocities:

Spring-Loaded Block Acceleration ($\ddot{k}'_1$):

$$\ddot{k}'_1 = -\frac{k}{m_g} (k'_1 - L_0) - g \quad (420)$$

Coupled Wheels and Rod Acceleration ($\ddot{\theta}$): Substituting the pendulum equation into the θ -equation eliminates $\ddot{\gamma}$, yielding the explicit expression:

$$\ddot{\theta} = \frac{\frac{1}{2} m_r g r_c \sin \theta + 2m_p R_2 l \sin \gamma \dot{\gamma}^2 + \frac{m_p g R_2 l^2 \sin(2\gamma)}{l^2 + \frac{w^2}{4}} - \frac{1}{2} \left(\frac{dM_\theta}{d\theta} \right) \dot{\theta}^2}{M_\theta - \frac{2m_p R_2^2 l^2 \cos^2 \gamma}{l^2 + \frac{w^2}{4}}} \quad (421)$$

T-Bar Pendulum Angular Acceleration ($\ddot{\gamma}$):

$$\ddot{\gamma} = \frac{-R_2 l \cos \gamma \ddot{\theta} - g l \sin \gamma}{l^2 + \frac{w^2}{4}} \quad (422)$$

Recovered Accelerations for Dependent Coordinates ($\ddot{\alpha}, \ddot{\phi}$): Differentiating the first-order constraint relations with respect to time provides the explicit individual accelerations for the connecting rod and left wheel:

$$\ddot{\alpha} = - \left(\frac{r_c \sin \theta}{L \cos \alpha} \right) \ddot{\theta} - \left(\frac{r_c \cos \theta}{L \cos \alpha} + \frac{r_c^2 \sin^2 \theta \sin \alpha}{L^2 \cos^3 \alpha} \right) \dot{\theta}^2 \quad (423)$$

$$\begin{aligned} \ddot{\phi} = & \left(\frac{R_2 - r_c (\cos \theta + \tan \alpha \sin \theta)}{R_1} \right) \ddot{\theta} \\ & + \frac{r_c}{R_1} \left[\sin \theta (1 - \tan \alpha \dot{\theta}) - \sec^2 \alpha \sin \theta \dot{\alpha} - \tan \alpha \cos \theta \dot{\theta} \right] \dot{\theta} \end{aligned} \quad (424)$$

22 Problem 22: Rocker-Coupled Flywheel with Spring-Piston and T-Bar Pendulum

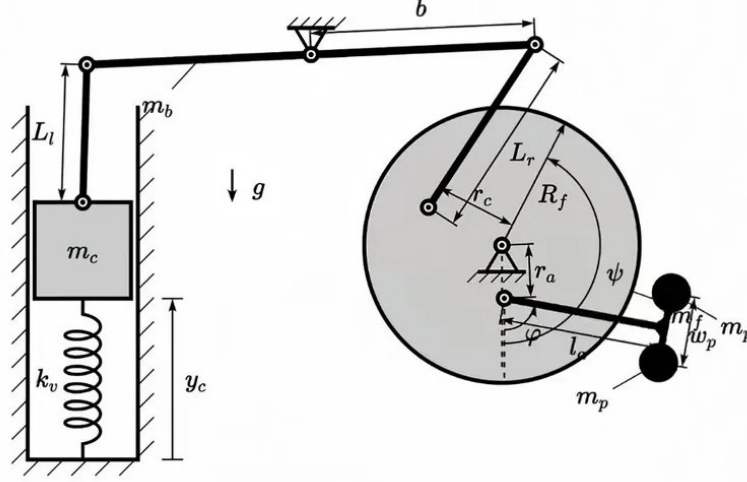


Figure 22: A symmetric rocker-lever mechanism coupling a spring-supported vertical piston m_c to a center-pinned flywheel m_f , which carries an eccentric T-bar pendulum with two tip masses m_p .

22.1 Kinematic Reduction and Generalized Coordinates

The mechanism consists of a closed kinematic chain linking the vertical piston, rocker beam, connecting rod, and flywheel, alongside an eccentric T-bar pendulum. Enforcing the kinematic loop closure relations leaves two dynamic degrees of freedom:

- $\theta(t)$: Angular rotation of the flywheel/disc of mass m_f and central moment of inertia I_f .
- $\alpha(t)$: Angular deflection of the suspended T-bar pendulum stem from the downward vertical.
- $k'_1(t)$: Vertical displacement of the piston of mass m_c restrained by vertical spring k (with unstretched length l_0).
- $\phi_1(t), \phi_2(t), \phi_3(t)$: Angular orientations of the piston connecting link (length L_1), rocker beam (half-span b), and flywheel drive link (length L_2).
- r_c, r_a : Radial offsets on the flywheel for the drive link pin and pendulum pivot pin, respectively.
- ϕ_4 : Fixed offset angle of the pendulum pivot pin on the flywheel.
- m_p, l, w : Two pendulum tip masses m_p mounted symmetrically on a crossbar of width w at stem length l .
- I_b, I_1, I_2 : Mass moments of inertia of the rocker beam and the two connecting links.

22.2 Kinematic Jacobians and Loop Relations

The closed vector loop equations relate the angular rates $\dot{\phi}_1, \dot{\phi}_2, \dot{\phi}_3$ and piston speed $\dot{k}'_1$ directly to the flywheel angular speed $\dot{\theta}$:

$$\dot{\phi}_1 = J_1 \dot{\theta}, \quad \dot{\phi}_2 = J_2 \dot{\theta}, \quad \dot{\phi}_3 = J_3 \dot{\theta}, \quad \dot{k}'_1 = J_4 \dot{\theta} \quad (425)$$

where the geometric transfer coefficients are:

$$J_2 = \frac{-v_{4x} \sin \phi_3 + v_{4y} \cos \phi_3}{b \cos(\phi_3 - \phi_2)} \quad (426)$$

$$J_1 = - \left(\frac{b \sin \phi_2}{L_1 \cos \phi_1} \right) J_2 \quad (427)$$

$$J_3 = \frac{v_{4x} \cos \phi_2 + v_{4y} \sin \phi_2}{L_2 \cos(\phi_3 - \phi_2)} \quad (428)$$

$$J_4 = -b \cos \phi_2 J_2 + L_1 \sin \phi_1 J_1 \quad (429)$$

Here (v_{4x}, v_{4y}) represent the normalized kinematic velocity components of the drive pin on the flywheel.

22.3 Kinematics of the Suspended T-Bar Pendulum

The Cartesian velocities of the two symmetric pendulum bobs m_p located at stem length l and transverse offsets $\pm w/2$ are:

$$v'_{1x} = r_a \dot{\theta} \cos(\theta + \phi_4) + l \dot{\alpha} \cos \alpha - \frac{w}{2} \dot{\alpha} \sin \alpha \quad (430)$$

$$v'_{1y} = r_a \dot{\theta} \sin(\theta + \phi_4) + l \dot{\alpha} \sin \alpha + \frac{w}{2} \dot{\alpha} \cos \alpha \quad (431)$$

$$v'_{2x} = r_a \dot{\theta} \cos(\theta + \phi_4) + l \dot{\alpha} \cos \alpha + \frac{w}{2} \dot{\alpha} \sin \alpha \quad (432)$$

$$v'_{2y} = r_a \dot{\theta} \sin(\theta + \phi_4) + l \dot{\alpha} \sin \alpha - \frac{w}{2} \dot{\alpha} \cos \alpha \quad (433)$$

Summing the squared bob velocities cancels the cross-terms in $\pm w/2$:

$$v'_{1x}{}^2 + v'_{1y}{}^2 + v'_{2x}{}^2 + v'_{2y}{}^2 = 2r_a^2 \dot{\theta}^2 + 2 \left(l^2 + \frac{w^2}{4} \right) \dot{\alpha}^2 + 4r_a l \cos(\theta + \phi_4 - \alpha) \dot{\theta} \dot{\alpha} \quad (434)$$

22.4 Kinetic and Potential Energy

The total system kinetic energy combines the constrained linkage mechanism and the T-bar pendulum:

$$\begin{aligned} KE = & \frac{1}{2} m_c (\dot{k}'_1)^2 + \frac{1}{2} [I_b + (m_1 + m_2) b^2] \dot{\phi}_2^2 + \frac{1}{2} \left(I_1 + \frac{1}{4} m_1 L_1^2 \right) \dot{\phi}_1^2 + \frac{1}{2} \left(I_2 + \frac{1}{4} m_2 L_2^2 \right) \dot{\phi}_3^2 \\ & - \frac{1}{2} m_1 L_1 b \sin(\phi_1 - \phi_2) \dot{\phi}_1 \dot{\phi}_2 + \frac{1}{2} m_2 L_2 b \sin(\phi_3 - \phi_2) \dot{\phi}_3 \dot{\phi}_2 \\ & + \frac{1}{2} (I_f + 2m_p r_a^2) \dot{\theta}^2 + m_p \left(l^2 + \frac{w^2}{4} \right) \dot{\alpha}^2 + 2m_p r_a l \cos(\theta + \phi_4 - \alpha) \dot{\theta} \dot{\alpha} \end{aligned} \quad (435)$$

Substituting the kinematic Jacobians J_1, J_2, J_3, J_4 condenses the mechanism inertia into an effective generalized inertia $M_\theta(\theta)$:

$$\begin{aligned} M_\theta(\theta) = & I_f + 2m_p r_a^2 + m_c J_4^2 + [I_b + (m_1 + m_2) b^2] J_2^2 + \left(I_1 + \frac{1}{4} m_1 L_1^2 \right) J_1^2 + \left(I_2 + \frac{1}{4} m_2 L_2^2 \right) J_3^2 \\ & - m_1 L_1 b \sin(\phi_1 - \phi_2) J_1 J_2 + m_2 L_2 b \sin(\phi_3 - \phi_2) J_3 J_2 \end{aligned} \quad (436)$$

The total gravitational and spring potential energy is:

$$PE = m_c g k'_1 + \frac{1}{2} k (k'_1 - l_0)^2 + (m_2 - m_1) g b \sin \phi_2 - \frac{1}{2} g (m_1 L_1 \cos \phi_1 + m_2 L_2 \cos \phi_3) - 2m_p g [r_a \cos(\theta + \phi_4) + l \cos \alpha] \quad (437)$$

Defining the generalized mechanism gravity and spring restoring force $X(\theta)$:

$$X(\theta) = [m_c g + k(k'_1 - l_0)] J_4 + (m_2 - m_1) g b \cos \phi_2 J_2 + \frac{1}{2} g (m_1 L_1 \sin \phi_1 J_1 + m_2 L_2 \sin \phi_3 J_3) + 2m_p g r_a \sin(\theta + \phi_4) \quad (438)$$

22.5 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$ for the two active coordinates $\{\alpha, \theta\}$:

For pendulum angle α :

$$2m_p \left(l^2 + \frac{w^2}{4} \right) \ddot{\alpha} + 2m_p r_a l \cos(\theta + \phi_4 - \alpha) \ddot{\theta} - 2m_p r_a l \sin(\theta + \phi_4 - \alpha) \dot{\theta}^2 + 2m_p g l \sin \alpha = 0 \quad (439)$$

For flywheel angle θ :

$$M_\theta \ddot{\theta} + 2m_p r_a l \cos(\theta + \phi_4 - \alpha) \ddot{\alpha} + \frac{1}{2} \left(\frac{dM_\theta}{d\theta} \right) \dot{\theta}^2 + 2m_p r_a l \sin(\theta + \phi_4 - \alpha) \dot{\alpha}^2 + X(\theta) = 0 \quad (440)$$

22.6 Explicit Equations of Motion

Decoupling the accelerations directly yields the explicit equations of motion:

Flywheel Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{-\frac{1}{2} \left(\frac{dM_\theta}{d\theta} \right) \dot{\theta}^2 - 2m_p r_a l \sin(\theta + \phi_4 - \alpha) \dot{\alpha}^2 - X(\theta) + \frac{2m_p r_a l \cos(\theta + \phi_4 - \alpha) [g l \sin \alpha - r_a l \sin(\theta + \phi_4 - \alpha) \dot{\theta}^2]}{l^2 + \frac{w^2}{4}}}{M_\theta - \frac{2m_p r_a^2 l^2 \cos^2(\theta + \phi_4 - \alpha)}{l^2 + \frac{w^2}{4}}} \quad (441)$$

T-Bar Pendulum Angular Acceleration ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{-r_a l \cos(\theta + \phi_4 - \alpha) \ddot{\theta} + r_a l \sin(\theta + \phi_4 - \alpha) \dot{\theta}^2 - g l \sin \alpha}{l^2 + \frac{w^2}{4}} \quad (442)$$

Kinematically Dependent Accelerations ($\ddot{k}'_1, \ddot{\phi}_1, \ddot{\phi}_2, \ddot{\phi}_3$): The accelerations of the intermediate links and piston are recovered directly via their kinematic Jacobian coefficients:

$$\ddot{k}'_1 = J_4 \ddot{\theta} + \dot{J}_4 \dot{\theta}, \quad \ddot{\phi}_i = J_i \ddot{\theta} + \dot{J}_i \dot{\theta} \quad (i \in \{1, 2, 3\}) \quad (443)$$

23 Problem 23: Rolling Ring on Parabolic Track with Linked Discs and Elastic Pendulum

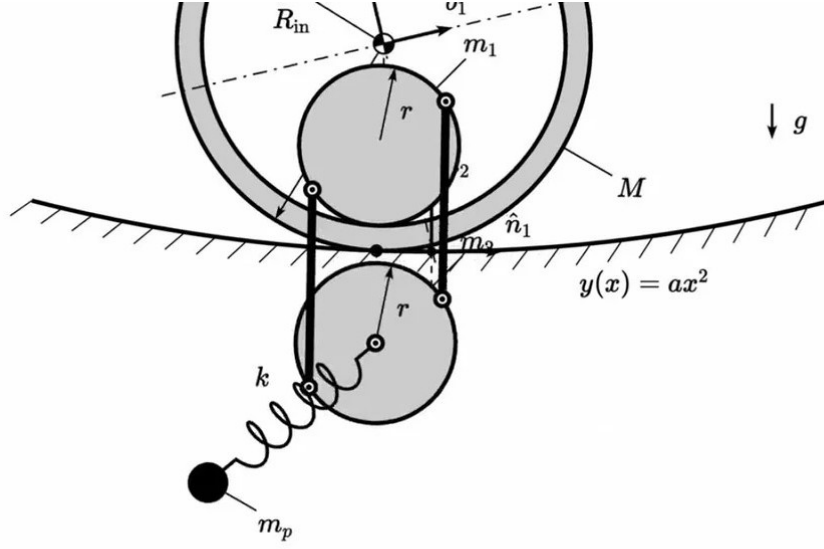


Figure 23: A hollow ring of mass M rolling on a parabolic substrate $y(x) = ax^2$, containing an internal rolling cylinder of mass m_1 , connected via a parallelogram link of length L to a second cylinder of mass m_2 , which suspends an elastic spring-pendulum of bob mass m_p and stiffness k .

23.1 Generalized Coordinates and Parameters

The planar multi-body system possesses five independent degrees of freedom:

- $x(t)$: Horizontal substrate contact coordinate of the outer rolling ring.
- $\phi_1(t)$: Relative angular position of the inner rolling cylinder m_1 within the ring cavity.
- $\phi_2(t)$: Angular orientation of the parallelogram suspension links of length L from the downward vertical.
- $k'_1(t)$: Instantaneous stretched length of the pendulum spring.
- $\phi_4(t)$: Angular deflection of the elastic pendulum from the downward vertical.
- M, I_c : Mass, central moment of inertia, outer radius R_0 , and inner cavity radius R_i of the rolling ring.
- m_1, I_1, r_1 : Mass, central moment of inertia, and radius of the internal rolling cylinder.
- m_2, I_2, r_2 : Mass, moment of inertia, and radius of the lower cylinder ($r_2 = r_1$, $\dot{\phi}_3 = \dot{\phi}_1$ via the parallelogram linkage).
- m_p, k, l_0 : Mass of the pendulum bob, linear spring stiffness, and unstretched spring length.
- $y(x) = ax^2$: Profile equation of the rigid parabolic track.

23.2 Kinematics and Velocity Cascade

The local slope angle $\beta(x)$ of the parabolic substrate satisfies:

$$\tan \beta = 2ax \implies \sin \beta = \frac{2ax}{\sqrt{1+4a^2x^2}}, \quad \cos \beta = \frac{1}{\sqrt{1+4a^2x^2}}, \quad \dot{\beta} = \frac{2a}{1+4a^2x^2} \dot{x} \quad (444)$$

Pure rolling of the outer ring along the parabolic profile dictates:

$$\dot{\theta} = \frac{\sqrt{1+4a^2x^2}}{R_0} \dot{x} \implies \ddot{\theta} = \frac{\sqrt{1+4a^2x^2}}{R_0} \ddot{x} + \frac{4a^2x}{R_0\sqrt{1+4a^2x^2}} \dot{x}^2 \quad (445)$$

Defining the geometric scale factor $\mu(x)$:

$$\mu(x) = 1 - \frac{2aR_0}{(1+4a^2x^2)^{3/2}} \quad (446)$$

The Cartesian velocities of each body in the kinematic chain are:

Outer Ring Center (v_{1x}, v_{1y}):

$$v_{1x} = \mu(x) \dot{x} \quad (447)$$

$$v_{1y} = 2ax \mu(x) \dot{x} = 2ax v_{1x} \quad (448)$$

Internal Cylinder Center (v_{2x}, v_{2y}):

$$v_{2x} = v_{1x} - (R_i - r_1)(\dot{\theta} + \dot{\phi}_1) \cos(\theta + \phi_1) \quad (449)$$

$$v_{2y} = v_{1y} + (R_i - r_1)(\dot{\theta} + \dot{\phi}_1) \sin(\theta + \phi_1) \quad (450)$$

where the spin angular velocity of the rolling inner cylinder is $\omega_{\text{spin}} = \dot{\theta} - \left(\frac{R_i - r_1}{r_1}\right) \dot{\phi}_1$.

Lower Suspended Cylinder Center (v_{3x}, v_{3y}):

$$v_{3x} = v_{2x} - L\dot{\phi}_2 \cos \phi_2 \quad (451)$$

$$v_{3y} = v_{2y} - L\dot{\phi}_2 \sin \phi_2 \quad (452)$$

Elastic Pendulum Bob (v_{4x}, v_{4y}):

$$v_{4x} = v_{3x} - \dot{k}'_1 \sin \phi_4 - k'_1 \dot{\phi}_4 \cos \phi_4 \quad (453)$$

$$v_{4y} = v_{3y} - \dot{k}'_1 \cos \phi_4 + k'_1 \dot{\phi}_4 \sin \phi_4 \quad (454)$$

23.3 Kinetic and Potential Energy Formulations

The total kinetic energy combines the outer ring, internal cylinder, lower cylinder, and pendulum bob:

$$\begin{aligned} KE = & \frac{1}{2}M \left[\dot{x}^2(1+4a^2x^2) + R_0^2\dot{\beta}^2 - 2R_0\dot{x}\dot{\beta}\sec\beta \right] + \frac{1}{2}I_c\dot{\theta}^2 \\ & + \frac{1}{2}m_1 \left[v_{1x}^2 + v_{1y}^2 + (R_i - r_1)^2(\dot{\theta} + \dot{\phi}_1)^2 - 2(R_i - r_1)(\dot{\theta} + \dot{\phi}_1)(v_{1x}\cos(\theta + \phi_1) - v_{1y}\sin(\theta + \phi_1)) \right] \\ & + \frac{1}{2}(I_1 + I_2) \left[\dot{\theta} - \left(\frac{R_i - r_1}{r_1}\right) \dot{\phi}_1 \right]^2 + \frac{1}{2}m_2 \left[v_{2x}^2 + v_{2y}^2 + L^2\dot{\phi}_2^2 - 2L\dot{\phi}_2(v_{2x}\cos\phi_2 + v_{2y}\sin\phi_2) \right] \\ & + \frac{1}{2}m_p \left[v_{3x}^2 + v_{3y}^2 + (\dot{k}'_1)^2 + (k'_1)^2\dot{\phi}_4^2 - 2\dot{k}'_1(v_{3x}\sin\phi_4 - v_{3y}\cos\phi_4) \right. \\ & \quad \left. - 2k'_1\dot{\phi}_4(v_{3x}\cos\phi_4 + v_{3y}\sin\phi_4) \right] \end{aligned} \quad (455)$$

Taking the vertex of the parabola as the gravitational potential datum:

$$\begin{aligned}
PE = (M + m_1 + m_2 + m_p)g \left(ax^2 + \frac{R_0}{\sqrt{1 + 4a^2x^2}} \right) - (m_1 + m_2 + m_p)g(R_i - r_1) \cos(\theta + \phi_1) \\
- (m_2 + m_p)gL \cos \phi_2 - m_pgk'_1 \cos \phi_4 + \frac{1}{2}k(k'_1 - l_0)^2
\end{aligned} \tag{456}$$

23.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

1. For pendulum angle ϕ_4 :

$$k'_1 \ddot{\phi}_4 + 2\dot{k}'_1 \dot{\phi}_4 - (\dot{v}_{3x} \cos \phi_4 + \dot{v}_{3y} \sin \phi_4) + g \sin \phi_4 = 0 \tag{457}$$

2. For spring length k'_1 :

$$\ddot{k}'_1 - k'_1 \dot{\phi}_4^2 - (\dot{v}_{3x} \sin \phi_4 - \dot{v}_{3y} \cos \phi_4) - g \cos \phi_4 + \frac{k}{m_p}(k'_1 - l_0) = 0 \tag{458}$$

3. For linkage angle ϕ_2 :

$$\begin{aligned}
(m_2 + m_p)L\ddot{\phi}_2 - (m_2 + m_p)(\dot{v}_{2x} \cos \phi_2 + \dot{v}_{2y} \sin \phi_2) + (m_2 + m_p)g \sin \phi_2 \\
+ m_p \left[(\ddot{k}'_1 - k'_1 \dot{\phi}_4^2) \sin(\phi_4 - \phi_2) + (k'_1 \ddot{\phi}_4 + 2\dot{k}'_1 \dot{\phi}_4) \cos(\phi_4 - \phi_2) \right] = 0
\end{aligned} \tag{459}$$

4. For cavity angle ϕ_1 :

$$\begin{aligned}
(m_1 + m_2 + m_p)(R_i - r_1)(\ddot{\theta} + \ddot{\phi}_1) + \frac{I_1 + I_2}{r_1} \left(\frac{R_i - r_1}{r_1} \right) \ddot{\phi}_1 - \frac{I_1 + I_2}{r_1} \ddot{\theta} \\
- (m_1 + m_2 + m_p) [\dot{v}_{1x} \cos(\theta + \phi_1) - \dot{v}_{1y} \sin(\theta + \phi_1)] + (m_1 + m_2 + m_p)g \sin(\theta + \phi_1) \\
+ (m_2 + m_p)L \left[\ddot{\phi}_2 \cos(\theta + \phi_1 - \phi_2) - \dot{\phi}_2^2 \sin(\theta + \phi_1 - \phi_2) \right] \\
+ m_p \left[(\ddot{k}'_1 - k'_1 \dot{\phi}_4^2) \sin(\theta + \phi_1 - \phi_4) + (k'_1 \ddot{\phi}_4 + 2\dot{k}'_1 \dot{\phi}_4) \cos(\theta + \phi_1 - \phi_4) \right] = 0
\end{aligned} \tag{460}$$

5. For substrate contact coordinate x :

$$\begin{aligned}
& \mu(x)(v_{1x} + 2ax v_{1y})(M + m_1 + m_2 + m_p) \\
& + \left(\frac{\sqrt{1 + 4a^2 x^2}}{R_0} \right) \left[I_c \ddot{\theta} + (I_1 + I_2) \left(\ddot{\theta} - \frac{R_i - r_1}{r_1} \ddot{\phi}_1 \right) \right] \\
& + \left(\frac{4a^2 x}{R_0 \sqrt{1 + 4a^2 x^2}} \right) \dot{x} \left[I_c \dot{\theta} + (I_1 + I_2) \left(\dot{\theta} - \frac{R_i - r_1}{r_1} \dot{\phi}_1 \right) \right] \\
& + (m_1 + m_2 + m_p)(R_i - r_1)\mu(x)(\ddot{\theta} + \ddot{\phi}_1) [\cos(\theta + \phi_1) - 2ax \sin(\theta + \phi_1)] \\
& - (m_1 + m_2 + m_p)(R_i - r_1)\mu(x)(\dot{\theta} + \dot{\phi}_1)^2 [\sin(\theta + \phi_1) + 2ax \cos(\theta + \phi_1)] \\
& - (m_2 + m_p)L\mu(x) \left[\ddot{\phi}_2 (\cos \phi_2 + 2ax \sin \phi_2) - \dot{\phi}_2^2 (\sin \phi_2 - 2ax \cos \phi_2) \right] \\
& - m_p \mu(x) \left[(\ddot{k}'_1 - k'_1 \dot{\phi}_4^2) (\sin \phi_4 - 2ax \cos \phi_4) + (k'_1 \ddot{\phi}_4 + 2\dot{k}'_1 \dot{\phi}_4) (\cos \phi_4 + 2ax \sin \phi_4) \right] \\
& + (M + m_1 + m_2 + m_p)g \left(2ax - \frac{2aR_0 x}{(1 + 4a^2 x^2)^{3/2}} \right) \\
& + (m_1 + m_2 + m_p)g(R_i - r_1) \left(\frac{\sqrt{1 + 4a^2 x^2}}{R_0} \right) \sin(\theta + \phi_1) = 0
\end{aligned} \tag{461}$$

23.5 Explicit Equations of Motion

Decoupling and isolating the accelerations hierarchically:

Pendulum Angular Acceleration ($\ddot{\phi}_4$):

$$\ddot{\phi}_4 = \frac{-2\dot{k}'_1 \dot{\phi}_4 + \dot{v}_{3x} \cos \phi_4 + \dot{v}_{3y} \sin \phi_4 - g \sin \phi_4}{k'_1} \tag{462}$$

Spring Length Acceleration ($\ddot{k}'_1$):

$$\ddot{k}'_1 = k'_1 \dot{\phi}_4^2 + \dot{v}_{3x} \sin \phi_4 - \dot{v}_{3y} \cos \phi_4 + g \cos \phi_4 - \frac{k}{m_p}(k'_1 - l_0) \tag{463}$$

Linkage Angular Acceleration ($\ddot{\phi}_2$):

$$\begin{aligned}
& (m_2 + m_p) (\dot{v}_{2x} \cos \phi_2 + \dot{v}_{2y} \sin \phi_2 - g \sin \phi_2) \\
& - m_p \left[(\ddot{k}'_1 - k'_1 \dot{\phi}_4^2) \sin(\phi_4 - \phi_2) + (k'_1 \ddot{\phi}_4 + 2\dot{k}'_1 \dot{\phi}_4) \cos(\phi_4 - \phi_2) \right] \\
& \ddot{\phi}_2 = \frac{(m_2 + m_p)L}{(m_2 + m_p)L}
\end{aligned} \tag{464}$$

Inner Cylinder Relative Angular Acceleration ($\ddot{\phi}_1$):

Defining the effective inertia $J_{\phi_1} = (m_1 + m_2 + m_p)(R_i - r_1) + \frac{I_1 + I_2}{r_1^2}(R_i - r_1)$:

$$\begin{aligned}
& \left[\frac{I_1 + I_2}{r_1} - (m_1 + m_2 + m_p)(R_i - r_1) \right] \ddot{\theta} \\
& + (m_1 + m_2 + m_p) [\dot{v}_{1x} \cos(\theta + \phi_1) - \dot{v}_{1y} \sin(\theta + \phi_1) - g \sin(\theta + \phi_1)] \\
& - (m_2 + m_p)L \left[\ddot{\phi}_2 \cos(\theta + \phi_1 - \phi_2) - \dot{\phi}_2^2 \sin(\theta + \phi_1 - \phi_2) \right] \\
& - m_p \left[(\ddot{k}'_1 - k'_1 \dot{\phi}_4^2) \sin(\theta + \phi_1 - \phi_4) + (k'_1 \ddot{\phi}_4 + 2\dot{k}'_1 \dot{\phi}_4) \cos(\theta + \phi_1 - \phi_4) \right] \\
& \ddot{\phi}_1 = \frac{J_{\phi_1}}{J_{\phi_1}}
\end{aligned} \tag{465}$$

Outer Ring Substrate Acceleration ($\ddot{x}$): Defining the total coordinate inertia M_{xx} :

$$M_{xx} = (M + m_1 + m_2 + m_p)[\mu(x)]^2(1 + 4a^2x^2) + [I_c + I_1 + I_2] \frac{1 + 4a^2x^2}{R_0^2} + 2(m_1 + m_2 + m_p)(R_i - r_1)\mu(x) \frac{\sqrt{1 + 4a^2x^2}}{R_0} [\cos(\theta + \phi_1) - 2ax \sin(\theta + \phi_1)] \quad (466)$$

The explicit second-order substrate acceleration is:

$$\begin{aligned} & - (M + m_1 + m_2 + m_p)g \left(2ax - \frac{2aR_0x}{(1 + 4a^2x^2)^{3/2}} \right) \\ & - (m_1 + m_2 + m_p)g(R_i - r_1) \left(\frac{\sqrt{1 + 4a^2x^2}}{R_0} \right) \sin(\theta + \phi_1) \\ & + \frac{(I_1 + I_2)(R_i - r_1)\sqrt{1 + 4a^2x^2}}{r_1R_0} \ddot{\phi}_1 \\ & - (m_1 + m_2 + m_p)(R_i - r_1)\mu(x) \ddot{\phi}_1 [\cos(\theta + \phi_1) - 2ax \sin(\theta + \phi_1)] \\ & + (m_1 + m_2 + m_p)(R_i - r_1)\mu(x)(\dot{\theta} + \dot{\phi}_1)^2 [\sin(\theta + \phi_1) + 2ax \cos(\theta + \phi_1)] \\ & + (m_2 + m_p)L\mu(x) \left[\ddot{\phi}_2(\cos \phi_2 + 2ax \sin \phi_2) - \dot{\phi}_2^2(\sin \phi_2 - 2ax \cos \phi_2) \right] \\ & + m_p\mu(x) \left[(\ddot{k}_1' - k_1'\dot{\phi}_4^2)(\sin \phi_4 - 2ax \cos \phi_4) + (k_1'\ddot{\phi}_4 + 2\dot{k}_1'\dot{\phi}_4)(\cos \phi_4 + 2ax \sin \phi_4) \right] \\ & - \left(\frac{4a^2x}{R_0\sqrt{1 + 4a^2x^2}} \right) \dot{x} \left[I_c\dot{\theta} + (I_1 + I_2) \left(\dot{\theta} - \frac{R_i - r_1}{r_1} \dot{\phi}_1 \right) \right] \\ \ddot{x} = & \frac{\text{Sum of the above terms}}{M_{xx}} \end{aligned} \quad (467)$$

24 Problem 24: Spring-Supported Platform with Rocking Semi-circular Shell, Dual Rolling Discs, and Pendulum

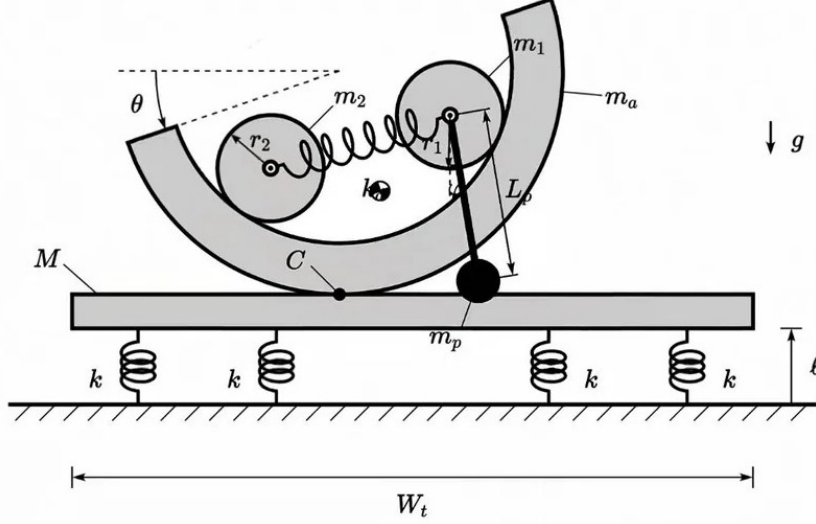


Figure 24: A horizontal platform of mass M supported vertically by four springs of stiffness k , carrying a rocking shell of mass M_s and outer radius R_0 , containing two internal rolling discs of masses m_1, m_2 connected by spring k_1 , with a simple pendulum of mass m_p suspended from disc m_1 .

24.1 Generalized Coordinates and System Parameters

The planar multi-body system possesses five independent generalized coordinates:

- $k'_2(t)$: Dynamic vertical displacement of the base platform (taken positive upward or downward along the spring axis).
- $\theta(t)$: Rocking rotation angle of the curved shell on the platform.
- $\phi_1(t)$: Relative angular position of disc 1 inside the shell cavity.
- $\phi_2(t)$: Relative angular position of disc 2 inside the shell cavity.
- $\alpha(t)$: Angular deflection of the suspended pendulum rod from the downward vertical.
- M : Mass of the horizontally translating/vertically vibrating base platform.
- M_s, I_s : Mass and centroidal moment of inertia of the curved shell (R_0 : outer radius, R : inner cavity radius).
- m_1, I_1, r_1 : Mass, moment of inertia, and radius of the first rolling disc.
- m_2, I_2, r_2 : Mass, moment of inertia, and radius of the second rolling disc.
- m_p, l : Mass of the pendulum bob and length of the pendulum rod.
- k_1, l_0 : Stiffness and unstretched length of the internal coupling spring between the two discs.

- k_2 : Stiffness of each of the four base-supporting foundation springs (total vertical stiffness $4k_2$).
- Aggregate mass definitions:

$$M_{\text{tot}} = M + M_s + m_1 + m_2 + m_p, \quad m_{1p} = m_1 + m_p, \quad M_r = M_s + m_1 + m_2 + m_p \quad (468)$$

- Composite angular orientations:

$$\psi_1 = \theta + \phi_1, \quad \psi_2 = \theta + \phi_2 \quad (469)$$

24.2 Kinematics and Spring Geometry

Enforcing pure rolling of the outer shell along the horizontal base platform yields the horizontal contact translation $x = R_0\theta$, so that $\dot{x} = R_0\dot{\theta}$ and $\ddot{x} = R_0\ddot{\theta}$.

The instantaneous extended length k'_1 of the spring coupling the centers of the two inner cylinders is:

$$k'_1 = \sqrt{(R - r_1)^2 + (R - r_2)^2 - 2(R - r_1)(R - r_2)\cos(\phi_2 - \phi_1)} \quad (470)$$

The generalized internal elastic spring force magnitude f_s acting between the discs is:

$$f_s = \frac{k_1(k'_1 - l_0)(R - r_1)(R - r_2)\sin(\phi_2 - \phi_1)}{k'_1} \quad (471)$$

The Cartesian velocity components of each body are:

Base Platform and Shell Center:

$$V_x = R_0\dot{\theta}, \quad V_y = \dot{k}'_2 \quad (472)$$

Inner Disc 1 Center (v_{1x}, v_{1y}):

$$v_{1x} = R_0\dot{\theta} + (R - r_1)\dot{\psi}_1 \cos \psi_1 \quad (473)$$

$$v_{1y} = \dot{k}'_2 + (R - r_1)\dot{\psi}_1 \sin \psi_1 \quad (474)$$

Inner Disc 2 Center (v_{2x}, v_{2y}):

$$v_{2x} = R_0\dot{\theta} + (R - r_2)\dot{\psi}_2 \cos \psi_2 \quad (475)$$

$$v_{2y} = \dot{k}'_2 + (R - r_2)\dot{\psi}_2 \sin \psi_2 \quad (476)$$

Suspended Pendulum Bob (v_{px}, v_{py}):

$$v_{px} = R_0\dot{\theta} + (R - r_1)\dot{\psi}_1 \cos \psi_1 - l\dot{\alpha} \cos \alpha \quad (477)$$

$$v_{py} = \dot{k}'_2 + (R - r_1)\dot{\psi}_1 \sin \psi_1 + l\dot{\alpha} \sin \alpha \quad (478)$$

24.3 Kinetic and Potential Energy Formulations

The total system kinetic energy combines the base platform, shell, both inner discs, and the pendulum bob:

$$\begin{aligned}
KE &= \frac{1}{2}M(\dot{k}_2')^2 + \frac{1}{2}M_s \left(R_0^2 \dot{\theta}^2 + (\dot{k}_2')^2 \right) + \frac{1}{2}I_s \dot{\theta}^2 + \frac{1}{2}m_1 (v_{1x}^2 + v_{1y}^2) + \frac{1}{2}I_1 \omega_1^2 \\
&\quad + \frac{1}{2}m_2 (v_{2x}^2 + v_{2y}^2) + \frac{1}{2}I_2 \omega_2^2 + \frac{1}{2}m_p (v_{px}^2 + v_{py}^2) \\
&= \frac{1}{2}M_{\text{tot}}(\dot{k}_2')^2 + \frac{1}{2}(I_s + M_r R_0^2) \dot{\theta}^2 + \frac{1}{2}m_{1p}(R - r_1)^2 \dot{\psi}_1^2 + \frac{1}{2}m_2(R - r_2)^2 \dot{\psi}_2^2 + \frac{1}{2}m_p l^2 \dot{\alpha}^2 \\
&\quad + m_{1p}(R - r_1) \dot{\psi}_1 \left(R_0 \dot{\theta} \cos \psi_1 + \dot{k}_2' \sin \psi_1 \right) + m_2(R - r_2) \dot{\psi}_2 \left(R_0 \dot{\theta} \cos \psi_2 + \dot{k}_2' \sin \psi_2 \right) \\
&\quad - m_p l \dot{\alpha} \left[R_0 \dot{\theta} \cos \alpha - \dot{k}_2' \sin \alpha + (R - r_1) \dot{\psi}_1 \cos(\psi_1 + \alpha) \right] + \frac{1}{2}I_1 \omega_1^2 + \frac{1}{2}I_2 \omega_2^2 \quad (479)
\end{aligned}$$

Taking the platform equilibrium datum as the reference for gravity:

$$\begin{aligned}
PE &= 2k_2(k_2')^2 + \frac{1}{2}k_1(k_1' - l_0)^2 + M_{\text{tot}}gk_2' + M_r g R_0 \\
&\quad - m_{1p}g(R - r_1) \cos \psi_1 - m_2g(R - r_2) \cos \psi_2 - m_p g l \cos \alpha \quad (480)
\end{aligned}$$

24.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$ across all generalized coordinates:

1. For base platform displacement k_2' :

$$\begin{aligned}
M_{\text{tot}} \ddot{k}_2' + m_{1p}(R - r_1) \left(\ddot{\psi}_1 \sin \psi_1 + \dot{\psi}_1^2 \cos \psi_1 \right) + m_2(R - r_2) \left(\ddot{\psi}_2 \sin \psi_2 + \dot{\psi}_2^2 \cos \psi_2 \right) \\
+ m_p l \left(\ddot{\alpha} \sin \alpha + \dot{\alpha}^2 \cos \alpha \right) + 4k_2 k_2' + M_{\text{tot}} g = 0 \quad (481)
\end{aligned}$$

2. For inner disc 1 angle ϕ_1 :

$$\begin{aligned}
m_{1p}(R - r_1)^2 \ddot{\psi}_1 + m_{1p}(R - r_1) \left(R_0 \ddot{\theta} \cos \psi_1 + \ddot{k}_2' \sin \psi_1 \right) \\
- m_p l (R - r_1) \left[\ddot{\alpha} \cos(\psi_1 + \alpha) - \dot{\alpha}^2 \sin(\psi_1 + \alpha) \right] + m_{1p} g (R - r_1) \sin \psi_1 - f_s = 0 \quad (482)
\end{aligned}$$

3. For inner disc 2 angle ϕ_2 :

$$m_2(R - r_2)^2 \ddot{\psi}_2 + m_2(R - r_2) \left(R_0 \ddot{\theta} \cos \psi_2 + \ddot{k}_2' \sin \psi_2 \right) + m_2 g (R - r_2) \sin \psi_2 + f_s = 0 \quad (483)$$

4. For pendulum angle α :

$$\begin{aligned}
l \ddot{\alpha} - R_0 \ddot{\theta} \cos \alpha + \ddot{k}_2' \sin \alpha - (R - r_1) \ddot{\psi}_1 \cos(\psi_1 + \alpha) \\
+ (R - r_1) \dot{\psi}_1^2 \sin(\psi_1 + \alpha) + g \sin \alpha = 0 \quad (484)
\end{aligned}$$

5. For shell rocking angle θ :

$$\begin{aligned}
(I_s + M_r R_0^2) \ddot{\theta} + m_{1p}(R - r_1) R_0 \ddot{\psi}_1 \cos \psi_1 + m_2(R - r_2) R_0 \ddot{\psi}_2 \cos \psi_2 - m_p l R_0 \ddot{\alpha} \cos \alpha \\
- R_0 \left[m_{1p}(R - r_1) \dot{\psi}_1^2 \sin \psi_1 + m_2(R - r_2) \dot{\psi}_2^2 \sin \psi_2 - m_p l \dot{\alpha}^2 \sin \alpha \right] \\
+ m_{1p} g (R - r_1) \sin \psi_1 + m_2 g (R - r_2) \sin \psi_2 = 0 \quad (485)
\end{aligned}$$

24.5 Explicit Equations of Motion

Isolating the explicit second-order accelerations for numerical implementation:

Base Platform Acceleration ($\ddot{k}'_2$):

$$\ddot{k}'_2 = \frac{-4k_2k'_2 - M_{\text{tot}}g - m_{1p}(R - r_1) \left(\ddot{\psi}_1 \sin \psi_1 + \dot{\psi}_1^2 \cos \psi_1 \right) - m_2(R - r_2) \left(\ddot{\psi}_2 \sin \psi_2 + \dot{\psi}_2^2 \cos \psi_2 \right) - m_pl \left(\ddot{\alpha} \sin \alpha + \dot{\alpha}^2 \cos \alpha \right)}{M_{\text{tot}}} \quad (486)$$

Right Disc Angular Acceleration ($\ddot{\phi}_1$):

$$\ddot{\phi}_1 = -\ddot{\theta} - \frac{R_0\ddot{\theta} \cos \psi_1 + \ddot{k}'_2 \sin \psi_1 + g \sin \psi_1}{R - r_1} + \frac{m_pl \left[\ddot{\alpha} \cos(\psi_1 + \alpha) - \dot{\alpha}^2 \sin(\psi_1 + \alpha) \right]}{m_{1p}(R - r_1)} + \frac{f_s}{m_{1p}(R - r_1)^2} \quad (487)$$

Left Disc Angular Acceleration ($\ddot{\phi}_2$):

$$\ddot{\phi}_2 = -\ddot{\theta} - \frac{R_0\ddot{\theta} \cos \psi_2 + \ddot{k}'_2 \sin \psi_2 + g \sin \psi_2}{R - r_2} - \frac{f_s}{m_2(R - r_2)^2} \quad (488)$$

Pendulum Angular Acceleration ($\ddot{\alpha}$):

$$\ddot{\alpha} = \frac{R_0\ddot{\theta} \cos \alpha - \ddot{k}'_2 \sin \alpha + (R - r_1)\ddot{\psi}_1 \cos(\psi_1 + \alpha) - (R - r_1)\dot{\psi}_1^2 \sin(\psi_1 + \alpha) - g \sin \alpha}{l} \quad (489)$$

Shell Rocking Angular Acceleration ($\ddot{\theta}$): Defining the effective direct rotational inertia M_1 :

$$M_1 = I_s + M_r R_0^2 \quad (490)$$

The explicit equation for $\ddot{\theta}$ is:

$$\ddot{\theta} = \frac{R_0 \left[m_{1p}(R - r_1)\dot{\psi}_1^2 \sin \psi_1 + m_2(R - r_2)\dot{\psi}_2^2 \sin \psi_2 - m_pl\dot{\alpha}^2 \sin \alpha \right] - m_{1p}g(R - r_1) \sin \psi_1 - m_2g(R - r_2) \sin \psi_2 - m_{1p}(R - r_1)R_0\ddot{\psi}_1 \cos \psi_1 - m_2(R - r_2)R_0\ddot{\psi}_2 \cos \psi_2 + m_plR_0\ddot{\alpha} \cos \alpha}{M_1} \quad (491)$$

25 Problem 25: Cable-Suspended Pendulum Coupled to Dual Spring-Mass Pulleys

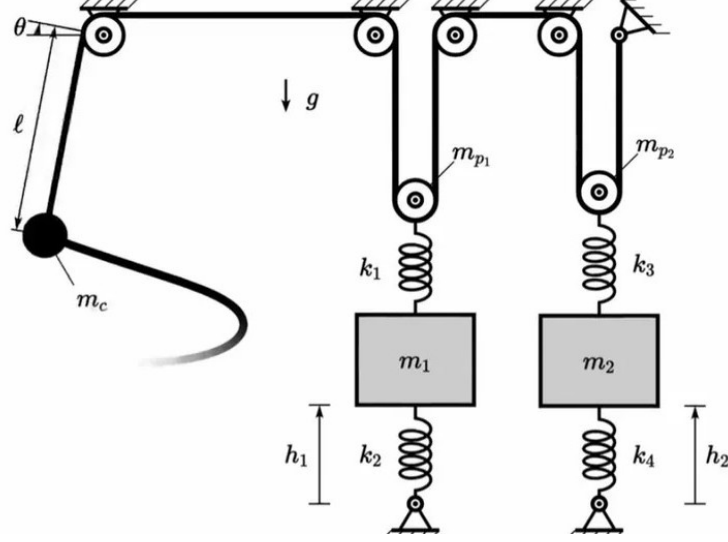


Figure 25: A cable-routed multi-body system: a single continuous inextensible cable suspends an oscillating pendulum bob m_c of variable length l at swing angle θ , passing across three fixed pulleys and looping under two vertically moving spring-mass pulleys m_{p1} and m_{p2} , which are coupled to guided blocks m_1 and m_2 through springs k_1, k_2, k_3, k_4 .

25.1 Generalized Coordinates and Parameters

The planar system possesses five independent generalized coordinates:

- $\theta(t)$: Swing angle of the cable-suspended pendulum bob m_c measured clockwise from the vertical.
- $y_{p1}(t)$: Downward vertical displacement of the first movable pulley of mass m_{p1} .
- $y_{p2}(t)$: Downward vertical displacement of the second movable pulley of mass m_{p2} .
- $x_1(t)$: Downward displacement of block 1 of mass m_1 along its vertical guide.
- $x_2(t)$: Downward displacement of block 2 of mass m_2 along its vertical guide.
- L_{tot} : Total length of the continuous cable routed across the pulley array.
- $l(t)$: Instantaneous suspended length of the pendulum cable segment.
- k_1, k_2, k_3, k_4 : Stiffness values of the four vertical springs.
- l_1, l_2, l_3, l_4 : Unstretched lengths of the four springs.
- h_1, h_2 : Nominal heights of blocks 1 and 2 from the fixed ground datum.
- H : Vertical distance between the ceiling pulley mount and the ground datum.

25.2 Kinematics and Cable Constraint

The continuous cable of total length L_{tot} spans across the fixed ceiling pulleys and loops under the two movable pulleys, enforcing the geometric loop closure constraint:

$$L_{\text{tot}} = l(t) + 2y_{p1}(t) + 2y_{p2}(t) + C \quad (492)$$

where C accounts for the constant horizontal spans between the fixed pulleys. The instantaneous pendulum cable length and its time derivative are:

$$l(t) = L_{\text{tot}} - 2(y_{p1} + y_{p2}) \quad (493)$$

$$\dot{l} = -2(\dot{y}_{p1} + \dot{y}_{p2}) \quad (494)$$

Taking the top-left suspension pulley as the reference origin, the Cartesian position and velocity of the pendulum bob m_c are:

$$x_c = -l \sin \theta \implies v_{cx} = 2(\dot{y}_{p1} + \dot{y}_{p2}) \sin \theta - l\dot{\theta} \cos \theta \quad (495)$$

$$y_c = l \cos \theta \implies v_{cy} = -2(\dot{y}_{p1} + \dot{y}_{p2}) \cos \theta - l\dot{\theta} \sin \theta \quad (496)$$

The squared velocity of the pendulum bob is:

$$v_c^2 = v_{cx}^2 + v_{cy}^2 = 4(\dot{y}_{p1} + \dot{y}_{p2})^2 + l^2\dot{\theta}^2 \quad (497)$$

The four vertical springs undergo the following dynamic extensions:

$$\Delta_1 = H - h_1 - l_1 - x_1 - y_{p1} \quad (498)$$

$$\Delta_2 = h_1 + x_1 - l_2 \quad (499)$$

$$\Delta_3 = H - h_2 - l_3 - x_2 - y_{p2} \quad (500)$$

$$\Delta_4 = h_2 + x_2 - l_4 \quad (501)$$

25.3 Kinetic and Potential Energy Formulations

The total system kinetic energy combines the pendulum bob, the two moving pulleys, and the two guided blocks:

$$KE = \frac{1}{2}m_c \left[4(\dot{y}_{p1} + \dot{y}_{p2})^2 + l^2\dot{\theta}^2 \right] + \frac{1}{2}m_{p1}\dot{y}_{p1}^2 + \frac{1}{2}m_{p2}\dot{y}_{p2}^2 + \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 \quad (502)$$

Taking the ceiling pulley plane as the gravitational datum (y positive downward):

$$\begin{aligned} PE = & -m_c g l \cos \theta - m_{p1} g y_{p1} - m_{p2} g y_{p2} + m_1 g x_1 + m_2 g x_2 \\ & + \frac{1}{2}k_1 (H - h_1 - l_1 - x_1 - y_{p1})^2 + \frac{1}{2}k_2 (h_1 + x_1 - l_2)^2 \\ & + \frac{1}{2}k_3 (H - h_2 - l_3 - x_2 - y_{p2})^2 + \frac{1}{2}k_4 (h_2 + x_2 - l_4)^2 \end{aligned} \quad (503)$$

25.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

1. For pendulum swing angle θ :

$$m_c l^2 \ddot{\theta} - 4m_c l (\dot{y}_{p1} + \dot{y}_{p2}) \dot{\theta} + m_c g l \sin \theta = 0 \quad (504)$$

2. For pulley 1 displacement y_{p1} :

$$(4m_c + m_{p1})\ddot{y}_{p1} + 4m_c\ddot{y}_{p2} + 2m_cl\dot{\theta}^2 + 2m_cg \cos \theta - m_{p1}g - k_1(H - h_1 - l_1 - x_1 - y_{p1}) = 0 \quad (505)$$

3. For pulley 2 displacement y_{p2} :

$$4m_c\ddot{y}_{p1} + (4m_c + m_{p2})\ddot{y}_{p2} + 2m_cl\dot{\theta}^2 + 2m_cg \cos \theta - m_{p2}g - k_3(H - h_2 - l_3 - x_2 - y_{p2}) = 0 \quad (506)$$

4. For block 1 displacement x_1 :

$$m_1\ddot{x}_1 + m_1g + k_2(h_1 - l_2 + x_1) - k_1(H - h_1 - l_1 - x_1 - y_{p1}) = 0 \quad (507)$$

5. For block 2 displacement x_2 :

$$m_2\ddot{x}_2 + m_2g + k_4(h_2 - l_4 + x_2) - k_3(H - h_2 - l_3 - x_2 - y_{p2}) = 0 \quad (508)$$

25.5 Explicit Decoupled Equations of Motion

Isolating the second-order accelerations into closed explicit form:

Pendulum Bob Angular Acceleration ($\ddot{\theta}$):

$$\ddot{\theta} = \frac{4(\dot{y}_{p1} + \dot{y}_{p2})\dot{\theta} - g \sin \theta}{L_{\text{tot}} - 2(y_{p1} + y_{p2})} \quad (509)$$

Movable Pulley Accelerations ($\ddot{y}_{p1}, \ddot{y}_{p2}$): The coupled 2×2 inertia subsystem for the two pulleys possesses the determinant:

$$\mathcal{D}_p = (4m_c + m_{p1})(4m_c + m_{p2}) - 16m_c^2 = m_{p1}m_{p2} + 4m_c(m_{p1} + m_{p2}) \quad (510)$$

Inverting the matrix yields the decoupled accelerations:

$$\ddot{y}_{p1} = \frac{-2m_cm_{p2}(\dot{l}\dot{\theta}^2 + g \cos \theta) + [4m_c(m_{p1} - m_{p2}) + m_{p1}m_{p2}]g + (4m_c + m_{p2})k_1(H - h_1 - l_1 - x_1 - y_{p1}) - 4m_ck_3(H - h_2 - l_3 - x_2 - y_{p2})}{\mathcal{D}_p} \quad (511)$$

$$\ddot{y}_{p2} = \frac{-2m_cm_{p1}(\dot{l}\dot{\theta}^2 + g \cos \theta) + [4m_c(m_{p2} - m_{p1}) + m_{p1}m_{p2}]g + (4m_c + m_{p1})k_3(H - h_2 - l_3 - x_2 - y_{p2}) - 4m_ck_1(H - h_1 - l_1 - x_1 - y_{p1})}{\mathcal{D}_p} \quad (512)$$

Guided Block 1 Acceleration ($\ddot{x}_1$):

$$\ddot{x}_1 = -g + \frac{k_1}{m_1}(H - h_1 - l_1 - x_1 - y_{p1}) - \frac{k_2}{m_1}(h_1 - l_2 + x_1) \quad (513)$$

Guided Block 2 Acceleration ($\ddot{x}_2$):

$$\ddot{x}_2 = -g + \frac{k_3}{m_2}(H - h_2 - l_3 - x_2 - y_{p2}) - \frac{k_4}{m_2}(h_2 - l_4 + x_2) \quad (514)$$

26 Problem 26: Spatial Tracking Mechanism on an Inclined Turntable with Rolling Disc and Spring-Loaded Arm

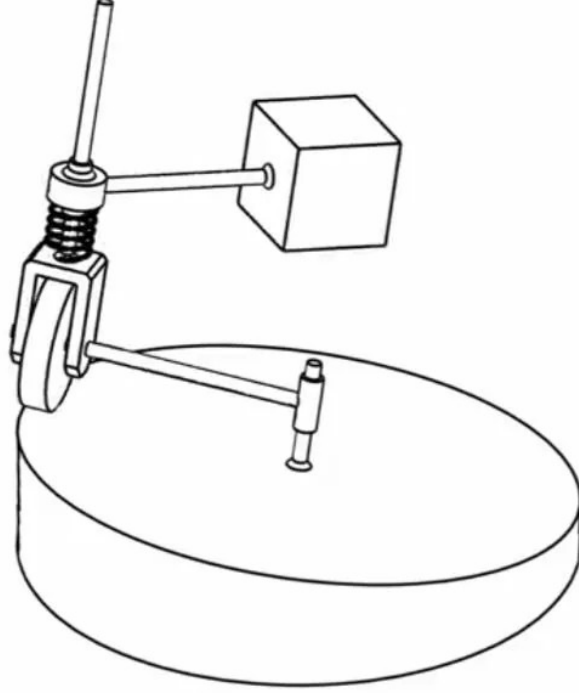


Figure 26: Spatial tracking mechanism on an inclined turntable rotating at rate $\dot{\Omega}$, supporting a rolling wheel m on an arm of length l , with a vertical spring-loaded post of length l_2 carrying a transverse arm l_1 and block mass m_1 .

26.1 Generalized Coordinates and Parameters

The planar/spatial coordinates and kinematic parameters defined in the user reference frame are:

- $\phi_1(t)$: Angular rotation of the main wheel arm of length l along the turntable face.
- $\phi_2(t)$: Angular orientation of the secondary transverse arm of length l_1 carrying block mass m_1 .
- $l_2(t)$: Dynamic axial extension of the spring-loaded post.
- $\dot{\Omega}$: Base turntable angular velocity.
- α : Inclination angle of the reference system.
- m, R, I_m : Mass, radius, and spin moment of inertia of the rolling wheel.
- $I_{m'}$: Mass moment of inertia of the wheel fork carrier about the pivot.
- m_1, I_{m1} : Mass and moment of inertia of the transverse block assembly.
- I_M : Mass moment of inertia of the main rotating turntable.

- k, l_{20} : Stiffness and unstretched length of the axial spring.
- g : Acceleration due to gravity.

26.2 Kinematics

The rolling contact condition relates the spin angular velocity of the wheel ω_{spin} to the relative sweep speed:

$$R\omega_{\text{spin}} = l(\dot{\phi}_1 - \dot{\Omega}) \implies \omega_{\text{spin}} = \frac{l}{R}(\dot{\phi}_1 - \dot{\Omega}) \quad (515)$$

The base turntable rotation rate is:

$$\omega_M = \dot{\Omega} \quad (516)$$

The Cartesian velocity components of the wheel center are:

$$v_{Mx} = \frac{d}{dt}(l \cos \phi_1 \cos \alpha) = -l\dot{\phi}_1 \sin \phi_1 \cos \alpha \quad (517)$$

$$v_{My} = \frac{d}{dt}(l \cos \phi_1 \sin \alpha) = -l\dot{\phi}_1 \sin \phi_1 \sin \alpha \quad (518)$$

$$v_{Mz} = \frac{d}{dt}(l \sin \phi_1) = l\dot{\phi}_1 \cos \phi_1 \quad (519)$$

The Cartesian velocity components of the transverse block m_1 located at arm offsets (l, l_1) and axial extension l_2 are:

$$\begin{aligned} v_{m1x} &= \frac{d}{dt} [(l_1 \cos \phi_2 - l \cos \phi_1) \cos \alpha - (l + l_2) \sin \alpha] \\ &= (-l_1\dot{\phi}_2 \sin \phi_2 + l\dot{\phi}_1 \sin \phi_1) \cos \alpha - \dot{l}_2 \sin \alpha \end{aligned} \quad (520)$$

$$\begin{aligned} v_{m1y} &= \frac{d}{dt} [(l_1 \cos \phi_2 - l \cos \phi_1) \sin \alpha + (l + l_2) \cos \alpha] \\ &= (-l_1\dot{\phi}_2 \sin \phi_2 + l\dot{\phi}_1 \sin \phi_1) \sin \alpha + \dot{l}_2 \cos \alpha \end{aligned} \quad (521)$$

$$v_{m1z} = \frac{d}{dt} (l \sin \phi_1 - l_1 \sin \phi_2) = l\dot{\phi}_1 \cos \phi_1 - l_1\dot{\phi}_2 \cos \phi_2 \quad (522)$$

26.3 Kinetic and Potential Energy

Summing the kinetic energy contributions of the turntable, rolling wheel, carrier fork, and sliding mass block:

$$\begin{aligned} KE &= \frac{1}{2}I_M\dot{\Omega}^2 + \frac{1}{2}ml^2\dot{\phi}_1^2 + \frac{1}{2}I_m\frac{l^2}{R^2}(\dot{\phi}_1 - \dot{\Omega})^2 + \frac{1}{2}I_{m'}\dot{\phi}_1^2 \\ &\quad + \frac{1}{2}m_1 \left[\dot{l}_2^2 + l^2\dot{\phi}_1^2 + l_1^2\dot{\phi}_2^2 - 2ll_1\dot{\phi}_1\dot{\phi}_2 \cos(\phi_1 - \phi_2) \right] + \frac{1}{2}I_{m1}\dot{\phi}_2^2 \end{aligned} \quad (523)$$

The total potential energy stored in gravity and the axial spring is:

$$PE = g \sin \alpha [(m - m_1)l \cos \phi_1 + m_1 l_1 \cos \phi_2] + m_1 g (l + l_2) \cos \alpha + \frac{1}{2}k(l_2 - l_{20})^2 \quad (524)$$

26.4 Euler–Lagrange Equations

Applying $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0$:

Define the system inertia coefficients:

$$J_1 = (m + m_1)l^2 + I_{m'} + I_m \left(\frac{l^2}{R^2} \right) \quad (525)$$

$$J_2 = m_1 l_1^2 + I_{m1} \quad (526)$$

$$Y = I_m \left(\frac{l^2}{R^2} \right) \ddot{\Omega} \quad (527)$$

1. For wheel sweep angle ϕ_1 :

$$J_1 \ddot{\phi}_1 - m_1 l l_1 \ddot{\phi}_2 \cos(\phi_1 - \phi_2) - m_1 l l_1 \dot{\phi}_2^2 \sin(\phi_1 - \phi_2) - (m - m_1)g \sin \alpha \sin \phi_1 - Y = 0 \quad (528)$$

2. For transverse block angle ϕ_2 :

$$J_2 \ddot{\phi}_2 - m_1 l l_1 \ddot{\phi}_1 \cos(\phi_1 - \phi_2) + m_1 l l_1 \dot{\phi}_1^2 \sin(\phi_1 - \phi_2) - m_1 g l_1 \sin \alpha \sin \phi_2 = 0 \quad (529)$$

3. For axial spring displacement l_2 :

$$m_1 \ddot{l}_2 + k(l_2 - l_{20}) + m_1 g \cos \alpha = 0 \quad (530)$$

26.5 Explicit Decoupled Equations of Motion

Defining the coupling determinant:

$$D = J_1 J_2 - m_1^2 l^2 l_1^2 \cos^2(\phi_1 - \phi_2) \quad (531)$$

The decoupled second-order accelerations are:

Wheel Angular Acceleration ($\ddot{\phi}_1$):

$$\ddot{\phi}_1 = \frac{J_2 \left[m_1 l l_1 \dot{\phi}_2^2 \sin(\phi_1 - \phi_2) + (m - m_1)g \sin \alpha \sin \phi_1 + Y \right] + m_1 l l_1 \cos(\phi_1 - \phi_2) \left[-m_1 l l_1 \dot{\phi}_1^2 \sin(\phi_1 - \phi_2) + m_1 g l_1 \sin \alpha \sin \phi_2 \right]}{D} \quad (532)$$

Transverse Arm Angular Acceleration ($\ddot{\phi}_2$):

$$\ddot{\phi}_2 = \frac{m_1 l l_1 \cos(\phi_1 - \phi_2) \left[m_1 l l_1 \dot{\phi}_2^2 \sin(\phi_1 - \phi_2) + (m - m_1)g \sin \alpha \sin \phi_1 + Y \right] + J_1 \left[-m_1 l l_1 \dot{\phi}_1^2 \sin(\phi_1 - \phi_2) + m_1 g l_1 \sin \alpha \sin \phi_2 \right]}{D} \quad (533)$$

Axial Spring Linear Acceleration ($\ddot{l}_2$):

$$\ddot{l}_2 = -\frac{k}{m_1}(l_2 - l_{20}) - g \cos \alpha \quad (534)$$